Trace-Tempered Choice

A Self-Contained Axiomatisation with Explicit Payoffs and Records

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Abstract

An action can matter twice: through the payoff it produces and through the visible trace it leaves. We separate these roles in the primitive domain. A finite channel maps actions into pairs consisting of a payoff-bearing outcome and an explicit record. Continuous weak order, coherent comparison environments, data processing on both sides of the channel, and a branchwise sure-thing principle characterize

$$\mathbb{E}[u(O)] + \lambda I(A; O, R), \qquad \lambda > 0.$$

No primitive payoff–trace separability axiom is required. It is derived from the temporal asymmetry of the domain: records may arrive first, while payoff is terminal. The proof isolates that derivation in a calibrated branch-recursion lemma. A complete proof of the mutual-information characterization for the pure-trace restriction is included as an appendix, so the paper is logically independent of earlier versions of the traceability paper. Every primitive axiom is shown to be independent of the others.

Contents

1 Introduction

An action can affect welfare and also leave evidence of its authorship. These roles are usually conflated because material consequences are also observable. This paper separates them. Payoff-bearing outcomes belong to a common set O; auxiliary visible records belong to a set R; and an unrestricted channel $K : A \to \Delta(O \times R)$ permits arbitrary correlation between them. The distinction allows one to ask whether an agent values the statistical trace of her action while retaining ordinary preferences over payoff risk.

Dostoevsky's Underground Man supplies a provocative limiting case: he is willing to risk material "cakes" so that his conduct can attest that he is not the mechanical implication of a table [21]. The evidentiary idea is useful even without that drama. An action may be valuable partly because a later public or private record makes its authorship legible. The model below takes this idea in a deliberately narrow statistical sense.

We call that motive *traceable agency.* It requires two ingredients. The realised action must be uncertain ex ante, and the visible consequence (O, R) must discriminate statistically among actions. Either can fail separately: a nondegenerate action lottery leaves no trace through a channel with identical rows, while a perfectly revealing record conveys no new information when the action was already certain. Mutual information captures exactly their conjunction,

$$I(A; O, R) = H(q) - \mathbb{E}\big[H(q(\cdot \mid O, R))\big].$$

The central question is whether behavioural restrictions identify this information term jointly with a material expected-utility term, rather than assuming their additivity.

The answer is affirmative. Preferences remain indexed by a fixed environment K and rank only lotteries over the agent's own actions. Weak order, continuity, coherent labelled-block comparisons, data processing on the visible and action sides of the channel, and branchwise continuation consistency imply

$$V(q, K) = \mathbb{E}_{q,K}[u(O)] + \lambda I_{q,K}(A; O, R), \qquad \lambda > 0.$$

Material relevance makes u nonconstant and positive trace orientation fixes the sign of λ. The pair is unique up to a common positive affine change of units. Thus a single information price is meaningful across priors, channels, and finite action alphabets.

The main identification issue is separability. If payoff and record were not distinguished in the primitive domain, data processing could garble a payoff and would conflict with ordinary material preferences. Once the two coordinates are explicit, record processing is permitted to read—but not alter—the realised payoff. A continuation axiom then uses the temporal asymmetry that a visible record may resolve first whereas payoff is terminal. Public-coin independence, pure-action consequentialism, common payoff units, and matching neutrality are all derived. In particular, no axiom directly asserts that payoff value and trace value add.

A donor example fixes the interpretation. A gift can be routed to a named project or to a pooled fund. Its material effect is the payoff outcome; a public attribution is an auxiliary record. Holding the distribution of material effects fixed, a record has positive trace value only if it is statistically informative about a genuinely uncertain route. An attribution that merely confirms an action known in advance carries zero mutual information. Conversely, if material consequences themselves differ across routes, they remain part of the trace even after the auxiliary record is deleted. The explicit (O, R) decomposition makes both observations transparent.

The representation also differs from expected utility in an experimentally sharp way. Suppose two action lotteries induce the same distribution of payoffs and the same marginal distribution of records. They need not induce the same joint relation between action and visible consequences. A lottery over two actions with distinct records may carry one bit of trace, while a lottery over two actions with identical record distributions carries none. Expected utility is indifferent; trace-tempered choice is not. A clean design therefore holds material consequences and the marginal record distribution fixed, varies only action–record informativeness, and includes a garbling treatment in which the record is drawn independently of the action.

Evidence on the intrinsic value of decision rights and payoff autonomy is substantial [6, 12, 41], but those designs do not independently vary what consequences reveal about the act. Experiments on high-agency environments come closer: under a uniform prior over two actions, their Jensen–Shannon divergence is mutual information [38, 40]. Yet those designs also permit instrumental flexibility, and a later model comparison favours dynamic expected utility [36]. The present theorem isolates the non-instrumental hypothesis and states what additional payoff and record variation is required to identify it.

The proof has two layers. The main text first quotients joint channels by their payoff–posterior terminal laws, derives mixture independence from branchwise continuation, calibrates a common expected-utility scale, and then uses a branch recursion to combine payoff and trace. The recursion reduces its information component to a pure-record submodel. The second layer, included in full in the appendix, proves from ordinal preferences that the only coherent positive pure-record ranking is mutual information. It constructs affine representations on posterior-law fibres, aligns their scales through compound experiments, derives the chain rule, and obtains Shannon entropy from Faddeev's recursion. Including this layer makes the present paper self-contained; no result from a discontinued version is used as a black box.

2 Related literature

The paper belongs to the decision-theoretic tradition that enriches the object of preference beyond final material payoffs. Menu models represent flexibility and uncertainty about future tastes [34, 19]; freedom-of-choice models rank opportunity sets [42]; and temporal lotteries permit the timing of information itself to matter [35, 28]. Commitment and temptation provide another preference extension [29]. Here the feasible actions are fixed. What varies is ex-post authorship: how strongly the realised visible consequence identifies the action. Deliberate randomisation has also been axiomatised through a hedging motive [16], but that motive does not have the observability "kill switch" implied by $I(A; O, R) = 0$ when the channel has identical rows.

The nearest behavioural neighbours lie in signalling and image models. Actions convey hidden characteristics in the tradition of Spence [48]; public and self-image can make attribution instrumentally valuable [27, 8, 9, 3]; and conformity can make legibility costly [10]. In those models, inference matters through esteem, sanctions, or self-regard. Here no observer response enters utility. Beliefs provide the statistical yardstick for the trace, and the preference can survive anonymity. Attribution and expressive value remain possible microfoundations, but are not theorem primitives. Models that directly value beliefs or belief paths offer a complementary route [15, 22].

The relation to axiomatic information theory is different. Classical characterisations begin with a numerical functional and impose continuity, grouping, additivity, or functoriality [46, 32, 23, 44, 1, 31, 7, 5, 18]. This paper begins with ordinal preferences. Mixture affinity, product coherence, the chain rule, and Faddeev's recursion are conclusions of behavioural restrictions. The exercise is therefore closer in form to expected-utility theory: the numerical information criterion is an output, not a primitive.

The value-of-information literature typically ranks signals through a downstream decision problem [26, 4, 13]. Rational-inattention and information-cost models similarly treat information as an instrumental input and often begin with a cardinal cost [11, 43, 14, 20, 24]. By contrast, the present prior is a lottery over the agent's own action; visible consequences reveal the action rather than a state revealing how to act; and data processing is a restriction on taste rather than a theorem that finer information can be ignored. Reversing its orientation yields the coherent deniability interpretation discussed below.

Finally, at zero payoff weight the maximised criterion is Shannon channel capacity, the empowerment index used in artificial intelligence and complex systems [33, 45]. The present contribution is a preference foundation for the entire map $q \mapsto I_{q,K}(A; O, R)$, not only its maximum, and a separation of the payoff and trace coordinates that permits payoff-tempered and constrained behaviour. Related free-energy approaches also use variational information objectives [25].

3 Environment

Fix a finite, common set O of *payoff-bearing outcomes*, with $|O| \geq 2$. The lower bound permits material relevance: two sure payoff outcomes can be strictly ranked. An environment consists of a nonempty finite action set A, a nonempty finite set R of explicit visible records, and an unrestricted joint channel

$$K : A \longrightarrow \Delta(O \times R).$$

The singleton record set $R = \{*\}$ represents the absence of any auxiliary record. We do not take R literally empty, because then $O \times R$ is empty and no probability channel into $\Delta(O \times R)$ exists. The agent chooses a lottery $q \in \Delta(A)$. The primitive behavioural datum is the family of binary relations

$$\succeq := \{\succeq_{A,R,K} : A, R \text{ finite and nonempty},\ K : A \to \Delta(O \times R)\},$$

where $\succeq_{A,R,K}$ is a binary relation on $\Delta(A)$. We suppress the redundant alphabet indices and write $\succeq_K$. Thus each relation compares lotteries while holding its channel fixed; neither the channel nor the environment is an object of choice.

Comparison environments. Preferences are indexed by fixed channels, so channels are compared by placing them in labelled blocks. If $K : A \to \Delta(O \times R)$ and $L : B \to \Delta(O \times S)$, then $K \sqcup L$ has action set $(A \times \{0\}) \cup (B \times \{1\})$, payoff set O, and record set $(R \times \{0\}) \cup (S \times \{1\})$. It applies the appropriate channel inside each block and writes the block tag only in the record coordinate. Write q^0 and p^1 for the block-supported copies of $q \in \Delta(A)$ and $p \in \Delta(B)$. We use the shorthand

$$(q, K) \succeq (p, L) \quad \Longleftrightarrow \quad q^0 \succeq_{K \sqcup L} p^1.$$

Finite multiple blocks are defined analogously. This notation does not make a channel an object of choice: it is a compact name for an ordinary comparison in one larger fixed environment.

Three operations. The axioms use the following observable operations.

First, a *record processor* is a stochastic kernel $T(r' \mid o, r)$. It may read the payoff outcome but cannot alter it:

$$(KT)(o, r' \mid a) := \sum_r K(o, r \mid a) T(r' \mid o, r). \tag{1}$$

Allowing T to depend on o is important. It permits an outcome already visible to be copied into, or deleted from, the explicit record; it does not permit a payoff to be changed.

Second, let $S : A \to \Delta(B)$ be a stochastic report of the realised action and let $(qS)(b) := \sum_a q(a)S(b \mid a)$. A channel $\widehat{K} : B \to \Delta(O \times R)$ is the corresponding *Bayesian action processing* when

$$(qS)(b)\widehat{K}(o, r \mid b) = \sum_a q(a)S(b \mid a)K(o, r \mid a) \quad \text{for all } (b, o, r). \tag{2}$$

Rows with $(qS)(b) = 0$ are unrestricted.

Third, a first-stage record channel $P : A \to \Delta(Y)$ may be followed, after record y, by a terminal channel $K^y : A \to \Delta(O \times R_y)$. The compound channel reports both stages and pays only at the terminal stage:

$$(P \triangleright \{K^y\})(o, (y, r) \mid a) := P(y \mid a)K^y(o, r \mid a). \tag{3}$$

For a prior q, write $m(y) := \sum_a q(a)P(y \mid a)$ and $q_y(a) := q(a)P(y \mid a)/m(y)$ on reached branches.

4 Axioms

The axioms below are restrictions on the entire family $\succeq$. The first three make the comparison language coherent. The next three say that descriptions cannot create agency and that sequential problems are evaluated branch by branch. The last two orient the payoff and trace components.

(A1) Weak order. For every finite joint channel K, $\succeq_K$ is complete and transitive on $\Delta(A)$.

(A2) Continuity. For each fixed triple of alphabets (A, O, R), the set

$$\{(K, q, p) : q \succeq_K p\} \subseteq \Delta(O \times R)^A \times \Delta(A)^2$$

is closed.

(A3) Block-comparison coherence. Duplicating a channel into two labelled blocks preserves its ranking:

$$q \succeq_K p \iff q^0 \succeq_{K \sqcup K} p^1.$$

Moreover, the comparison between two block-supported lotteries is unchanged when unused blocks are added or deleted. Thus, for distinct i, j in a finite block environment $\bigsqcup_k K_k$,

$$q_i^i \succeq_{\bigsqcup_k K_k} q_j^j \iff q_i^0 \succeq_{K_i \sqcup K_j} q_j^1.$$

(A4) Record data processing. For every record processor T as in (1),

$$(q, K) \succeq (q, KT).$$

Rewriting what is visible, after the payoff has been fixed, cannot create a stronger trace of the action.

(A5) Action data processing. For every stochastic action report S and every completion $\widehat{K}$ satisfying (2),

$$(q, K) \succeq (qS, \widehat{K}).$$

A noisy relabelling of the realised action cannot make the relation between action and visible consequences stronger. Because (2) preserves the O-marginal, this comparison holds payoff fixed.

(A6) Branchwise continuation consistency. Fix a first-stage record channel P, a prior q, and two continuation profiles $\{K^y\}$ and $\{L^y\}$ with the same terminal payoff set O. If

$$(q_y, K^y) \succeq (q_y, L^y) \quad \text{for every reached } y,$$

then

$$(q, P \triangleright \{K^y\}) \succeq (q, P \triangleright \{L^y\}).$$

If one reached branch comparison is strict, the compound comparison is strict.

Only records may precede the branch. Payoff is terminal. The axiom is therefore an observable sure-thing principle: holding the first-stage record technology fixed, a branchwise improvement is an aggregate improvement.

(A7) Material relevance. For some $o^+, o^- \in O$, the one-action channel paying o^+ for sure is strictly preferred to the one paying o^- for sure.

(A8) Positive trace orientation. For every finite A with $|A| \geq 2$, every full-support $q \in \Delta(A)$, and every $o \in O$, full revelation of the action in the record is strictly preferred to an uninformative record when the payoff is constantly o:

$$(q, K_o^{\mathrm{id}}) \succ (q, K_o^0),$$

where $K_o^{\mathrm{id}}(o, a \mid a) = 1$ (with record alphabet A) and $K_o^0(o, * \mid a) = 1$.

Remark 1 (Why there is no separability axiom)**.** There is no primitive assertion that payoff and trace enter additively, and no primitive public-coin independence axiom. Record data processing identifies experiments with the same payoff–posterior terminal law. Branchwise consistency then makes continuation values affine. Finally, action data processing makes independent dummy actions neutral and aligns the affine scales across priors. These facts force a common material scale and permit the pure-trace scale to calibrate every continuation branch. Lemma 13 gives the argument.

5 Main result

For $q \in \Delta(A)$ and $K : A \to \Delta(O \times R)$, let

$$p_{qK}(o, r) := \sum_a q(a) K(o, r \mid a)$$

be the induced distribution of visible consequences, and define

$$I_{qK}(A; O, R) := \sum_{a,o,r} q(a) K(o, r \mid a) \log \frac{K(o, r \mid a)}{p_{qK}(o, r)}. \tag{4}$$

Summands with $q(a)K(o, r \mid a) = 0$ are set to zero, the logarithm has any fixed base, and expectations below are taken under the joint distribution $q(a)K(o, r \mid a)$.

Theorem 2 (Trace-tempered choice)**.** *The family $\succeq$ satisfies Axioms* (A1)–(A8) *if and only if there are a nonconstant function $u : O \to \mathbb{R}$ and a number $\lambda > 0$ such that, for every nonempty finite A, R, every $K : A \to \Delta(O \times R)$, and every $q, p \in \Delta(A)$,*

$$q \succeq_K p \iff V(q, K) \geq V(p, K), \qquad V(q, K) := \mathbb{E}_{q,K}[u(O)] + \lambda I_{qK}(A; O, R). \tag{5}$$

The same formula represents every block-supported cross-channel comparison.

The pair (u, λ) is unique up to a common positive affine change of units:

$$\widetilde{u} = \alpha u + \beta, \qquad \widetilde{\lambda} = \alpha\lambda, \qquad \alpha > 0.$$

If two payoff outcomes are assigned numerical utilities in advance, λ is uniquely measured in those payoff units per unit of information.

For the following consequence, write

$$\ell_{qK}(o) := \sum_r p_{qK}(o, r),$$
$$\pi^{qK}_{or}(a) := \frac{q(a)K(o, r \mid a)}{p_{qK}(o, r)} \quad \text{when } p_{qK}(o, r) > 0,$$
$$\mu_{qK} := \sum_{(o,r):p_{qK}(o,r)>0} p_{qK}(o, r)\, \delta_{\pi^{qK}_{or}}.$$

Thus ℓ_{qK} is the payoff marginal and μ_{qK} is the distribution of reached posteriors. The familiar entropy identity is

$$I_{qK}(A; O, R) = H(q) - \int H(\pi)\, d\mu_{qK}(\pi).$$

Corollary 3 (Derived matching neutrality)**.** *Suppose Axioms* (A1)–(A8) *hold. Fix an action set A and prior q. If two channels K, L have equally valued payoff lotteries and the same law of terminal posteriors,*

$$\sum_o \ell_{qK}(o)u(o) = \sum_o \ell_{qL}(o)u(o), \qquad \mu_{qK} = \mu_{qL},$$

then $(q, K) \sim (q, L)$. In particular, how payoff labels are matched with posterior atoms has no additional value.

Thus separating O from R is sufficient for identification, but an additional separability axiom is not necessary. The separation is in the domain and in the timing restriction of Axiom (A6): records can resolve first; payoff cannot.

The proof of Theorem 2, including the derivation of Corollary 3, is in Appendix A.

6 Choice implications

The representation determines optimal stochastic choice as well as rankings. Let $Z := O \times R$, write $K_Z(z \mid a) := K(o, r \mid a)$ for $z = (o, r)$, and define

$$\bar{u}(a) := \sum_{o,r} K(o, r \mid a)u(o), \qquad m_{q,K}(z) := \sum_a q(a)K_Z(z \mid a).$$

Proposition 4 (Distinctiveness condition)**.** *Let $\lambda > 0$. The objective $q \mapsto V(q, K)$ is concave and attains a maximum. A lottery q^* is optimal if and only if there is a constant c such that*

$$\bar{u}(a) + \lambda D_{\mathrm{KL}}(K_Z(\cdot \mid a) \,\|\, m_{q^*,K}) = c \quad \textit{for } a \in \operatorname{supp}(q^*), \qquad \leq c \quad \textit{for } a \notin \operatorname{supp}(q^*),$$

where divergence is interpreted in the extended sense.

Proof. Expected utility is linear in q, while mutual information is concave in the input distribution of a fixed channel [17]. Continuity on the compact simplex gives existence. Using

$$I_{q,K}(A;Z) = \sum_a q(a) D_{\mathrm{KL}}(K_Z(\cdot \mid a) \,\|\, m_{q,K}) \,,$$

its directional derivative in the mass on action a equals the displayed divergence up to a term common to every action. The result is the Kuhn–Tucker condition for a concave programme. At constant u it reduces to the classical capacity condition. □

The proposition gives a *distinctiveness premium*. On the support, expected payoff plus λ times the divergence between an action's visible-consequence distribution and the endogenous average is equalised. An action can therefore receive more probability because it pays more or because its consequences are more distinctive. Neither nondegeneracy nor full support is guaranteed: sufficiently large payoff differences can make a pure action optimal even under an informative channel. The condition mirrors rational-inattention first-order conditions with the channel direction and sign reversed [47, 37].

The trace motive has an observability kill switch. If all rows of K_Z are identical, then $I_{q,K} = 0$ for every q, and choice reduces to expected utility. Garbling only the auxiliary record need not switch the motive off, because payoff-bearing outcomes can themselves reveal the action. An empirical zero-information baseline must make the whole visible pair (O, R) independent of A, while preserving the desired payoff comparison.

The explicit payoff–record model also makes λ empirically meaningful. Standard payoff lotteries can calibrate u up to its usual affine normalisation. A separate treatment can then hold the payoff lottery fixed, vary a non-payoff record, and introduce payoff offsets to elicit willingness to pay for trace. The uniqueness part of Theorem 2 ensures that the same λ is transported across finite environments once payoff units and the logarithm base are fixed.

At constant material utility, maximising V is equivalent to maximising $I(A;Z)$. The indirect value is therefore the Shannon capacity of the visible-consequence channel, or empowerment. The theorem strengthens that connection: it represents every lottery, including constrained and payoff-tempered choices, rather than positing capacity only as an objective over environments.

7 Interpretation and scope

Traceability and deniability. The appendix records a sign trichotomy for the pure-record system. Reversing both positive orientation and data processing yields $-I$, a preference for deniability; leaving both unoriented yields indifference. In the present main theorem the maintained directions select $\lambda > 0$. A negative trace coefficient would require the corresponding reversal of the trace-facing axioms, not merely a negative parameter inserted into the conclusion. With unconstrained choice, a trace-averse agent can always eliminate information by choosing a pure action, so deniability has its sharpest content for comparisons at a fixed nondegenerate prior or under constrained randomisation.

Two readings of the lottery. The primitive q can describe a mixing device deliberately operated by the agent. Then trace value can generate deliberate randomisation because a degenerate lottery has zero information. Alternatively, q can describe an observer's prior or a population frequency against which a realised action is evaluated. This reading covers settings where a donor or worker does not literally toss a coin. The protocols are not interchangeable: under the second, q is exogenous rather than chosen. The theorem evaluates trace against the specified ensemble; it does not explain how an observer forms that prior. The first reading connects to evidence on deliberate randomisation [2]; the second connects to preferences over belief paths [22]. Both are prospective and evidentiary, unlike a purely retrospective reinterpretation of a fixed act as willed [39].

Trace and control. Mutual information is symmetric. Within this domain, a taste for being learned about—visible consequences revealing actions—is observationally equivalent to a taste for statistical influence—actions predicting visible consequences. This explains why the decoder-side language of traceable agency and the encoder-side language of empowerment reach the same functional. It also limits causal interpretation: distinguishing attribution from control would require intervention or counterfactual primitives absent from the channel-indexed preference domain.

Empowerment is exercised, not latent, control. Capacity is sometimes glossed as a value of keeping options open. That gloss is not delivered by the representation. The criterion values the lottery actually used. Actions assigned zero probability are inert, and deleting every action outside a capacity-achieving support leaves indirect value unchanged. The axioms therefore identify a taste for exercised distinguishability of consequences, not a strict premium for an unexercised menu. A genuine option value would require preferences over opportunity sets or future decision problems.

Why explicit records matter. The distinction between O and R is not cosmetic. It permits record DPI without permitting the agent's payoff to be garbled, and permits terminal payoff to calibrate continuation scales. Yet the information term properly includes both coordinates because payoff realisations are visible too. Separating records from outcomes is thus required for the behavioural identification strategy, while recombining them as $Z = (O, R)$ in mutual information is required for the substantive notion of trace.

8 Independence of the axioms

The next proposition treats each displayed axiom as one behavioural group. The countermodels use a fixed nonconstant u and write $U(q, K) := \mathbb{E}_{q,K}[u(O)]$, $Z := (O, R)$, and $I := I(A; Z)$. All scores are extended to block and compound channels by the same formula.

Proposition 5 (Individual independence)**.** *For each $i \in \{1, \ldots, 8\}$ there is a family satisfying the other seven axioms and violating Axiom* (Ai)*.*

Proof. Failure of Axiom (A1)*.* Use the Pareto relation on the two coordinates (U, I):

$$(q, K) \succeq (p, L) \quad \Longleftrightarrow \quad U(q, K) \geq U(p, L) \text{ and } I(q, K) \geq I(p, L).$$

It is transitive but incomplete whenever one alternative has more payoff and the other more information. Its graph is closed. Both data-processing axioms are coordinatewise monotonic, and both chain rules make branchwise weak and strict improvements aggregate. The two relevance axioms and block coherence hold.

Failure of Axiom (A2)*.* Order alternatives lexicographically by (U, I), with payoff first. This is a weak order and satisfies all structural, processing, continuation, and relevance requirements. It is not continuous even within one fixed channel. Let a three-action channel reveal the action fully and let only action 2 pay the high outcome. Put

$$p = (1/4, 1/2, 1/4), \qquad q = (0.49, 0.50, 0.01), \qquad q_n = (0.49 - \varepsilon_n, 0.50 + \varepsilon_n, 0.01),$$

where $\varepsilon_n \downarrow 0$. The lotteries p and q have the same expected utility but $H(q) < H(p)$, so $p \succ_K q$. Every q_n has strictly higher expected utility than p, so $q_n \succ_K p$. Thus the graph in Axiom (A2) is not closed.

Failure of Axiom (A3)*.* Treat block labels as syntactic tags, with the outermost tag taking precedence. Put $w(a^0) = 1$, $w(a^i) = 0$ for $i \neq 0$, and $w(a) = 0$ on untagged actions. For $\varepsilon > 0$, use

$$V_\varepsilon(q, K) := U(q, K) + I(q, K) + \varepsilon \sum_a q(a) w(a).$$

Every record- or action-processing comparison places the original alternative in block 0, so the data-processing inequality receives the additional ε. If Δ_y is the base-score difference in continuation branch y, its block comparison is weak exactly when $\Delta_y + \varepsilon \geq 0$, while the aggregate block difference is

$$\sum_y m(y)\Delta_y + \varepsilon = \sum_y m(y)(\Delta_y + \varepsilon).$$

Thus Axiom (A6), including strictness, holds; the remaining axioms are immediate. Duplication coherence fails. Take constant payoff, K revealing a binary action, a degenerate q, and a nearby nondegenerate p with $0 < H(p) < \varepsilon$. Then $p \succ_K q$, but $q^0 \succ_{K \sqcup K} p^1$.

Failure of Axiom (A4). For $\varepsilon > 0$, use

$$V(q, K) := U(q, K) + I(A; Z) - \varepsilon H(Z \mid A). \tag{6}$$

This score is continuous and branch-additive, since both mutual information and conditional entropy obey the chain rule. Under action processing $A \to B$, the score difference between the original and processed alternatives is

$$\big[I(A; Z) - I(B; Z)\big] + \varepsilon\big[H(Z \mid B) - H(Z \mid A)\big] = (1 + \varepsilon)I(A; Z \mid B) \geq 0,$$

so (6) satisfies Axiom (A5). It also satisfies both relevance axioms. Record data processing fails: with constant payoff and a fair record independent of A, the original value is $-\varepsilon$; deleting that noise gives value zero.

Failure of Axiom (A5). Let $G(q) := 1 - \sum_a q(a)^2$ be Gini entropy and define its posterior reduction

$$J_G(q, K) := G(q) - \int G(\pi)\, d\mu_{qK}(\pi), \qquad V := U + J_G.$$

Concavity of G gives record data processing, and the potential-difference form gives the exact branch chain rule. Full revelation is strictly valuable. Action processing can nevertheless increase J_G. Take $q = (1/8, 1/8, 3/4)$, a binary record with

$$\Pr(0 \mid a_1) = \Pr(0 \mid a_2) = 0, \qquad \Pr(0 \mid a_3) = 1/4,$$

and merge a_1, a_2 into one reported action. Direct calculation gives $J_G = 9/416$ before the merge and $J_G = 3/104$ afterwards. Hence Axiom (A5) fails.

Failure of Axiom (A6). Use $V := U + \min\{I, 1\}$, with information measured in bits. It is continuous, respects both data-processing axioms, and has both strict orientations. Let q be uniform on four actions. A first-stage record reveals which of two pairs contains the action. In every branch, revealing the action within the pair strictly improves the continuation from zero to one bit. The two compounds have respectively one and two bits before capping and are therefore indifferent. The strict clause of Axiom (A6) fails.

Failure of Axiom (A7). Use $V := I$. All other axioms hold, but all payoff lotteries are indifferent.

Failure of Axiom (A8). Use $V := U$. All other axioms hold, but revealing the action at a fixed payoff is valueless. □

Remark 6 (What the independence argument says about separability)**.** The record-DPI countermodel (6) and the Gini countermodel separate the two sides of data processing. The capped-information countermodel shows that continuation consistency is a substantive restriction. None of them requires a primitive matching-neutrality axiom. Under all eight axioms, matching neutrality is a theorem; dropping either the temporal or processing discipline permits non-Shannon or nonseparable behaviour.

9 Conclusion

The payoff–record split is doing identification work, but it does not do that work by mechanically assuming additivity. It supplies two independently variable directions and a meaningful temporal order. Record processing removes irrelevant descriptions, action processing fixes the value of uninformative payoff lotteries across priors, and terminal-payoff branch consistency turns those comparisons into a common affine scale. Separability follows; the pure trace theorem then identifies the residual with mutual information.

References

[1] János Aczél and Zoltán Daróczy. *On Measures of Information and Their Characterizations*. Academic Press, New York, 1975.

[2] Marina Agranov and Pietro Ortoleva. Stochastic choice and preferences for randomization. *Journal of Political Economy*, 125(1):40–68, 2017.

[3] James Andreoni and B. Douglas Bernheim. Social image and the 50–50 norm: A theoretical and experimental analysis of audience effects. *Econometrica*, 77(5):1607–1636, 2009.

[4] Yaron Azrieli and Ehud Lehrer. The value of a stochastic information structure. *Games and Economic Behavior*, 63(2):679–693, 2008.

[5] John C. Baez and Tobias Fritz. A Bayesian characterization of relative entropy. *Theory and Applications of Categories*, 29(16):421–456, 2014.

[6] Björn Bartling, Ernst Fehr, and Holger Herz. The intrinsic value of decision rights. *Econometrica*, 82(6):2005–2039, 2014.

[7] John C. Baez, Tobias Fritz, and Tom Leinster. A characterization of entropy in terms of information loss. *Entropy*, 13(11):1945–1957, 2011.

[8] Roland Bénabou and Jean Tirole. Self-confidence and personal motivation. *Quarterly Journal of Economics*, 117(3):871–915, 2002.

[9] Roland Bénabou and Jean Tirole. Incentives and prosocial behavior. *American Economic Review*, 96(5):1652–1678, 2006.

[10] B. Douglas Bernheim. A theory of conformity. *Journal of Political Economy*, 102(5):841–877, 1994.

[11] David Blackwell. Equivalent comparisons of experiments. *Annals of Mathematical Statistics*, 24(2):265–272, 1953.

[12] Justin Buffat, Matthias Praxmarer, and Matthias Sutter. The intrinsic value of decision rights: A replication and an extension to team decision making. *Journal of Economic Behavior & Organization*, 209:560–571, 2023.

[13] Antonio Cabrales, Olivier Gossner, and Roberto Serrano. Entropy and the value of information for investors. *American Economic Review*, 103(1):360–377, 2013.

[14] Andrew Caplin, Mark Dean, and John Leahy. Rationally inattentive behavior: Characterizing and generalizing Shannon entropy. *Journal of Political Economy*, 130(6):1676–1715, 2022.

[15] Andrew Caplin and John Leahy. Psychological expected utility theory and anticipatory feelings. *Quarterly Journal of Economics*, 116(1):55–79, 2001.

[16] Simone Cerreia-Vioglio, David Dillenberger, Pietro Ortoleva, and Gil Riella. Deliberately stochastic. *American Economic Review*, 109(7):2425–2445, 2019.

[17] Thomas M. Cover and Joy A. Thomas. *Elements of Information Theory.* Wiley-Interscience, Hoboken, second edition, 2006.

[18] Imre Csiszár. Axiomatic characterizations of information measures. *Entropy*, 10(3):261–273, 2008.

[19] Eddie Dekel, Barton L. Lipman, and Aldo Rustichini. Representing preferences with a unique subjective state space. *Econometrica*, 69(4):891–934, 2001.

[20] Tommaso Denti. Posterior separable cost of information. *American Economic Review*, 112(10):3215–3259, 2022.

[21] Fyodor Dostoevsky. *Notes from the Underground*, Part I. 1864. Translated by Constance Garnett, 1918.

[22] Jeffrey Ely, Alexander Frankel, and Emir Kamenica. Suspense and surprise. *Journal of Political Economy*, 123(1):215–260, 2015.

[23] D. K. Faddeev. On the concept of entropy of a finite probabilistic scheme. *Uspekhi Matematicheskikh Nauk*, 11(1):227–231, 1956. In Russian.

[24] Alexander Frankel and Emir Kamenica. Quantifying information and uncertainty. *American Economic Review*, 109(10):3650–3680, 2019.

[25] Karl Friston. The free-energy principle: A unified brain theory? *Nature Reviews Neuroscience*, 11(2):127–138, 2010.

[26] Itzhak Gilboa and Ehud Lehrer. The value of information—an axiomatic approach. *Journal of Mathematical Economics*, 20(5):443–459, 1991.

[27] Amihai Glazer and Kai A. Konrad. A signaling explanation for charity. *American Economic Review*, 86(4):1019–1028, 1996.

[28] Simon Grant, Atsushi Kajii, and Ben Polak. Intrinsic preference for information. *Journal of Economic Theory*, 83(2):233–259, 1998.

[29] Faruk Gul and Wolfgang Pesendorfer. Temptation and self-control. *Econometrica*, 69(6):1403–1435, 2001.

[30] I. N. Herstein and John Milnor. An axiomatic approach to measurable utility. *Econometrica*, 21(2):291–297, 1953.

[31] Arthur Hobson. A new theorem of information theory. *Journal of Statistical Physics*, 1(3):383–391, 1969.

[32] Aleksandr I. Khinchin. *Mathematical Foundations of Information Theory.* Dover, New York, 1957.

[33] Alexander S. Klyubin, Daniel Polani, and Chrystopher L. Nehaniv. Empowerment: A universal agent-centric measure of control. In *Proceedings of the 2005 IEEE Congress on Evolutionary Computation*, volume 1, pages 128–135. IEEE, 2005.

[34] David M. Kreps. A representation theorem for preference for flexibility. *Econometrica*, 47(3):565–577, 1979.

[35] David M. Kreps and Evan L. Porteus. Temporal resolution of uncertainty and dynamic choice theory. *Econometrica*, 46(1):185–200, 1978.

[36] Mimi Liljeholm. Flexible control as surrogate reward or dynamic reward maximization. *Cognition*, 229:105262, 2022.

[37] Filip Matějka and Alisdair McKay. Rational inattention to discrete choices: A new foundation for the multinomial logit model. *American Economic Review*, 105(1):272–298, 2015.

[38] Prachi Mistry and Mimi Liljeholm. Instrumental divergence and the value of control. *Scientific Reports*, 6:36295, 2016.

[39] Friedrich Nietzsche. *Also sprach Zarathustra*, Part II, “Von der Erlösung”. Ernst Schmeitzner, Chemnitz, 1883.

[40] Kaitlyn G. Norton and Mimi Liljeholm. The rostrolateral prefrontal cortex mediates a preference for high-agency environments. *Journal of Neuroscience*, 40(22):4401–4409, 2020.

[41] David Owens, Zachary Grossman, and Ryan Fackler. The control premium: A preference for payoff autonomy. *American Economic Journal: Microeconomics*, 6(4):138–161, 2014.

[42] Prasanta K. Pattanaik and Yongsheng Xu. On ranking opportunity sets in terms of freedom of choice. *Recherches Économiques de Louvain*, 56(3–4):383–390, 1990.

[43] Luciano Pomatto, Philipp Strack, and Omer Tamuz. The cost of information: The case of constant marginal costs. *American Economic Review*, 113(5):1360–1393, 2023.

[44] Alfréd Rényi. On measures of entropy and information. In *Proceedings of the Fourth Berkeley Symposium on Mathematical Statistics and Probability*, volume 1, pages 547–561. University of California Press, 1961.

[45] Christoph Salge, Cornelius Glackin, and Daniel Polani. Empowerment—an introduction. In Mikhail Prokopenko, editor, *Guided Self-Organization: Inception*, pages 67–114. Springer, Berlin, 2014.

[46] Claude E. Shannon. A mathematical theory of communication. *Bell System Technical Journal*, 27:379–423, 623–656, 1948.

[47] Christopher A. Sims. Implications of rational inattention. *Journal of Monetary Economics*, 50(3):665–690, 2003.

[48] Michael Spence. Job market signaling. *Quarterly Journal of Economics*, 87(3):355–374, 1973.

Proofs of the main results

A Proof of the representation theorem

The sufficiency proof is modular. The first lemma removes inessential record labels. The second records the precise consequence of Axiom (A6) that was an axiom in many earlier formulations. The next lemmas calibrate material payoffs, import the pure-trace engine, and align continuation scales. Sequentialising an arbitrary joint channel then yields the representation in one line.

Proof notation. For a prior q and joint channel K, define the *terminal law*

$$\eta_{qK} := \sum_{(o,r):p_{qK}(o,r)>0} p_{qK}(o,r)\,\delta_{(o,\pi^{qK}_{or})}. \tag{7}$$

It records the payoff and the posterior about the realised action after everything visible has been observed; its posterior marginal is μ_{qK}, as defined immediately after the main theorem. Bayes plausibility gives

$$\int_{O\times\Delta(A)} \pi\, d\eta_{qK}(o,\pi) = q.$$

Let $\mathcal{E}_q$ be the set of finite-support laws η on $O \times \Delta(A)$ satisfying this barycentre condition. Every $\eta = \sum_i m_i\delta_{(o_i,\pi_i)} \in \mathcal{E}_q$ is implementable: take records i and, for $q(a) > 0$, set

$$K_\eta(o, i \mid a) := \mathbf{1}_{\{o=o_i\}}\frac{m_i\pi_i(a)}{q(a)}. \tag{8}$$

Bayes plausibility makes the rows sum to one; null rows may be completed arbitrarily. Thus terminal laws range over the entire finite Bayes-plausible domain.

For channels K, L on the same action set and $t \in [0, 1]$, let $tK \oplus (1 - t)L$ denote the channel that draws a public coin independent of the action, applies the selected channel, and includes the coin in the record. Then

$$\eta_{q,tK\oplus(1-t)L} = t\eta_{qK} + (1-t)\eta_{qL}. \tag{9}$$

A.1 Terminal-law sufficiency and a derived public coin

Lemma 7 (Terminal-law sufficiency)**.** *Assume Axioms* (A1), (A3), (A4)*. Fix a full-support prior q on A. If $\eta_{qK} = \eta_{qL}$, then $(q, K) \sim (q, L)$. Boundary priors satisfy the same conclusion after deleting null actions under Axiom* (A5)*.*

Proof. For every reached (o, r), replace r by the posterior π^{qK}_{or}. This is a deterministic record processor allowed to depend on o. The resulting canonical channel is

$$K^\eta(o,\pi \mid a) = \frac{\eta_{qK}(o,\pi)\pi(a)}{q(a)}.$$

Conversely, conditional on (o, π), the canonical record can draw an original record r with probability $p_{qK}(o, r)/\eta_{qK}(o, \pi)$ among those records inducing π. This draw is independent of a: Bayes' formula gives $q(a)K(o, r \mid a) = p_{qK}(o, r)\pi(a)$. Hence K and K^η are record processings of one another. Axiom (A4) ranks each weakly above the other; weak order and block coherence give indifference. The same canonical channel is obtained from L when the terminal laws agree. For a boundary q, Axiom (A5) projects to $\operatorname{supp} q$ and embeds back, making null rows irrelevant. □

Corollary 8 (Derived pure-action consequentialism)**.** *Let $q = \delta_a$ and let $\ell_a(o) := \sum_r K(o, r \mid a)$. Then (δ_a, K) is indifferent to the one-action, uninformative-record channel that pays according to ℓ_a. Thus no separate pure-action axiom is needed.*

Proof. Axiom (A5) first restricts the problem to the singleton support $\{a\}$ and embeds it back. On a singleton, Axiom (A4) may delete the record conditional on o; in the reverse direction it redraws r from the conditional distribution induced by $K(\cdot \mid a)$. The two record processings and the two action maps give indifference by weak order and block coherence. $\square$

Lemma 9 (Derived public-coin independence)**.** *Assume Axioms* (A1), (A3), (A4), (A6)*. For a full-support q, channels K, L, M on A, and $t \in (0,1)$,*

$$(q, K) \succeq (q, L) \quad \Longleftrightarrow \quad (q, tK \oplus (1-t)M) \succeq (q, tL \oplus (1-t)M).$$

Proof. Use as first stage an action-independent binary record with probabilities $(t, 1-t)$. Both branch posteriors equal q. Put K versus L on the first branch and the common continuation M on the second. Axiom (A6) gives the forward implication. The compounds have the terminal laws in (9), so Lemma 7 identifies them with the stated public mixtures.

For the converse, if $K \not\succeq_q L$, completeness gives $L \succ_q K$. Apply the strict clause of Axiom (A6) with the continuations exchanged. It makes the reverse public mixture strictly preferred, which contradicts the displayed weak comparison. $\square$

This is the old public-coin axiom derived from the continuation axiom. It is not a ninth primitive.

A.2 Material and product calibration

Lemma 10 (Common material scale and dummy neutrality)**.** *Assume Axioms* (A1)–(A7)*. There is a nonconstant vNM index $u : O \to \mathbb{R}$ such that every channel whose payoff lottery ℓ is independent of the action and whose record is uninformative is ranked by $\sum_o \ell(o)u(o)$, independently of its prior and action alphabet.*

Moreover, independent dummy actions are neutral. If $p \in \Delta(B)$ and $K^{\uparrow}(o, r \mid a, b) := K(o, r \mid a)$, then

$$(q \otimes p, K^{\uparrow}) \sim (q, K). \tag{10}$$

Proof. At a singleton action, Lemma 9 gives mixture independence over payoff lotteries. The public coin is visible, but Lemma 7 flattens it to the ordinary mixed payoff lottery because every terminal posterior is the singleton prior. Axiom (A2) supplies the Archimedean condition, so the finite mixture theorem gives expected utility. Axiom (A7) makes the index nonconstant.

For arbitrary priors q and p, let an action report ignore its input and draw the reported action from p. Applied to an action-independent payoff channel, (2) produces the same payoff lottery at prior p. Axiom (A5) gives one weak comparison; repeating the construction from p to q gives the reverse. Thus the payoff-lottery order is independent of the prior. Block coherence and two-way relabelling transport it across finite action alphabets. Inserting two payoff lotteries into the same reached branch and using the weak and strict parts of Axiom (A6) shows that the insertion is an order isomorphism. The two distinct payoff anchors supplied by nonconstant u will later fix its cardinal multiplier.

For dummy neutrality, project (a, b) to a. Equation (2) and Axiom (A5) give $(q \otimes p, K^{\uparrow}) \succeq (q, K)$. In the other direction, report (a, b) from a by independently drawing $b \sim p$; the Bayesian pushforward is $K^{\uparrow}$, so Axiom (A5) gives the reverse comparison. Weak order and block coherence yield (10). $\square$

Lemma 11 (Affine terminal representation and block bridge)**.** *For every full-support prior q, preferences on $\mathcal{E}_q$ have a continuous affine representative W_q. These representatives may be normalized so that*

$$W_q\left(\sum_o \ell(o)\delta_{(o,q)}\right) = \sum_o \ell(o)u(o). \tag{11}$$

With this normalization, dummy lifting preserves numerical value and the family $\{W_q\}$ represents every block-supported comparison, including comparisons whose priors or action alphabets differ.

Proof. Terminal-law sufficiency turns the order at q into an order on the mixture set $\mathcal{E}_q$. Public-coin independence is mixture independence there. Along any mixture segment the union of the two finite supports is fixed, so Axiom (A2) gives mixture continuity. The mixture-space theorem therefore supplies a continuous affine representative directly on all of $\mathcal{E}_q$. Normalize it by the two distinct payoff values in Lemma 10, which gives (11).

Dummy lifting is an affine ordinal isomorphism between a q-face and its copy in the $q \otimes p$-fibre. Affine uniqueness initially permits a transformation $x \mapsto cx + d$. The two payoff anchors have the same numerical values on both faces by (11); hence $c = 1$ and $d = 0$.

Finally, lift an alternative at $q \in \Delta(A)$ by an independent $p \in \Delta(B)$ and lift an alternative at p by an independent q. Both lifted alternatives now have prior $q \otimes p$ on $A \times B$. Dummy neutrality, block coherence, and transitivity in a common multiple-block environment allow each original alternative to be replaced by its lift. Their block comparison is therefore represented by the single affine index $W_{q \otimes p}$, whose two restrictions equal W_q and W_p. Boundary priors follow by support restriction and a nondegenerate dummy lift. □

A.3 The pure-trace engine

Lemma 12 (Pure-record reduction and affine scale)**.** *Fix $o_0 \in O$ and restrict attention to channels*

$$K_P(o, r \mid a) := \mathbf{1}_{\{o=o_0\}} P(r \mid a).$$

Under Axioms (A1)–(A6), (A8)*, every within-channel and block-supported comparison on this restriction is ordered by $I_{qP}(A; R)$. After the common material scale and dummy normalisation of Lemmas 10–11, the affine value of this restriction is*

$$W(q, K_P) = u(o_0) + \lambda I_{qP}(A; R) \tag{12}$$

for one $\lambda > 0$, common to all priors and finite action sets.

Proof. On a constant-payoff restriction, Axioms (A1)–(A6), (A8) induce Conditions (P1)–(P5) of the pure-record model developed in Appendix B: weak order and positive orientation, continuity, block coherence, record and action data processing, and branchwise continuation. The self-contained pure-record characterization, Theorem 16, therefore gives the ordinal conclusion $I(A; R)$.

By Lemma 11, $W - u(o_0)$ is a common-scale affine representation of the ordinal I order on the constant-payoff domain, so it is $\varphi(I)$ for a strictly increasing φ with $\varphi(0) = 0$. Publicly mixing two experiments at the same prior makes both their posterior laws and their I values mix linearly; affinity gives

$$\varphi(tx + (1-t)y) = t\varphi(x) + (1-t)\varphi(y).$$

Erasure channels at the uniform prior on n actions attain the whole interval $[0, \log n]$. These intervals are nested and unbounded as n grows, so continuity gives $\varphi(x) = \lambda x$ on $\mathbb{R}_+$. Axiom (A8) gives $\lambda > 0$, proving (12). □

A.4 Calibrated branch recursion

Lemma 13 (Calibrated branch recursion)**.** *The common-scale representative W of Lemma 11 satisfies*

$$W(q, P \triangleright \{K^y\}) = \lambda I_q(A; Y) + \sum_{y: m(y) > 0} m(y) W(q_y, K^y). \tag{13}$$

For a continuation that pays o and reveals nothing further, $W = u(o)$.

Proof. Hold all continuations except one reached branch y fixed. Axiom (A6) and its strict converse say that branch insertion is an order isomorphism. Between affine faces, the change in compound value must therefore equal a positive multiplier c_y times the change in continuation value.

This multiplier is independent of the fixed continuations in the other branches. Indeed, if their terminal sublaw is η_0 and the reached branch has mass $m = m(y)$, affinity gives

$$W_q(\eta_0 + m\eta_K) - W_q(\eta_0 + m\eta_L) = m\, L_q(\eta_K - \eta_L),$$

where L_q is the linear part of W_q; the background η_0 cancels.

By the background-independence just shown, reset every other continuation to the deterministic payoff o_0 fixed in Lemma 12 and an uninformative record. Now choose in branch y two continuations with that same payoff and different, nontrivial record experiments. Both the local and aggregate comparisons lie in the pure-record restriction. Equation (12) and the mutual-information chain rule give

$$\Delta W_{\text{compound}} = \lambda m(y)\Delta I_{\text{branch}} = m(y)\Delta W_{\text{branch}}.$$

Thus $c_y = m(y)$. The multiplier belongs to the insertion map, so it applies to every pair of continuations, including payoff changes and mixed payoff–record changes. If q_y is degenerate and has no nontrivial trace direction, append an independent nondegenerate dummy action, use the preceding calibration there, and project it away using (10); the conclusion is unchanged.

Changing branches one at a time now shows that the two sides of (13) differ by a term depending only on the fixed first stage. Evaluate that term with every continuation paying o_0 and revealing nothing. The compound then lies in the pure-record restriction and has value $u(o_0) + \lambda I_q(A;Y)$, while the weighted continuation values sum to $u(o_0)$. The missing term is therefore exactly $\lambda I_q(A;Y)$. □

A.5 Completion of the representation

Proof of Theorem 2. Assume Axioms (A1)–(A8) and fix an arbitrary joint channel K. Write $Z := O \times R$ and sequentialise K as follows. The first-stage record channel is

$$P_K(z \mid a) := K(o, r \mid a), \qquad z = (o, r).$$

After the record $z = (o, r)$, the continuation pays o for sure and reveals nothing further. This does not mean that payoff is paid early: the first record announces the payoff that the terminal continuation subsequently delivers.

The compound channel is indifferent to K. Starting from K, a record processor can copy the already observed pair (o, r) into the compound record; starting from the compound, another processor deletes the copy. Both processors leave the payoff coordinate fixed, so Axiom (A4) applies in both directions. The calibrated branch recursion now gives

$$W(q, K) = \lambda I_q(A;Z) + \sum_z p_{qK}(z)u(o_z) = \lambda I_{qK}(A;O,R) + \mathbb{E}_{q,K}[u(O)].$$

This is (5), including block comparisons by Lemma 11. Axiom (A7) makes u nonconstant and Axiom (A8) makes $\lambda > 0$.

Conversely, let preferences be represented by (5). Weak order and continuity are immediate. Block tags are constant under a block-supported lottery, so they change neither expected utility nor mutual information, proving block coherence. Record processing preserves the payoff marginal and weakly reduces mutual information by data processing. Equation (2) also preserves the payoff marginal, and the Markov chain $(O, R) - A - B$ gives $I(A;O,R) \geq I(B;O,R)$. Thus Axioms (A4) and (A5) hold.

For a compound channel, the two chain rules are

$$\mathbb{E}[u(O)] = \sum_y m(y)\mathbb{E}_{q_y,K^y}[u(O)],$$
$$I(A;Y,O,R) = I(A;Y) + \sum_y m(y)I_{q_y,K^y}(A;O,R).$$

The first-stage term $\lambda I(A;Y)$ is common to the two continuation profiles. Their aggregate value difference is therefore the positive-probability weighted sum of their branch value differences. This proves both clauses of Axiom (A6). Nonconstant u gives Axiom (A7), and $\lambda > 0$ gives Axiom (A8).

For uniqueness, restrict first to uninformative records. Standard expected-utility uniqueness gives $\widetilde{u} = \alpha u + \beta$ with $\alpha > 0$. Subtract β and divide the second representation by α; write its information coefficient as $c := \widetilde{\lambda}/\alpha$. At a nondegenerate prior, an erasure record can produce any sufficiently small information value $t > 0$. Since u is nonconstant, choose payoff lotteries ℓ, m with $\mathbb{E}_m u - \mathbb{E}_\ell u = \lambda t$. The prospect $(\ell, \text{trace } t)$ is indifferent to $(m, \text{no trace})$ under the first representation and hence under the second. Therefore $\mathbb{E}_\ell u + ct = \mathbb{E}_m u$, so $c = \lambda$ and $\widetilde{\lambda} = \alpha\lambda$. $\square$

Corollary 3 follows immediately from the representation and $I = H(q) - \int H(\pi)d\mu$.

Self-contained foundation for the trace component

This appendix contains the pure-record characterization on which Lemma 12 rests. It is included in full so that the present paper has no logical dependency on an earlier manuscript. In this auxiliary model, X is a finite visible-record alphabet, $P : A \to \Delta(X)$ is a record channel, and the primitive relation ranks action lotteries by the trace they leave in X. Conditions (P1)–(P5) below are exactly the restrictions induced on the constant-payoff subdomain by, respectively, (A1),(A8), (A2), (A3), (A4),(A5), and (A6) of the main model.

B The pure-record characterization

Let A be a finite set of *actions* and X a finite set of *records*. A channel is a stochastic kernel $P : A \to \Delta(X)$. For each finite channel P, the primitive behavioural datum is a weak order

$$\succeq_P \quad \text{on} \quad \Delta(A).$$

The interpretation is that $q \succeq_P q'$ means: in the fixed environment P, lottery q displays at least as much traceability as q'. The environment P is not an object of choice. A lottery is the procedure being evaluated; the channel is the fixed observation technology through which the realised act may be recorded by records.

Comparison environments. To compare lotteries carried by different channels without abandoning channel-indexed preferences, place the channels in labelled, disjoint blocks of one larger environment. For channels $P : A \to \Delta(X)$ and $Q : B \to \Delta(Y)$, write $P \sqcup Q$ for the block-diagonal channel with action alphabet $(A \times \{0\}) \cup (B \times \{1\})$ and record alphabet $(X \times \{0\}) \cup (Y \times \{1\})$. It sends $(a, 0)$ to $(x, 0)$ with probability $P(x \mid a)$ and $(b, 1)$ to $(y, 1)$ with probability $Q(y \mid b)$; all cross-block probabilities are zero. For $q \in \Delta(A)$ and $p \in \Delta(B)$, define lotteries on the tagged action alphabet by

$$\begin{aligned} q^0(a,0) &:= q(a), & q^0(b,1) &:= 0, \\ p^1(a,0) &:= 0, & p^1(b,1) &:= p(b), \end{aligned} \qquad (a \in A,\ b \in B).$$

The superscript is a block label, not a power. Accordingly, $q^0 \succeq_{P \sqcup Q} p^1$ is an ordinary preference comparison inside one fixed environment.[1] When the two blocks have the same action alphabet, q^0 and q^1 are the two tagged copies of the same lottery q. For a finite family $\{P_i : A_i \to \Delta(X_i)\}_{i \in K}$, write $\bigsqcup_{i \in K} P_i$ for the analogous finite block environment and q_i^i for the lottery that assigns $q_i(a)$ to each tagged action (a, i) and zero to every other block.

Pure-record conditions. The conditions are imposed directly on the family $\{\succeq_P\}_P$. Five conditions are imposed on the family. The first three are structural: each environment carries a continuous weak order with a fixed orientation (P1, P2), and labelled blocks provide the comparison framework (P3). The substantive restrictions are two: garbling cannot increase traceability (P4), and sequential observation is evaluated branch by branch (P5).

[1] For a complete numerical example, let $A = \{a_1, a_2\}$, $B = \{b_1, b_2\}$, $X = \{x_0, x_1\}$, and $Y = \{y_0, y_1\}$. With rows indexed by actions and columns by records, take $P = \begin{pmatrix} 0.1 & 0.9 \\ 0.8 & 0.2 \end{pmatrix}$ and $Q = \begin{pmatrix} 0.4 & 0.6 \\ 0.6 & 0.4 \end{pmatrix}$. Order the tagged actions as $(a_1, 0), (a_2, 0), (b_1, 1), (b_2, 1)$ and the tagged records as $(x_0, 0), (x_1, 0), (y_0, 1), (y_1, 1)$. Then $P \sqcup Q = \operatorname{diag}(P, Q)$: the first two actions generate only the first two records, and the last two actions only the last two. If $q = (0.3, 0.7)$ and $p = (0.8, 0.2)$, their embeddings are $q^0 = (0.3, 0.7\,;\,0, 0)$ and $p^1 = (0, 0\,;\,0.8, 0.2)$, where the semicolon separates the two blocks. Thus $q^0 \succeq_{P \sqcup Q} p^1$ says precisely that lottery q, observed through P, is at least as traceable as lottery p, observed through Q; it does not treat either channel as an object of choice. Under the representation proved below, the two sides yield approximately 0.330 and 0.019 bits of mutual information, respectively, and hence the comparison is strict in this example.

(P1) Weak order and local non-triviality. For every finite channel $P : A \to \Delta(X)$, $\succeq_P$ is complete and transitive on $\Delta(A)$. The family is locally non-trivial: for every finite action set A with $|A| \geq 2$ and every full-support $q \in \Delta(A)$,

$$q^0 \succ_{\mathrm{Id}_A \sqcup U_A} q^1,$$

where $\mathrm{Id}_A : A \to \Delta(A)$ is full revelation and $U_A : A \to \Delta(\{*\})$ is the one-record uninformative channel.

This condition gives each observation environment an ordinary weak order over act lotteries and fixes the orientation of the comparison. When the realised act is genuinely uncertain, observing it directly is strictly more traceable than observing nothing about it.

(P2) Continuity. For every pair of finite alphabets (A, X), the set

$$\{(P, q, q') \in \Delta(X)^A \times \Delta(A) \times \Delta(A) : q \succeq_P q'\}$$

is closed in the usual Euclidean product topology. Thus within-environment rankings vary continuously with both the lotteries and the fixed channel.

Continuity says that traceability comparisons are stable under small changes in the procedure and the environment. Small modelling perturbations should not create abrupt reversals. The condition is stated entirely in the primitives: it quantifies over channels and lotteries, not over derived Bayesian objects. Continuity of block comparisons in posterior-law convergence at a fixed full-support prior is not assumed; it is derived below (Lemma 24).

(P3) Block-comparison coherence. First, duplicating the same environment into two labelled blocks does not change the original ranking of lotteries: for every channel $P : A \to \Delta(X)$ and every $q, q' \in \Delta(A)$,

$$q \succeq_P q' \quad \Longleftrightarrow \quad q^0 \succeq_{P \sqcup P} (q')^1.$$

Second, block comparisons are invariant to adding or deleting irrelevant blocks. For every finite block environment $\bigsqcup_{k \in K} P_k$, every two distinct blocks $i, j \in K$, and every $q_i \in \Delta(A_i)$ and $q_j \in \Delta(A_j)$,

$$q_i^i \succeq_{\bigsqcup_{k \in K} P_k} q_j^j \quad \Longleftrightarrow \quad q_i^0 \succeq_{P_i \sqcup P_j} q_j^1.$$

The pair (i, j) is ordered. Taking $(i, j) = (1, 0)$ in a two-block environment $P_0 \sqcup P_1$ therefore yields the label-swap identity

$$q_1^1 \succeq_{P_0 \sqcup P_1} q_0^0 \quad \Longleftrightarrow \quad q_1^0 \succeq_{P_1 \sqcup P_0} q_0^1,$$

which is how reversed block comparisons are brought to canonical form throughout.

Labelled blocks are a way to place procedures carried by different channels in a common comparison problem. The labels should not themselves create traceability, alter the ranking inside a duplicated environment, or let unused blocks affect the comparison between active blocks.

(P4) Data-processing aversion. Let $P : A \to \Delta(X)$ be a channel, let $S : A \to \Delta(A')$ and $T : X \to \Delta(X')$ be stochastic kernels, and fix $q \in \Delta(A)$. Write

$$(qS)(a') := \sum_{a \in A} q(a) S(a' \mid a)$$

for the induced distribution of the reported action. Let $P' : A' \to \Delta(X')$ be any channel satisfying

$$(qS)(a')\, P'(x' \mid a') = \sum_{a \in A} \sum_{x \in X} q(a) P(x \mid a) S(a' \mid a) T(x' \mid x) \qquad \text{for every } (a', x') \in A' \times X'. \tag{DP}$$

Then

$$q^0 \succeq_{P \sqcup P'} (qS)^1.$$

Two special cases are used repeatedly and are named for reference.

(DP-rec) *Record post-processing.* For every $P : A \to \Delta(X)$, every stochastic $T : X \to \Delta(X')$, and every $q \in \Delta(A)$, define

$$(PT)(x' \mid a) := \sum_{x \in X} P(x \mid a) T(x' \mid x).$$

Then

$$q^0 \succeq_{P \sqcup PT} q^1.$$

(DP-act) *Action coarsening.* For every $P : A \to \Delta(X)$, every stochastic $S : A \to \Delta(A')$, every $q \in \Delta(A)$, and every $\widehat{P} : A' \to \Delta(X)$ satisfying (DP) with $T = \mathrm{Id}_X$ (the identity kernel on X) and $P' = \widehat{P}$,

$$q^0 \succeq_{P \sqcup \widehat{P}} (qS)^1.$$

When qS has full support, this equation determines $\widehat{P}$ uniquely; otherwise the clause applies to every channel consistent with it. In particular, bijective relabellings are neutral by applying (DP-act) in both directions.

The act whose trace is evaluated may be reported through a possibly noisy description, and the visible record may be garbled after it is generated. Neither operation can create observable trace: the original procedure is weakly more traceable than any procedure obtained from it by processing the act description, the record, or both. The two named clauses are the two ends of a single chain. Equation (DP) says that the compared procedure is obtained from the joint law $q(a)P(x \mid a)$ by garbling the act coordinate through S, the record coordinate through T, or both.

Remark 14. If $(qS)(a') = 0$, equation (DP) leaves the corresponding row of P' unrestricted; null reports therefore require no convention. In the presence of Conditions (P1) and (P3), Condition (P4) is equivalent to the conjunction of its two named clauses. Each clause is a specialisation; conversely, the two clauses chain in a common three-block environment (Lemma 20). Every proof below cites whichever clause it needs by name.

Reached records and posteriors. For a lottery $q \in \Delta(A)$ and a channel $P : A \to \Delta(X)$, write

$$m_{q,P}(x) := \sum_{a \in A} q(a)P(x \mid a), \qquad r_x^{q,P}(a) := \frac{q(a)P(x \mid a)}{m_{q,P}(x)} \quad \text{if } m_{q,P}(x) > 0$$

for, respectively, the probability of record x and the posterior belief about the realised action. Call x reached when $m_{q,P}(x) > 0$; all branch comparisons below are restricted to reached records.

(P5) Branchwise continuation monotonicity. Let $P_1 : A \to \Delta(X_1)$ be a first-stage channel and, after each first-stage record x, let $Q^x, R^x : A \to \Delta(X_2^x)$ be two alternative continuation channels. Write $P_1 \triangleright \{Q^x\}$, and analogously $P_1 \triangleright \{R^x\}$, for the compound channel with record alphabet $\bigcup_{x \in X_1} (\{x\} \times X_2^x)$ that reports both stages:

$$(P_1 \triangleright \{Q^x\})(x, z \mid a) := P_1(x \mid a) Q^x(z \mid a).$$

Fix $q \in \Delta(A)$ and let $m(x) := m_{q,P_1}(x)$ and $r_x := r_x^{q,P_1}$ for all branches with $m(x) > 0$.

If

$$r_x^0 \succeq_{Q^x \sqcup R^x} r_x^1 \qquad \text{for every } x \text{ with } m(x) > 0,$$

then

$$q^0 \succeq_{(P_1 \triangleright \{Q^x\}) \sqcup (P_1 \triangleright \{R^x\})} q^1.$$

If at least one reached branch comparison is strict, then the aggregate comparison is strict.

Sequential observation is evaluated branch by branch. After a first-stage signal, if one continuation leaves at least as much trace as another at every posterior that can be reached, then replacing the continuation branch by branch should not reduce overall traceability. If the improvement is strict on a reached branch, the aggregate comparison is strict.

Remark 15. Condition (P5) plays in this system the role the sure-thing principle plays in Savage's: it is a dominance condition across the branches of a resolving observation, with zero-probability branches unconstrained exactly as Savage's null events are. It is also where all cardinal content resides. The other conditions are ordinal, environment-by-environment conditions; it is (P5), through its derived consequences—public-coin independence (Lemma 23), background inertness (Lemma 30), and the chain rule (Lemma 35)—that aligns the initially separate scales across priors and forces branch-additive aggregation. The strictness clause is not decorative: it is what allows the converse directions of Lemmas 23 and 30 to be extracted from a dominance condition. The independence examples (Proposition 19) include a satiation criterion—mutual information capped at a fixed level—that satisfies (P1)–(P4) and fails exactly (P5). Compound observation environments are accordingly the natural locus of experimental testing: they are where branch-additive and non-additive criteria come apart.

B.1 Pure-record result

Throughout, logarithms are base two, so information is measured in bits; changing the base merely rescales all information values. For a finite distribution x, write

$$H(x) := -\sum_z x(z) \log x(z), \qquad 0 \log 0 := 0,$$

and, for a prior–channel pair (q, P), define

$$I_{q,P}(A; X) := H(m_{q,P}) - \sum_{a \in A} q(a) H(P(\cdot \mid a)).$$

When the alphabets are clear, write $I_{q,P}$; write I for the map $(q, P) \mapsto I_{q,P}$.

Theorem 16 (Mutual-information characterisation)**.** *For a family $\{\succeq_P\}_P$ as described above, the following are equivalent.*

(i) *The family satisfies Conditions* (P1)–(P5)*.*

(ii) *For every finite channel $P : A \to \Delta(X)$ and every $q, q' \in \Delta(A)$,*

$$q \succeq_P q' \iff I_{q,P}(A; X) \geq I_{q',P}(A; X).$$

Moreover, under these equivalent conditions, block-supported cross-channel comparisons are represented on the same scale: for every finite block environment $\bigsqcup_{k \in K} P_k$, every two distinct blocks $i, j \in K$, and every $q_i \in \Delta(A_i)$, $q_j \in \Delta(A_j)$,

$$q_i^i \succeq_{\bigsqcup_{k \in K} P_k} q_j^j \iff I_{q_i,P_i}(A_i; X_i) \geq I_{q_j,P_j}(A_j; X_j).$$

The proof of the equivalence of (i) and (ii) is in the Appendix; the moreover clause is immediate from (ii), since a block-supported lottery has constant block label. The sufficiency argument is the hard part and has three movements. First, *quotient*: block coherence, record post-processing, and a prior-specific Blackwell argument show that at each full-support prior only the induced law of posterior beliefs about the act matters, and that this quotient inherits closedness from Condition (P2) (Lemmas 20–24). Second, *cardinalisation*: public-coin independence—derived from Condition (P5), not assumed—delivers a continuous affine representative on each posterior-law fibre by the mixture-space theorem; derived background inertness of independent components yields a quasi-additive product form; and a tangent-space argument makes branch coefficients path-independent, giving a cocycle, a chain rule, and coherent scales across support faces (Lemmas 25–36). Third, *identification*: the key step is scale alignment. The conditions derive it through data processing and branchwise continuation, rather than through an assumption on product environments. The initially fibrewise affine indices are thereby placed on one cardinal scale; a two-grouping argument eliminates the possible product interaction, and Faddeev's recursion then identifies the value of full revelation with Shannon entropy, hence the index with mutual information (Lemmas 37–39).

To state uniqueness, it is useful to make the common scale explicit. A real-valued index V on all finite prior–channel pairs is a *common-scale representation* if, for every channel $P : A \to \Delta(X)$ and every $q, q' \in \Delta(A)$,

$$q \succeq_P q' \quad \Longleftrightarrow \quad V(q, P) \geq V(q', P),$$

and, for every finite block environment $\bigsqcup_{k \in K} P_k$, every two distinct blocks $i, j \in K$, and all $q_i \in \Delta(A_i)$ and $q_j \in \Delta(A_j)$,

$$q_i^i \succeq_{\bigsqcup_{k \in K} P_k} q_j^j \quad \Longleftrightarrow \quad V(q_i, P_i) \geq V(q_j, P_j).$$

Call a common-scale representation *branch-additive* if, for every finite A, every $q \in \Delta(A)$, every first-stage channel $P_1 : A \to \Delta(X_1)$, and every continuation profile $\{Q^x : A \to \Delta(X_2^x)\}_{x \in X_1}$,

$$V(q, P_1 \triangleright \{Q^x\}) = V(q, P_1) + \sum_{x : m(x) > 0} m(x)\, V(r_x, Q^x),$$

where $m(x) = m_{q,P_1}(x)$ and $r_x = r_x^{q,P_1}$ on every reached branch.

Proposition 17 (Uniqueness)**.** *Let $\{\succeq_P\}_P$ satisfy the equivalent conditions of Theorem 16. For any real-valued index V on all finite prior–channel pairs:*

(i) *V is a common-scale representation if and only if $V = \varphi \circ I$ for a single strictly increasing $\varphi : [0, \infty) \to \mathbb{R}$.*

(ii) *A common-scale representation V is branch-additive if and only if $V = aI$ for some $a > 0$.*

The primitive remains ordinal: part (i) permits any common strictly increasing transformation. Once block-supported procedures are required to be scored by their component-pair values, however, the block comparisons permit only one transformation across environments. Branch additivity is not another behavioural condition; it is an exact accounting requirement on a numerical index. Part (ii) shows that this requirement selects the linear gauge, just as mixture affinity selects the expected-utility index within its ordinal class. It fixes both the zero and the curvature of the information scale, leaving only a positive choice of unit, with bits and nats the standard base-two and base-e conventions.

This normalization is what makes the additive combination in the main model meaningful. If the trace component were represented by a nonlinear transformation $\varphi \circ I$, a coefficient multiplying it would have no stable interpretation. Proposition 17 shows that the branch-additive gauge is $V = aI$; the factor a is absorbed into the main coefficient λ. Once the payoff-utility scale and the information unit are fixed, one effective trade-off parameter can therefore be carried across environments. The proof is at the end of this appendix.

Uniqueness concerns the scale; a complementary observation concerns the sign. The condition system splits into a structural part and an oriented part. Conditions (P2) and (P3) are equivalences or closedness conditions and are invariant under reversal of every preference in the family; Condition (P5) is self-dual, since branchwise monotonicity for the reversed family is the original condition with the two continuation profiles exchanged, and strictness survives the exchange. Only the orientation clause of Condition (P1) and the data-processing condition (P4) carry a sign.

Call an *orientation* of the system a choice, for each of the two signed clauses—the non-triviality clause of (P1) and Condition (P4)—of either its original or its reversed form, or the replacement of both by total indifference on the comparisons of the non-triviality clause.

Corollary 18 (Sign trichotomy)**.** *Let $\{\succeq_P\}_P$ satisfy the weak-order clause of* (P1) *and Conditions* (P2)*,* (P3)*,* (P5)*. Then:*

(i) *under the original orientation of both signed clauses, the family is represented by $I_{q,P}$ (Theorem 16);*

(ii) *under the reversal of both signed clauses, the family is represented by $-I_{q,P}$;*

(iii) *the two mixed orientations are inconsistent: no family satisfies the original* (P4) *together with reversed non-triviality, or reversed* (P4) *together with original non-triviality.*

The totally indifferent family satisfies the structural conditions with both signed clauses replaced by indifference. Hence the structural conditions admit exactly three coherent orientations, represented by I, $-I$, and 0.

Proof. For (ii), the reversed family $\{\preceq_P\}_P$ satisfies (P1)–(P5)—the structural conditions are reversal-invariant and (P5) is self-dual as noted above—so Theorem 16 represents it by $I_{q,P}$, hence the original family by $-I_{q,P}$. For (iii), fix a full-support q on a non-singleton A and apply (P4) to $P = \mathrm{Id}_A$, $S = \mathrm{Id}_A$, and a constant record kernel $T : A \to \Delta(\{*\})$. Then $qS = q$ and the processed channel is U_A, so the original orientation gives

$$q^0 \succeq_{\mathrm{Id}_A \sqcup U_A} q^1,$$

contradicting reversed non-triviality. The reversed orientation of (P4) gives the opposite weak comparison, contradicting original non-triviality. The final claim is the necessity verification in the Appendix for (i), its reversal for (ii), and direct verification for the indifferent family. ☐

Proposition 19 (Independence of the pure-record conditions)**.** *None of Conditions* (P1)–(P5) *is implied by the other four: for each condition there is a family of weak orders that satisfies the remaining four and violates it.*

The five families are constructed in the Appendix. Two of them carry behavioural content of their own and mark the substantive frontier of the system: the Shannon-entropy-scaled criterion $(1 + H(q))I_{q,P}$ satisfies every condition except data processing, valuing the uncertainty of a procedure beyond its trace, and the satiation criterion $\min\{I_{q,P}, c\}$, for any fixed $c > 0$, satisfies every condition except branchwise continuation monotonicity, ceasing to value revelation once c bits are secured.

Taken together, the results measure the freedom the conditions leave, and it amounts to a sign and a unit. Behaviour selects the sign: a family satisfying the structural conditions together with an orientation of the two signed clauses is represented by I, by $-I$, or by total indifference (Corollary 18), and none of the five conditions securing this rigidity is redundant (Proposition 19). Convention selects the unit: within an orientation, the branch-additive representations are exactly the positive multiples of I (Proposition 17). In this sense the formula is less the content of Theorem 16 than the trichotomy is: the conditions leave open not which functional measures traceable agency, but whether the agent seeks it, shuns it, or is indifferent to it. The two oriented arms are not symmetric in behavioural content, however; the Discussion takes up what the negative orientation does and does not deliver.

Embedding into the payoff–record model. Fix any $o_0 \in O$ and identify an auxiliary record channel $P : A \to \Delta(X)$ with the joint channel $K_P(o, x \mid a) = \mathbf{1}_{\{o=o_0\}} P(x \mid a)$. Main Axiom (A4) is (DP-rec), main Axiom (A5) is (DP-act), and main Axiom (A6) is (P5) because every payoff is constant. Thus Theorem 16 applies directly to the restriction used in Lemma 12.

C Proof of the pure-record characterization

C.1 Necessity

For every finite channel $P : A \to \Delta(X)$, define

$$q \succeq^I_P q' \quad \iff \quad I_{q,P}(A; X) \geq I_{q',P}(A; X).$$

We verify that the family $\{\succeq^I_P\}_P$ satisfies Conditions (P1)–(P5): this is the implication (ii)⇒(i) of Theorem 16, and it establishes the consistency of the condition system.

The *posterior law* induced by (q, P) is the finite distribution

$$\mu_{q,P} := \sum_{x: m_{q,P}(x)>0} m_{q,P}(x)\, \delta_{r_x^{q,P}}$$

over posterior beliefs. Bayes plausibility gives $\int r\, d\mu_{q,P}(r) = q$. We repeatedly use the elementary identities

$$I_{q,P} = H(m_{q,P}) - \sum_{a \in A} q(a) H(P(\cdot \mid a)), \tag{N1}$$

and

$$I_{q,P} = H(q) - \int_{\Delta(A)} H(r)\, d\mu_{q,P}(r). \tag{N2}$$

Both formulae are read on the support of the relevant distributions; zero-probability rows and records do not affect the value.

Condition (P1). For each fixed P, the relation $\succeq^I_P$ is complete and transitive because it is represented by the real-valued function $q \mapsto I_{q,P}$. If $|A| \geq 2$ and q has full support, then full revelation gives

$$I_{q,\mathrm{Id}_A} = H(q) > 0,$$

whereas the one-record uninformative channel U_A gives $I_{q,U_A} = 0$. Hence

$$q^0 \succ^I_{\mathrm{Id}_A \sqcup U_A} q^1.$$

Condition (P2). For fixed finite alphabets, (N1) shows that $(P, q) \mapsto I_{q,P}$ is continuous, since Shannon entropy is continuous on every finite simplex. Therefore the set

$$\{(P, q, q') : q \succeq^I_P q'\} = \{(P, q, q') : I_{q,P} \geq I_{q',P}\}$$

is closed. Posterior-law continuity of block comparisons is not an condition; for the mutual-information family it also holds directly, by (N2) and continuity of H on the compact simplex $\Delta(A)$, consistently with its derivation from the conditions in Lemma 24 below.

Condition (P3). For every channel P and lotteries $q, q' \in \Delta(A)$,

$$I_{q^0, P \sqcup P} = I_{q,P}, \qquad I_{(q')^1, P \sqcup P} = I_{q',P},$$

because a block-supported lottery puts probability one on its own block and the block label is constant under that lottery. Hence

$$q \succeq^I_P q' \quad \iff \quad q^0 \succeq^I_{P \sqcup P} (q')^1.$$

The same observation proves invariance to adding or deleting irrelevant blocks. If q_i^i and q_j^j are supported on blocks i and j, their mutual-information values in $\bigsqcup_{k\in K} P_k$ are exactly

$$I_{q_i,P_i} \quad \text{and} \quad I_{q_j,P_j},$$

independently of all other blocks.

Condition (P4). Let $S : A \to \Delta(A')$ and $T : X \to \Delta(X')$ be stochastic, and let P' satisfy (DP). Generate $A \sim q$; then, conditionally independently given A, generate A' from $S(\cdot \mid A)$ and X from $P(\cdot \mid A)$; finally generate X' from $T(\cdot \mid X)$. The joint law of (A, A', X, X') is

$$q(a)S(a' \mid a)P(x \mid a)T(x' \mid x),$$

so A' is conditionally independent of (X, X') given A, and X' is conditionally independent of (A, A') given X. Hence

$$A' \longleftarrow A \longrightarrow X \longrightarrow X'$$

is a Markov chain in the sense that $A' \to A \to X \to X'$ satisfies the required conditional independences. Data processing applied to the fork $A' \leftarrow A \to X$ gives $I(A; X) \geq I(A'; X)$, and data processing applied to $A' \to X \to X'$ gives $I(A'; X) \geq I(A'; X')$. Therefore

$$I_{q,P}(A; X) \geq I(A'; X').$$

By construction the joint law of (A', X') is exactly $(qS)(a')P'(x' \mid a')$, the left-hand side of (DP), so $I(A'; X') = I_{qS,P'}(A'; X')$. Rows a' with $(qS)(a') = 0$ carry no mass under qS, so this value is the same for every P' satisfying (DP), however the null rows are completed. Using the block computation of (P3) above to evaluate the two blocks separately,

$$q^0 \succeq^I_{P \sqcup P'} (qS)^1.$$

The first named clause is the case $S = \mathrm{Id}_A$ and gives $I_{q,P}(A; X) \geq I_{q,PT}(A; X')$. For later use, in the action-only case $T = \mathrm{Id}_X$ write $S^q P$ for the conditional channel from the reported action to the record on every reached row:

$$S^q P(x \mid a') := \frac{\sum_{a\in A} q(a)S(a' \mid a)P(x \mid a)}{(qS)(a')} \qquad \text{when } (qS)(a') > 0.$$

A *completion* of $S^q P$ is any channel $\widehat{P} : A' \to \Delta(X)$ that agrees with this formula on reached rows; its values on null rows are unrestricted. The second named clause gives $I_{q,P}(A; X) \geq I_{qS,\widehat{P}}(A'; X)$ for every completion $\widehat{P}$ of $S^q P$.

Condition (P5). Fix a first-stage channel $P_1 : A \to \Delta(X_1)$. Let

$$m(x) = m_{q,P_1}(x), \qquad r_x = r_x^{q,P_1}.$$

For a continuation profile $\mathcal{Q} = \{Q^x\}_x$, the chain rule for mutual information gives

$$I_{q,P_1 \triangleright \mathcal{Q}}(A; X_1, X_2) = I_{q,P_1}(A; X_1) + \sum_{x:m(x)>0} m(x)\, I_{r_x,Q^x}(A; X_2^x). \tag{N3}$$

Therefore, if

$$r_x^0 \succeq^I_{Q^x \sqcup R^x} r_x^1 \quad \text{for every positive-probability branch } x,$$

then $I_{r_x,Q^x} \geq I_{r_x,R^x}$ for every such x. Using (N3),

$$I_{q,P_1 \triangleright \{Q^x\}} \geq I_{q,P_1 \triangleright \{R^x\}},$$

so

$$q^0 \succeq^I_{(P_1 \triangleright \{Q^x\}) \sqcup (P_1 \triangleright \{R^x\})} q^1.$$

If one branch comparison is strict and that branch has positive probability, then the corresponding summand in (N3) is strictly larger, so the aggregate comparison is strict.

No product condition is imposed, so the verification ends here. For later use, write $q_1 \otimes q_2$ for the product lottery and define the product of channels by

$$(P_1 \otimes P_2)(x_1, x_2 \mid a_1, a_2) := P_1(x_1 \mid a_1)P_2(x_2 \mid a_2).$$

The mutual-information family satisfies the background-inertness property derived in Lemma 30 below, since

$$I_{q_1 \otimes q_2, P_1 \otimes R_2}(A_1, A_2; X_1, X_2) = I_{q_1,P_1}(A_1; X_1) + I_{q_2,R_2}(A_2; X_2), \tag{N4}$$

so that the comparison of $P_1 \otimes R_2$ and $Q_1 \otimes R_2$ at $q_1 \otimes q_2$ reduces to the comparison of P_1 and Q_1 at q_1, whatever the common background R_2. Consistently with Lemma 30, this is a consequence of the chain rule (N3) and not an independent requirement.

C.2 Sufficiency

Assume Conditions (P1)–(P5). We prove the implication (i)⇒(ii) in Theorem 16. The proof is written first for full-support lotteries. Boundary cases are obtained by restricting to the support and then using continuity. The key point is that posterior-law sufficiency is not assumed; it is derived from block comparisons and record post-processing, and so is its continuity (Lemma 24).

Proof roadmap. Lemmas 20–21 derive posterior-law sufficiency from block comparisons and Blackwell equivalence. Lemma 23 derives public-coin independence from branchwise continuation monotonicity, before its first use. Lemma 24 derives continuity of block comparisons in posterior-law convergence from the primitive closedness of Condition (P2). Lemmas 22–27 give fibrewise affine posterior-separable representations and basic invariances. Lemma 28 records undetectable-distinctions neutrality as a derived consequence of action coarsening, and Lemma 29 identifies comparisons at a boundary prior with comparisons on its support face. Lemma 30 then derives the only product-environment property the argument uses: an independent background component is behaviourally inert, because a product channel is a two-stage channel whose first stage reveals nothing about the foreground act. Lemma 31 chooses a coherent product gauge and proves quasi-additivity, allowing a possible bilinear interaction term. Lemma 33 moves cross-prior block comparisons into these unscaled values. Lemma 34 derives branch aggregation, and Lemma 35 derives the cocycle, the beta-form, and the normalised chain rule. Lemma 36 aligns the chain scales across support faces. Lemma 37 uses the chain rule and a two-grouping argument to eliminate the interaction term and prove the scale a_q is universal. Lemma 38 then proves entropy reduction, Faddeev recursion, and the mutual-information formula. Lemma 39 gives the fixed-environment representation.

Lemma 20 (Coherent block comparisons)**.** *Assume Conditions* (P1) *and* (P3).

(a) *Let* (q_1, P_1), (q_2, P_2), (q_3, P_3) *be pairs of a lottery* $q_k \in \Delta(A_k)$ *and a channel* $P_k : A_k \to \Delta(X_k)$*, with action sets, record sets, and lotteries not necessarily equal and the lotteries not necessarily of full support. Then the two-block comparisons between such pairs are complete,*

$$q_1^0 \succeq_{P_1 \sqcup P_2} q_2^1 \quad \text{or} \quad q_2^0 \succeq_{P_2 \sqcup P_1} q_1^1,$$

satisfy the reverse-block equivalence

$$q_2^1 \succeq_{P_1 \sqcup P_2} q_1^0 \quad \iff \quad q_2^0 \succeq_{P_2 \sqcup P_1} q_1^1,$$

and are transitive:

$$q_1^0 \succeq_{P_1 \sqcup P_2} q_2^1 \quad \text{and} \quad q_2^0 \succeq_{P_2 \sqcup P_3} q_3^1 \quad \Longrightarrow \quad q_1^0 \succeq_{P_1 \sqcup P_3} q_3^1.$$

In particular, for a fixed full-support $q \in \Delta(A)$ the comparisons $q^0 \succeq_{P \sqcup Q} q^1$ define a weak order over channels on A at prior q, and pairwise comparisons between block-supported lotteries are independent of the block environment in which the two blocks are embedded.

(b) *Under* (P2)*, the comparisons in* (a) *are also closed in the channels at fixed alphabets and fixed lotteries: if $P_n \to P$ in $\Delta(X_1)^{A_1}$, $Q_n \to Q$ in $\Delta(X_2)^{A_2}$, $q_1 \in \Delta(A_1)$ and $q_2 \in \Delta(A_2)$ are fixed, and*

$$q_1^0 \succeq_{P_n \sqcup Q_n} q_2^1 \quad \textit{for all } n,$$

then

$$q_1^0 \succeq_{P \sqcup Q} q_2^1.$$

Fixedness of the alphabets is inherited from Condition (P2)*, which is stated at a fixed pair of alphabets.*

Proof. (a) Completeness follows from Condition (P1) applied to the two-block environment $P_1 \sqcup P_2$: either $q_1^0 \succeq_{P_1 \sqcup P_2} q_2^1$ or $q_2^1 \succeq_{P_1 \sqcup P_2} q_1^0$. In the second case, applying Condition (P3) to the two-block environment with blocks P_1 and P_2, using the ordered pair $(i, j) = (1, 0)$, gives the canonical reverse comparison

$$q_2^1 \succeq_{P_1 \sqcup P_2} q_1^0 \quad \Longleftrightarrow \quad q_2^0 \succeq_{P_2 \sqcup P_1} q_1^1,$$

which is the reverse-block equivalence.

For transitivity, form the three-block environment $\bigsqcup_{k=1}^3 P_k$. By the irrelevant-block clause of Condition (P3) with block pair $(1, 2)$, the first hypothesis is equivalent to

$$q_1^1 \succeq_{\bigsqcup_{k=1}^3 P_k} q_2^2,$$

and by the same clause with block pair $(2, 3)$, the second hypothesis is equivalent to

$$q_2^2 \succeq_{\bigsqcup_{k=1}^3 P_k} q_3^3.$$

Transitivity of the weak order given by Condition (P1) in this common environment gives

$$q_1^1 \succeq_{\bigsqcup_{k=1}^3 P_k} q_3^3,$$

and applying Condition (P3) again, now deleting the irrelevant middle block, yields $q_1^0 \succeq_{P_1 \sqcup P_3} q_3^1$. Nothing in these arguments requires the pairs to share alphabets or lotteries, or the lotteries to have full support. The specialisation to a fixed full-support q and channels on a common A is immediate.

(b) The block channel $P_n \sqcup Q_n$ has fixed tagged copies of the alphabets A_1, A_2, X_1, X_2 and converges to $P \sqcup Q$ in the finite-dimensional product simplex. Since the block-supported lotteries q_1^0, q_2^1 are fixed, Condition (P2) applied at this pair of alphabets gives

$$q_1^0 \succeq_{P \sqcup Q} q_2^1. \qquad \square$$

Lemma 21 (Prior-specific Blackwell equivalence)**.** *Let $q \in \Delta(A)$ have full support. Let $P : A \to \Delta(X)$ and $Q : A \to \Delta(Y)$ be finite channels. If*

$$\mu_{q,P} = \mu_{q,Q},$$

then there are stochastic kernels $T : X \to \Delta(Y)$ and $T' : Y \to \Delta(X)$ such that

$$Q = PT, \qquad P = QT'.$$

Proof. It is enough to construct T; the construction of T' is symmetric. Let

$$m_P(x) := m_{q,P}(x), \qquad m_Q(y) := m_{q,Q}(y).$$

If $m_P(x) = 0$, then $P(x \mid a) = 0$ for every $a \in A$, because q has full support. Such records do not affect PT, so define the corresponding row of T arbitrarily. The same observation applies to null records of Q.

Let $\mathcal{R}$ be the finite set of posterior vectors that occur with positive probability under either channel. For $r \in \mathcal{R}$, define

$$X_r := \{x \in X : m_P(x) > 0,\ r_x^{q,P} = r\}, \qquad Y_r := \{y \in Y : m_Q(y) > 0,\ r_y^{q,Q} = r\}.$$

The sets $\{X_r\}_{r\in\mathcal{R}}$ partition the positive-probability records of P, and the sets $\{Y_r\}_{r\in\mathcal{R}}$ partition the positive-probability records of Q, by posterior vector. Equality of posterior laws implies that the total probability attached to posterior r is the same for the two channels:

$$\lambda_r := \sum_{x\in X_r} m_P(x) = \sum_{y\in Y_r} m_Q(y) > 0.$$

For $x \in X_r$ and $y \in Y_r$, set

$$T(y \mid x) := \frac{m_Q(y)}{\lambda_r},$$

and set $T(y \mid x) = 0$ when $x \in X_r$ and $y \notin Y_r$. This defines a stochastic row for every positive-probability record x, since the probabilities over Y_r sum to one. Rows corresponding to null records of P were already chosen arbitrarily.

Fix $a \in A$ and $y \in Y_r$. Since all records in X_r have posterior r, Bayes' rule gives

$$q(a)P(x \mid a) = m_P(x)r(a) \qquad (x \in X_r).$$

Therefore

$$\sum_{x\in X_r} P(x \mid a) = \frac{\lambda_r r(a)}{q(a)}.$$

Using the definition of T,

$$(PT)(y \mid a) = \sum_{x\in X_r} P(x \mid a)\frac{m_Q(y)}{\lambda_r} = \frac{m_Q(y)r(a)}{q(a)}.$$

But $y \in Y_r$ also has posterior r, so Bayes' rule for Q gives

$$Q(y \mid a) = \frac{m_Q(y)r(a)}{q(a)}.$$

Thus $(PT)(y \mid a) = Q(y \mid a)$ for every positive-probability y. If $m_Q(y) = 0$, then full support of q implies $Q(y \mid a) = 0$ for every a. No positive-probability record of P sends mass to such a y under the construction; rows of T indexed by null records of P may be chosen arbitrarily, but those records have $P(x \mid a) = 0$ for every a, so they contribute nothing to PT. Hence $(PT)(y \mid a) = 0 = Q(y \mid a)$ for every a, and therefore $Q = PT$. $\square$

Lemma 22 (Posterior-law sufficiency)**.** *Fix a full-support $q \in \Delta(A)$. If $P : A \to \Delta(X)$ and $Q : A \to \Delta(Y)$ generate the same posterior law at q,*

$$\mu_{q,P} = \mu_{q,Q},$$

then

$$q^0 \sim_{P\sqcup Q} q^1.$$

Consequently, by Lemma 20, at a fixed full-support prior the block comparison of channels depends only on the induced posterior law.

Proof. By Lemma 21, there are stochastic kernels T and T' such that $Q = PT$ and $P = QT'$. Clause (DP-rec) of Condition (P4) gives

$$q^0 \succeq_{P \sqcup PT} q^1, \qquad q^0 \succeq_{Q \sqcup QT'} q^1.$$

Using $PT = Q$, the first comparison is

$$q^0 \succeq_{P \sqcup Q} q^1.$$

Using $QT' = P$, the second comparison is

$$q^0 \succeq_{Q \sqcup P} q^1.$$

By the reverse-block equivalence in Lemma 20 (equivalently, Condition (P3) applied to $P \sqcup Q$ with ordered pair $(i, j) = (1, 0)$), this is equivalent to

$$q^1 \succeq_{P \sqcup Q} q^0.$$

Thus $q^0 \sim_{P \sqcup Q} q^1$. $\square$

For channels $P : A \to \Delta(X_P)$ and $R : A \to \Delta(X_R)$ and $\lambda \in [0, 1]$, write $\lambda P \oplus (1 - \lambda)R$ for the channel that publicly selects P with probability λ and R otherwise, and reports the selected channel together with its record. Thus

$$(\lambda P \oplus (1-\lambda)R)((z, i) \mid a) = \begin{cases} \lambda P(z \mid a), & i = 0, \\ (1-\lambda)R(z \mid a), & i = 1. \end{cases}$$

Because the coin is observed,

$$\mu_{q,\lambda P \oplus (1-\lambda)R} = \lambda \mu_{q,P} + (1 - \lambda)\mu_{q,R}.$$

Lemma 23 (Derived public-coin independence)**.** *Assume Conditions* (P1) *and* (P3)*, clause* (DP-rec) *of Condition* (P4)*, and Condition* (P5)*. Fix a finite action set A, a full-support $q \in \Delta(A)$, and channels P, Q, R on A. For every $\lambda \in (0, 1)$,*

$$q^0 \succeq_{P \sqcup Q} q^1 \quad \iff \quad q^0 \succeq_{(\lambda P \oplus (1-\lambda)R) \sqcup (\lambda Q \oplus (1-\lambda)R)} q^1.$$

Proof. Throughout, for channels V, W on A write $V \succeq_q W$ for the block comparison $q^0 \succeq_{V \sqcup W} q^1$, and $V \succ_q W$ for the corresponding strict comparison. By Lemma 20 the comparison $\succeq_q$ is complete and transitive, and by Lemma 22 it depends only on the pair of posterior laws $(\mu_{q,V}, \mu_{q,W})$. In particular, channels with equal posterior laws at q may be substituted on either side of any comparison; we refer to this as *substitution.* Substitution also preserves strict comparisons: if $\mu_{q,V} = \mu_{q,V'}$ and $\mu_{q,W} = \mu_{q,W'}$, then $V \sim_q V'$ and $W \sim_q W'$ by Lemma 22, and transitivity of the complete relation $\succeq_q$ gives $V \succ_q W$ if and only if $V' \succ_q W'$.

Step 1: the public coin is an uninformative first stage. Define the constant channel $C : A \to \Delta(\{0, 1\})$ by

$$C(0 \mid a) := \lambda, \qquad C(1 \mid a) := 1 - \lambda \qquad (a \in A).$$

Since $\lambda \in (0, 1)$, both branches have positive mass: $m_{q,C}(0) = \lambda$ and $m_{q,C}(1) = 1 - \lambda$. Because the rows of C are constant, Bayes' rule gives, for each $x \in \{0, 1\}$,

$$r_x^{q,C}(a) = \frac{q(a)C(x \mid a)}{m_{q,C}(x)} = q(a),$$

so both branch posteriors equal the prior: $r_0 = r_1 = q$.

Step 2: padding the continuations to a common alphabet. Condition (P5) requires the two continuation channels on a given branch to share a record alphabet, whereas P and Q may not. Let $X_{PQ} := X_P \sqcup X_Q$ and define $\widetilde{P}, \widetilde{Q} : A \to \Delta(X_{PQ})$ by

$$\widetilde{P}(z \mid a) := \begin{cases} P(z \mid a) & z \in X_P, \\ 0 & z \in X_Q, \end{cases} \qquad \widetilde{Q}(z \mid a) := \begin{cases} 0 & z \in X_P, \\ Q(z \mid a) & z \in X_Q. \end{cases}$$

Padding adds only zero-probability records, so the positive-mass records and their posteriors are unchanged: $\mu_{q,\widetilde{P}} = \mu_{q,P}$ and $\mu_{q,\widetilde{Q}} = \mu_{q,Q}$. By substitution,

$$q^0 \succeq_{P \sqcup Q} q^1 \iff \widetilde{P} \succeq_q \widetilde{Q},$$

and the same equivalence holds for the strict comparisons.

Step 3: the compounds implement the public-coin mixtures. Consider the two-stage channels with first stage C, continuation pair $\widetilde{P}$ versus $\widetilde{Q}$ on branch 0, and the common continuation R on branch 1:

$$C \triangleright \{\widetilde{P}, R\} \qquad \text{and} \qquad C \triangleright \{\widetilde{Q}, R\},$$

where the profile notation lists the branch-0 and branch-1 continuations in order. By the definition of two-stage composition,

$$(C \triangleright \{\widetilde{P}, R\})(0, z \mid a) = \lambda\, \widetilde{P}(z \mid a), \qquad (C \triangleright \{\widetilde{P}, R\})(1, z \mid a) = (1 - \lambda)\, R(z \mid a).$$

A record $(0, z)$ has marginal mass $\lambda\, m_{q,\widetilde{P}}(z)$, and when this is positive the constant factor λ cancels in Bayes' rule, so its posterior is $r_z^{q,\widetilde{P}}$; symmetrically for records $(1, z)$. Hence

$$\mu_{q,C \triangleright \{\widetilde{P},R\}} = \lambda \mu_{q,\widetilde{P}} + (1 - \lambda)\mu_{q,R} = \lambda \mu_{q,P} + (1 - \lambda)\mu_{q,R} = \mu_{q,\lambda P \oplus (1-\lambda) R},$$

using the public-coin mixing identity, and likewise $\mu_{q,C \triangleright \{\widetilde{Q},R\}} = \mu_{q,\lambda Q \oplus (1-\lambda)R}$. By substitution, the right-hand comparison of the lemma is therefore equivalent to

$$C \triangleright \{\widetilde{P}, R\} \succeq_q C \triangleright \{\widetilde{Q}, R\}. \tag{$*$}$$

Step 4: forward implication. Assume $q^0 \succeq_{P \sqcup Q} q^1$. We verify the branch hypotheses of Condition (P5) for the first stage C and the continuation profiles $\{\widetilde{P}, R\}$ versus $\{\widetilde{Q}, R\}$. Both branches have positive mass and both branch posteriors equal q (Step 1). On branch 0 the hypothesis is $q^0 \succeq_{\widetilde{P} \sqcup \widetilde{Q}} q^1$, which holds by Step 2. On branch 1 the two profiles share the continuation R, and the hypothesis $q^0 \succeq_{R \sqcup R} q^1$ holds because Condition (P1) gives $q \succeq_R q$ and the duplication clause of Condition (P3) converts it into the two-block form. Condition (P5) then yields $(*)$, hence the mixture comparison by Step 3.

Step 5: converse, via the strictness clause. Assume the mixture comparison holds, so that $(*)$ holds by Step 3, and suppose towards a contradiction that $q^0 \succeq_{P \sqcup Q} q^1$ fails. By completeness and the reverse-block equivalence of Lemma 20, $Q \succ_q P$, and Step 2 gives $\widetilde{Q} \succ_q \widetilde{P}$. Now apply Condition (P5) with the roles of the two profiles exchanged: the branch-0 comparison $\widetilde{Q} \succ_q \widetilde{P}$ is strict on a branch of mass $\lambda > 0$, and the branch-1 comparison $R \succeq_q R$ holds as in Step 4. The strictness clause of Condition (P5) yields

$$C \triangleright \{\widetilde{Q}, R\} \succ_q C \triangleright \{\widetilde{P}, R\},$$

which contradicts $(*)$. Hence $q^0 \succeq_{P \sqcup Q} q^1$. □

The lemma shows that public-coin independence is not an independent behavioural commitment: a public coin is exactly a first-stage observation that reveals nothing about the act, so invariance under common public mixtures is branchwise continuation monotonicity applied through trivial branch posteriors. The statement is given at full-support priors, which is the only case used below (Lemma 25).

For a finite action set A, let X_A be the set of finite-support probability measures on $\Delta(A)$. Write

$$b(\mu) := \int_{\Delta(A)} r \, d\mu(r), \qquad \mathcal{M}_q := \{\mu \in \mathsf{X}_A : b(\mu) = q\}.$$

Thus $\mathcal{M}_q$ is the set of finite posterior laws with prior q as their mean, and $\mu_{q,P} \in \mathcal{M}_q$ for every channel P.

Lemma 24 (Posterior-law continuity)**.** *Fix a full-support $q \in \Delta(A)$. Let P_n, Q_n, P, Q be finite channels on A with*

$$\mu_{q,P_n} \Rightarrow \mu_{q,P}, \qquad \mu_{q,Q_n} \Rightarrow \mu_{q,Q}.$$

If

$$q^0 \succeq_{P_n \sqcup Q_n} q^1 \qquad \textit{for all } n,$$

then

$$q^0 \succeq_{P \sqcup Q} q^1.$$

Proof. Throughout the proof, for channels V, W on A write $V \succeq_q W$ for the block comparison $q^0 \succeq_{V \sqcup W} q^1$. By Lemma 20 this comparison is complete and transitive, and by Lemma 22 it depends only on the pair of posterior laws $(\mu_{q,V}, \mu_{q,W})$; in particular, channels with equal posterior laws at q may be substituted on either side of any comparison. We refer to this as *substitution.*

For a finite-support Bayes-plausible law

$$\mu = \sum_{k \in K} \lambda_k \delta_{r_k} \in \mathcal{M}_q,$$

with atoms listed so that $\lambda_k > 0$, define the canonical implementing channel $C_\mu : A \to \Delta(K)$ by

$$C_\mu(k \mid a) := \frac{\lambda_k r_k(a)}{q(a)}.$$

Full support of q and $\sum_k \lambda_k r_k = q$ make C_μ a channel with $\mu_{q,C_\mu} = \mu$.

Step 1: merging posteriors moves weakly down. Let $\mu = \sum_{k \in K} \lambda_k \delta_{r_k} \in \mathcal{M}_q$ and let $\{K_1, \dots, K_m\}$ partition K into nonempty groups. The merged law is

$$\mathrm{mrg}(\mu) := \sum_{j=1}^{m} w_j \, \delta_{s_j}, \qquad w_j := \sum_{k \in K_j} \lambda_k, \qquad s_j := \frac{1}{w_j} \sum_{k \in K_j} \lambda_k r_k.$$

Barycentres are preserved, so $\mathrm{mrg}(\mu) \in \mathcal{M}_q$. Let $T : K \to \Delta(\{1, \dots, m\})$ be the deterministic kernel with $T(j \mid k) = 1$ if and only if $k \in K_j$. Then

$$(C_\mu T)(j \mid a) = \sum_{k \in K_j} \frac{\lambda_k r_k(a)}{q(a)} = \frac{w_j s_j(a)}{q(a)} = C_{\mathrm{mrg}(\mu)}(j \mid a),$$

so clause (DP-rec) of Condition (P4) gives

$$C_\mu \succeq_q C_{\mathrm{mrg}(\mu)}.$$

Conversely, say that $\mu' \in \mathcal{M}_q$ is a *spread* of μ if μ' is obtained by replacing each atom (λ_k, r_k) of μ by finitely many atoms $(\lambda_k w_{kj}, v_{kj})_j$ with $w_{kj} \geq 0$, $\sum_j w_{kj} = 1$, and $\sum_j w_{kj} v_{kj} = r_k$. Then $\mu = \mathrm{mrg}(\mu')$ for the grouping by k, so

$$C_{\mu'} \succeq_q C_\mu.$$

Atoms with $w_{kj} = 0$ are deleted from the list; and if the same posterior appears in several list entries, any two listings of the same measure induce the same posterior law at q, so the corresponding canonical channels are interchangeable by substitution. Merging records can only obscure which act was realised; splitting an atom into a barycentric family is the reverse operation.

Step 2: bounded supports. We claim the lemma holds under the additional hypothesis that all laws μ_{q,P_n} and μ_{q,Q_n} have support of size at most K, for some fixed K. Write

$$\mu_{q,P_n} = \sum_{k=1}^{K} \lambda_{n,k}\, \delta_{r_{n,k}},$$

padding with zero-mass atoms ($\lambda_{n,k} = 0$, $r_{n,k} := q$) so that exactly K terms appear, and define the K-record channel $C_n : A \to \Delta(\{1, \dots, K\})$ by

$$C_n(k \mid a) := \frac{\lambda_{n,k}\, r_{n,k}(a)}{q(a)}.$$

Each row sums to one because $\sum_k \lambda_{n,k} r_{n,k} = q$, zero-mass columns contribute nothing, and $\mu_{q,C_n} = \mu_{q,P_n}$. Define D_n from μ_{q,Q_n} in the same way. By substitution,

$$C_n \succeq_q D_n \qquad \text{for all } n.$$

The parameter sequences of C_n and D_n jointly live in the compact metric space $([0,1] \times \Delta(A))^{2K}$; pass to a single subsequence along which $\lambda_{n,k} \to \lambda_k^*$ and $r_{n,k} \to r_k^*$ for every k, simultaneously for both channels. Then $C_n \to C^*$ entrywise, where $C^*(k \mid a) = \lambda_k^* r_k^*(a)/q(a)$; each row of C^* sums to one as an entrywise limit of rows summing to one, so C^* is a channel. For every continuous f on $\Delta(A)$,

$$\int f\, d\mu_{q,P_n} = \sum_{k=1}^{K} \lambda_{n,k} f(r_{n,k}) \longrightarrow \sum_{k=1}^{K} \lambda_k^* f(r_k^*),$$

so by uniqueness of weak limits

$$\mu_{q,P} = \sum_{k=1}^{K} \lambda_k^*\, \delta_{r_k^*} = \mu_{q,C^*},$$

where the second equality holds because record k of C^* has probability λ_k^* and, when $\lambda_k^* > 0$, posterior r_k^*. Similarly $D_n \to D^*$ entrywise with $\mu_{q,D^*} = \mu_{q,Q}$. The closedness clause, Lemma 20(b), applied along the subsequence gives

$$C^* \succeq_q D^*,$$

and substitution returns $q^0 \succeq_{P \sqcup Q} q^1$. This proves the bounded-support case; note that only the primitive closedness of Condition (P2) has been used, through Lemma 20.

Step 3: the general case, by an ε-grid sandwich. Write $\mu_n := \mu_{q,P_n}$, $\nu_n := \mu_{q,Q_n}$, $\mu := \mu_{q,P}$, $\nu := \mu_{q,Q}$. The only remaining gap is that the supports of μ_n, ν_n need not be bounded. Fix $\varepsilon > 0$. Two finite partitions of $\Delta(A)$ are used, one for each surrogate; they play independent roles and need not be related. For the spread, fix a finite triangulation $\mathcal{T}$ of $\Delta(A)$ into simplices of diameter at most ε, together with a measurable partition $\{\widetilde{S} : S \in \mathcal{T}\}$ of $\Delta(A)$ with $\widetilde{S} \subseteq S$, assigning each shared face to exactly one incident cell. For the merge, fix a finite Borel partition $\{E_1, \dots, E_M\}$ of $\Delta(A)$ into sets of diameter at most ε whose relative boundaries carry no ν-mass: cover $\Delta(A)$ by finitely many open balls of radius at most $\varepsilon/2$, choosing the radii so that no atom of ν lies on a bounding sphere (ν has finitely many atoms, so all but finitely many radii work), and disjointify.

For finite-support $\rho \in \mathcal{M}_q$ define two bounded-support surrogates. Both remain in $\mathcal{M}_q$: the spread operation replaces each posterior by a barycentric decomposition of itself, while the merge operation replaces a group of posteriors by their conditional mean, weighted by the total mass of the group. Cells with zero mass are ignored (or, equivalently, padded by an arbitrary posterior with zero weight when a fixed record alphabet is convenient). The *grid spread* $\mathrm{sp}_\varepsilon(\rho)$ replaces each atom (λ, β) with $\beta \in \widetilde{S}$ by the atoms $\big(\lambda\, w_v(\beta),\, v\big)_v$, where v ranges over the vertices of S and $(w_v(\beta))_v$ are the barycentric coordinates of β in S. It is a spread of ρ in the sense of Step 1, is supported on the fixed vertex set of $\mathcal{T}$, and satisfies $C_{\mathrm{sp}_\varepsilon(\rho)} \succeq_q C_\rho$. The *cell merge* $\mathrm{mg}_\varepsilon(\rho)$ merges the atoms of ρ within each class E_j of the Borel partition; it has at most one atom per class and satisfies $C_\rho \succeq_q C_{\mathrm{mg}_\varepsilon(\rho)}$ by Step 1, since Step 1 allows any grouping of atoms.

By substitution and transitivity (Lemma 20), the hypothesis $P_n \succeq_q Q_n$ gives the chain

$$C_{\mathrm{sp}_\varepsilon(\mu_n)} \succeq_q C_{\mu_n} \succeq_q C_{\nu_n} \succeq_q C_{\mathrm{mg}_\varepsilon(\nu_n)}.$$

Both surrogate sequences converge weakly on fixed alphabets. For the spread side, let f be continuous on $\Delta(A)$. The triangulation defines a continuous piecewise-affine interpolant $I_{\mathcal{T}} f$ by

$$I_{\mathcal{T}} f(r) := \sum_{v \in \mathrm{vert}(S)} w_v(r) f(v) \qquad \text{for } r \in S.$$

This is well-defined: on a shared face the barycentric coordinates computed in the two incident simplices put zero weight on vertices outside the face and coincide with the intrinsic barycentric coordinates on that face. Hence $I_{\mathcal{T}} f$ is continuous on all of $\Delta(A)$. Moreover, for every finite-support ρ,

$$\int f \, d\,\mathrm{sp}_\varepsilon(\rho) = \int I_{\mathcal{T}} f \, d\rho,$$

because each posterior is replaced by the barycentric lottery over the vertices of its simplex. Thus $\mu_n \Rightarrow \mu$ gives $\mathrm{sp}_\varepsilon(\mu_n) \Rightarrow \mathrm{sp}_\varepsilon(\mu)$. For the merge side, each class E_j has relative boundary contained in the chosen bounding spheres, which carry no ν-mass, so E_j is a ν-continuity set; therefore $\nu_n(E_j) \to \nu(E_j)$ and, since the truncated map $r \mapsto r\,\mathbf{1}_{E_j}(r)$ is bounded and its discontinuity set lies in the relative boundary of E_j, which is ν-null, $\int_{E_j} r \, d\nu_n \to \int_{E_j} r \, d\nu$ for every class. Classes with $\nu(E_j) > 0$ thus have converging merged masses and merged locations. If $\nu(E_j) = 0$, then the merged mass tends to zero, so the chosen location in that class is immaterial for weak convergence; padding such a class by any fixed posterior only changes a zero-weight atom. Hence $\mathrm{mg}_\varepsilon(\nu_n) \Rightarrow \mathrm{mg}_\varepsilon(\nu)$.

The laws $\mathrm{sp}_\varepsilon(\mu_n)$ and $\mathrm{mg}_\varepsilon(\nu_n)$ have supports of size at most $N(\varepsilon)$, the maximum of the number of vertices of $\mathcal{T}$ and the number M of classes of the Borel partition. Step 2 applied to the chain's endpoints therefore yields

$$C_{\mathrm{sp}_\varepsilon(\mu)} \succeq_q C_{\mathrm{mg}_\varepsilon(\nu)}.$$

Finally, send $\varepsilon \to 0$ along a sequence $\varepsilon_m \downarrow 0$ with triangulations and Borel partitions chosen as above. The law $\mathrm{sp}_{\varepsilon_m}(\mu)$ has at most $k_P\,|A|$ atoms, where k_P is the number of atoms of μ: each atom splits among the $|A|$ vertices of one cell. Each new atom lies within ε_m of the atom it came from and group masses are preserved, so $\mathrm{sp}_{\varepsilon_m}(\mu) \Rightarrow \mu$. The law $\mathrm{mg}_{\varepsilon_m}(\nu)$ has at most k_Q atoms, where k_Q is the number of atoms of ν: only classes meeting the support of ν carry mass. Once ε_m is smaller than the minimal distance between distinct atoms of ν, each class contains at most one atom of ν and $\mathrm{mg}_{\varepsilon_m}(\nu) = \nu$ exactly. The comparisons $C_{\mathrm{sp}_{\varepsilon_m}(\mu)} \succeq_q C_{\mathrm{mg}_{\varepsilon_m}(\nu)}$ hold for every m, the supports are uniformly bounded, and the laws converge weakly to μ and ν; one further application of Step 2 gives $C_\mu \succeq_q C_\nu$. Substitution returns

$$q^0 \succeq_{P \sqcup Q} q^1. \qquad \square$$

The lemma shows that continuity of block comparisons in posterior-law convergence is not an independent behavioural commitment: it is the primitive closedness of Condition (P2) seen through the quotient that Condition (P3) and clause (DP-rec) of Condition (P4) impose, with the convex order supplying compactness where record alphabets grow without bound. Only weak preference is closed in the limit; no strictness claim is made or used.

Lemma 25 (Posterior-separable representation). *Fix a full-support $q \in \Delta(A)$. There is a continuous affine functional*

$$F_q : \mathcal{M}_q \to \mathbb{R}$$

such that, for any two channels P, Q on A,

$$q^0 \succeq_{P \sqcup Q} q^1 \quad \Longleftrightarrow \quad F_q(\mu_{q,P}) \geq F_q(\mu_{q,Q}).$$

Moreover, for some continuous $\phi_q : \Delta(A) \to \mathbb{R}$,

$$F_q(\mu) = \int_{\Delta(A)} \phi_q(r)\, d\mu(r),$$

and therefore

$$F_q(\mu_{q,P}) = \sum_{x: m_{q,P}(x) > 0} m_{q,P}(x)\, \phi_q(r_x^{q,P}).$$

The affine functional is unique up to positive affine transformations. The integrand ϕ_q is not pointwise unique; adding an affine function of r leaves its integral on the fibre $\mathcal{M}_q$ changed only by a constant.

Proof. First note that every finite-support Bayes-plausible law is implemented by a finite channel. If

$$\mu = \sum_{k=1}^{n} \lambda_k \delta_{r_k} \in \mathcal{M}_q,$$

define

$$P(k \mid a) = \frac{\lambda_k r_k(a)}{q(a)}.$$

Since q has full support and $\sum_k \lambda_k r_k = q$, this is a channel. Its record probability at k is λ_k, and the posterior after k is r_k; hence $\mu_{q,P} = \mu$.

Define a comparison on $\mathcal{M}_q$ as follows: for $\mu, \nu \in \mathcal{M}_q$, choose any channels P, Q implementing them and compare the two block-supported copies of q in $P \sqcup Q$. This comparison is independent of the chosen implementations. Indeed, if P, P' both implement μ and Q, Q' both implement ν, then Lemma 22 gives

$$q^0 \sim_{P \sqcup P'} q^1, \qquad q^0 \sim_{Q \sqcup Q'} q^1.$$

Placing the relevant blocks in common block environments and using Lemma 20, transitivity gives

$$q^0 \succeq_{P \sqcup Q} q^1 \quad \Longleftrightarrow \quad q^0 \succeq_{P' \sqcup Q'} q^1.$$

Thus the quotient comparison on $\mathcal{M}_q$ is well defined; denote it by $\succeq_q$. This construction is only at the fixed full-support prior q, and every law in $\mathcal{M}_q$ is finite-support and, by the preceding construction, implemented by a finite channel. Lemma 20 also gives completeness and transitivity of this quotient comparison.

The set $\mathcal{M}_q$ is a mixture space under ordinary convex combination. It is convex because barycentres are preserved by convex combination. The quotient relation $\succeq_q$ is a weak order by Lemma 20. Public-coin mixtures of implementing channels correspond exactly to convex mixtures of posterior laws:

$$\mu_{q, \lambda P \oplus (1-\lambda) R} = \lambda \mu_{q,P} + (1-\lambda) \mu_{q,R}.$$

Therefore the derived public-coin independence property (Lemma 23) gives mixture independence: for every $\mu, \nu, \rho \in \mathcal{M}_q$ and $\lambda \in (0,1)$,

$$\mu \succeq_q \nu \iff \lambda\mu + (1-\lambda)\rho \succeq_q \lambda\nu + (1-\lambda)\rho.$$

The relative weak topology on $\mathcal{M}_q$ is inherited from the weak topology on probability measures over the compact metric simplex $\Delta(A)$, and convex-mixture paths are continuous in this topology. Lemma 24 gives relatively closed upper and lower contour sets. Indeed, if $\mu_n \Rightarrow \mu$ with $\mu_n, \mu, z \in \mathcal{M}_q$ and $\mu_n \succeq_q z$, choose finite implementing channels for these laws and apply Lemma 24 (with the constant sequence implementing z) to get $\mu \succeq_q z$; the argument for $\{\, \mu : z \succeq_q \mu \,\}$ is the same. Therefore, for fixed $\mu, \nu, \rho \in \mathcal{M}_q$, the sets

$$\{\lambda \in [0,1] : \lambda\mu + (1-\lambda)\nu \succeq_q \rho\} \quad \text{and} \quad \{\lambda \in [0,1] : \rho \succeq_q \lambda\mu + (1-\lambda)\nu\}$$

are closed.

By the Herstein–Milnor mixture-space theorem [30], there is a mixture-preserving representation $F_q : \mathcal{M}_q \to \mathbb{R}$:

$$\mu \succeq_q \nu \iff F_q(\mu) \geq F_q(\nu),$$

and

$$F_q(\lambda\mu + (1-\lambda)\nu) = \lambda F_q(\mu) + (1-\lambda)F_q(\nu).$$

If the quotient order is nonconstant, this representative is unique up to positive affine transformations. If the quotient order is constant, take $F_q \equiv 0$.

Moreover, F_q is continuous in the relative weak topology. For any $z \in \mathcal{M}_q$, the sets

$$\{\mu : F_q(\mu) \geq F_q(z)\} = \{\mu : \mu \succeq_q z\} \quad \text{and} \quad \{\mu : F_q(\mu) \leq F_q(z)\} = \{\mu : z \succeq_q \mu\}$$

are relatively closed by the preceding Lemma 24 argument. Since F_q is affine, its range on the convex set $\mathcal{M}_q$ is an interval. Hence every upper and lower level set of F_q is relatively closed: levels outside the range are trivial, and levels inside the range equal $F_q(z)$ for some $z \in \mathcal{M}_q$. Thus F_q is both upper and lower semicontinuous, and therefore continuous. For any channels P, Q on A,

$$q^0 \succeq_{P \sqcup Q} q^1 \iff F_q(\mu_{q,P}) \geq F_q(\mu_{q,Q}).$$

The no-information posterior law at q is δ_q. Write

$$\chi_q := \sum_{a \in A} q(a)\delta_{e_a}$$

for the full-revelation posterior law, where e_a assigns probability one to action a.

It remains to write the affine functional in posterior-separable form, without leaving the finite-support domain. Let $X := \Delta(A)$. Choose

$$0 < \varepsilon < \min_{a \in A} q(a).$$

For each $r \in X$, define

$$s_r := \frac{q - \varepsilon r}{1 - \varepsilon} \in \Delta(A)$$

and

$$\eta_r := \varepsilon\delta_r + (1-\varepsilon)\sum_{a \in A} s_r(a)\delta_{e_a}.$$

Then $\eta_r \in \mathcal{M}_q$, because its barycentre is

$$\varepsilon r + (1-\varepsilon)s_r = q.$$

Choose constants c_a such that

$$\sum_{a\in A} q(a)c_a = F_q(\chi_q),$$

for instance $c_a = F_q(\chi_q)$ for every a. Define

$$\phi_q(r) := \frac{F_q(\eta_r) - (1-\varepsilon)\sum_{a\in A} s_r(a)c_a}{\varepsilon}.$$

The map $r \mapsto \eta_r$ is weakly continuous and F_q is continuous, so ϕ_q is continuous.

Now take any finite-support law

$$\mu = \sum_{k=1}^{n} \lambda_k \delta_{r_k} \in \mathcal{M}_q.$$

Since $\sum_k \lambda_k r_k = q$, we have

$$\sum_{k=1}^{n} \lambda_k s_{r_k} = q.$$

Therefore

$$\sum_{k=1}^{n} \lambda_k \eta_{r_k} = \varepsilon\mu + (1-\varepsilon)\chi_q.$$

By affinity of F_q,

$$\sum_{k=1}^{n} \lambda_k F_q(\eta_{r_k}) = F_q(\varepsilon\mu + (1-\varepsilon)\chi_q) = \varepsilon F_q(\mu) + (1-\varepsilon)F_q(\chi_q).$$

Using $\sum_k \lambda_k s_{r_k} = q$ and $\sum_a q(a)c_a = F_q(\chi_q)$ gives

$$\sum_{k=1}^{n} \lambda_k \phi_q(r_k) = F_q(\mu).$$

This is exactly

$$F_q(\mu) = \int_X \phi_q(r)\, d\mu(r)$$

for every finite-support $\mu \in \mathcal{M}_q$. Applying this to $\mu_{q,P}$ gives the stated finite-sum formula. The affine representative F_q is unique up to positive affine transformations by Herstein–Milnor. The integrand is not pointwise unique on a fixed barycentre fibre: if ℓ is affine in r, then $\int \ell(r)\, d\mu(r) = \ell(q)$ for every $\mu \in \mathcal{M}_q$. $\square$

Lemma 26 (Refinement monotonicity, convexity, and normalisation)**.** *Fix a full-support $q \in \Delta(A)$ and let F_q, ϕ_q be as in Lemma 25. If a finite law $\mu \in \mathcal{M}_q$ is obtained from another finite law $\nu \in \mathcal{M}_q$ by replacing each posterior of ν by a finite Bayes-plausible continuation experiment, then*

$$F_q(\mu) \geq F_q(\nu).$$

Consequently, ϕ_q may be chosen convex. After replacing ϕ_q by an observationally equivalent representative and subtracting a constant from F_q, one may impose

$$\phi_q(q) = F_q(\delta_q) = 0, \qquad \phi_q(r) \geq 0 \quad \textit{for every } r \in \Delta(A),$$

and therefore

$$F_q(\mu) \geq 0 \quad \textit{for every } \mu \in \mathcal{M}_q.$$

Proof. Write

$$\nu = \sum_{i=1}^{n} m_i \delta_{r_i}, \qquad \mu = \sum_{i=1}^{n} m_i \nu_i, \qquad \nu_i \in \mathcal{M}_{r_i}.$$

Choose a channel $P : A \to \Delta(\{1, \ldots, n\})$ implementing ν at prior q. For each i, choose a finite continuation channel $Q^i : A \to \Delta(Z_i)$ implementing ν_i at prior r_i; on actions outside the support of r_i, define the rows arbitrarily. Those rows are never reached under prior r_i and do not affect ν_i; moreover, if $r_i(a) = 0$, then $q(a)P(i \mid a) = m_i r_i(a) = 0$, so they do not affect the compound posterior law at q. The compound channel

$$C := P \triangleright \{Q^i\}_{i=1}^{n}$$

implements μ at prior q. Let $T : \bigsqcup_{i=1}^{n}(\{i\} \times Z_i) \to \Delta(\{1, \ldots, n\})$ be the kernel $T(i \mid (i, z)) = 1$; then $CT = P$. Clause (DP-rec) of Condition (P4) applied to C and T gives

$$q^0 \succeq_{C \sqcup P} q^1.$$

By Lemma 25,

$$F_q(\mu) \geq F_q(\nu).$$

To prove convexity, take $r, s_1, \ldots, s_k \in \Delta(A)$ with

$$r = \sum_{j=1}^{k} \lambda_j s_j, \qquad \lambda_j \geq 0, \quad \sum_j \lambda_j = 1.$$

Choose $0 < \varepsilon < \min_a q(a)$ and set

$$t := \frac{q - \varepsilon r}{1 - \varepsilon}.$$

This belongs to $\Delta(A)$: for each a, $q(a) - \varepsilon r(a) \geq q(a) - \varepsilon > 0$, and the coordinates sum to one. The law

$$\varepsilon \sum_{j=1}^{k} \lambda_j \delta_{s_j} + (1 - \varepsilon)\delta_t$$

is a refinement of

$$\varepsilon \delta_r + (1 - \varepsilon)\delta_t,$$

and both belong to $\mathcal{M}_q$. Refinement monotonicity and the integral representation give

$$\varepsilon \sum_{j=1}^{k} \lambda_j \phi_q(s_j) + (1 - \varepsilon)\phi_q(t) \geq \varepsilon \phi_q(r) + (1 - \varepsilon)\phi_q(t).$$

Cancelling the common term and dividing by ε yields

$$\phi_q(r) \leq \sum_{j=1}^{k} \lambda_j \phi_q(s_j),$$

so ϕ_q is convex.

Since q is in the relative interior of the finite-dimensional simplex, the supporting-hyperplane theorem for finite convex functions gives a relative subgradient $g \in \mathbb{R}^A$ of the continuous convex function ϕ_q at q. Define

$$\widetilde{\phi}_q(r) := \phi_q(r) - g \cdot (r - q).$$

For every $\mu \in \mathcal{M}_q$,

$$\int g \cdot (r - q)\, d\mu(r) = g \cdot (b(\mu) - q) = 0,$$

so $\widetilde{\phi}_q$ represents the same affine functional on $\mathcal{M}_q$. The subgradient inequality gives

$$\widetilde{\phi}_q(r) \geq \widetilde{\phi}_q(q) \qquad \text{for all } r \in \Delta(A).$$

Finally replace

$$F_q(\mu) \quad \text{by} \quad F_q(\mu) - F_q(\delta_q)$$

and replace $\widetilde{\phi}_q$ by

$$\widetilde{\phi}_q - \widetilde{\phi}_q(q).$$

Before this final constant shift, the representation with $\widetilde{\phi}_q$ gives

$$F_q(\delta_q) = \int \widetilde{\phi}_q(r)\, d\delta_q(r) = \widetilde{\phi}_q(q).$$

Thus the same constant is subtracted from both sides of the integral representation, and subtracting a constant from F_q preserves all ordinal comparisons on $\mathcal{M}_q$. The final normalisation gives

$$\phi_q(q) = F_q(\delta_q) = 0,$$

and makes ϕ_q non-negative on the whole simplex. Hence

$$F_q(\mu) = \int \phi_q(r)\, d\mu(r) \geq 0$$

for every $\mu \in \mathcal{M}_q$. $\square$

For a finitely supported posterior law $\mu = \sum_i \lambda_i \delta_{r_i}$ and a bijection $\pi : A \to A'$, write

$$\mu\pi := \sum_i \lambda_i \delta_{r_i \pi}$$

for the posterior law obtained by relabelling actions through π.

Lemma 27 (Cross-fibre ordinal equivalence)**.** *Action coarsening implies ordinal cross-fibre equivalence for bijections. In particular, all no-information experiments have the same normalised value zero, and for every bijection $\pi : A \to A'$ and full-support $q \in \Delta(A)$, there exists $c_{\pi,q} > 0$ such that*

$$F_{q\pi}(\mu\pi) = c_{\pi,q} F_q(\mu)$$

for all $\mu \in \mathcal{M}_q$. The exact equality $c_{\pi,q} = 1$ for every q will be derived as Corollary 32 after product normalisation is established.

Proof. Ordinal equivalence. Let $\pi : A \to A'$ be a bijection. View π as the deterministic kernel $S(a' \mid a) = \mathbf{1}_{a'=\pi(a)}$. For any channel $P : A \to \Delta(X)$ and full-support $q \in \Delta(A)$, the Bayesian pushforward satisfies $S^q P = P\pi^{-1}$ (explicitly: $(S^q P)(x \mid a') = P(x \mid \pi^{-1}(a'))$). Clause (DP-act) of Condition (P4) gives

$$q^0 \succeq_{P \sqcup P\pi^{-1}} (q\pi)^1.$$

For the reverse inequality, apply (DP-act) to the bijection $\pi^{-1} : A' \to A$, the prior $q\pi \in \Delta(A')$, and the channel $P\pi^{-1} : A' \to \Delta(X)$. Then $(\pi^{-1})^{q\pi}(P\pi^{-1}) = P$ (the inverse bijection undoes the relabeling), and $(q\pi)\pi^{-1} = q$. Hence

$$(q\pi)^0 \succeq_{P\pi^{-1} \sqcup P} q^1.$$

By the second part of Condition (P3), block comparisons depend only on the pair of channels involved, not on numeric labels. The comparison $(q\pi)^0 \succeq_{P\pi^{-1} \sqcup P} q^1$ asks whether $(q\pi, P\pi^{-1})$ is at least as good as (q, P). Swapping the block labels (so that P becomes block 0 and $P\pi^{-1}$

becomes block 1) gives the equivalent statement $q^0 \preceq_{P \sqcup P\pi^{-1}} (q\pi)^1$. Combining with the first inequality gives

$$q^0 \sim_{P \sqcup P\pi^{-1}} (q\pi)^1.$$

If P implements $\mu \in \mathcal{M}_q$, then $P\pi^{-1}$ implements $\mu\pi \in \mathcal{M}_{q\pi}$: for every positive-probability record x,

$$m_{q\pi, P\pi^{-1}}(x) = m_{q,P}(x), \qquad r_x^{q\pi, P\pi^{-1}} = r_x^{q,P}\pi.$$

Now take arbitrary $\mu, \nu \in \mathcal{M}_q$, and choose channels P, Q implementing them at prior q. Let E be the four-block environment with blocks P, $P\pi^{-1}$, Q, and $Q\pi^{-1}$. Write x, x', y, y' for the block-supported lotteries q, $q\pi$, q, and $q\pi$ on these four blocks, respectively. By Lemma 20, the two two-block indifferences give $x \sim_E x'$ and $y \sim_E y'$. Hence, by transitivity of the weak order on E,

$$x \succeq_E y \iff x' \succeq_E y'.$$

Applying Lemma 20 again to delete irrelevant blocks gives

$$q^0 \succeq_{P \sqcup Q} q^1 \iff (q\pi)^0 \succeq_{P\pi^{-1} \sqcup Q\pi^{-1}} (q\pi)^1.$$

By Lemma 25 on the two fibres, this is equivalent to

$$F_q(\mu) \geq F_q(\nu) \iff F_{q\pi}(\mu\pi) \geq F_{q\pi}(\nu\pi).$$

The map $\mu \mapsto \mu\pi$ is an affine bijection from $\mathcal{M}_q$ to $\mathcal{M}_{q\pi}$, with inverse induced by π^{-1}. Therefore $G(\mu) := F_{q\pi}(\mu\pi)$ is a continuous affine representative of the same weak order on $\mathcal{M}_q$ as F_q. Herstein–Milnor uniqueness gives $G = aF_q + b$ with $a > 0$. Since $\delta_q\pi = \delta_{q\pi}$ and Lemma 26 gives

$$F_q(\delta_q) = F_{q\pi}(\delta_{q\pi}) = 0,$$

we have $b = 0$. Setting $c_{\pi,q} := a$ yields

$$F_{q\pi}(\mu\pi) = c_{\pi,q} F_q(\mu)$$

for all $\mu \in \mathcal{M}_q$.

No-information equivalence. By Lemma 26, $F_q(\delta_q) = 0$ for every full-support q on every finite action set, independently of cardinality. $\square$

Lemma 28 (Undetectable distinctions neutrality)**.** *Let $S : A \to A'$ be an onto deterministic partition map, and let $P : A \to \Delta(X)$ be S-measurable:*

$$S(a) = S(b) \quad \Rightarrow \quad P(\cdot \mid a) = P(\cdot \mid b).$$

Then for every full-support $q \in \Delta(A)$,

$$q^0 \sim_{P \sqcup S^q P} (qS)^1.$$

In particular, if $G_S : A \to \Delta(A')$ reveals the block $S(a)$, then

$$(q, G_S) \sim (qS, \mathrm{Id}_{A'}).$$

Proof. Since S is onto and q has full support, qS has full support on A'. Thus the Bayesian pushforward $S^q P$ has no zero-row ambiguity in this lemma. View the deterministic map S as its associated stochastic kernel. One direction is exactly clause (DP-act) of Condition (P4):

$$q^0 \succeq_{P \sqcup S^q P} (qS)^1.$$

For the reverse direction, define the Bayesian refinement kernel $R : A' \to \Delta(A)$ by

$$R(a \mid a') = \begin{cases} q(a)/(qS)(a'), & S(a) = a', \\ 0, & \text{otherwise.} \end{cases}$$

For each $a' \in A'$, $R(\cdot \mid a')$ is stochastic because

$$\sum_{a \in A} R(a \mid a') = \frac{\sum_{a:S(a)=a'} q(a)}{(qS)(a')} = 1.$$

Moreover, for every $a \in A$,

$$((qS)R)(a) = \sum_{a' \in A'} (qS)(a')R(a \mid a') = (qS)(S(a))\frac{q(a)}{(qS)(S(a))} = q(a) > 0.$$

Thus $(qS)R = q$.

Since P is S-measurable, the row of $S^q P$ at each block a' equals the common P-row on that block: if $S(a) = a'$, then

$$S^q P(x \mid a') = \frac{\sum_{b:S(b)=a'} q(b)P(x \mid b)}{(qS)(a')} = P(x \mid a).$$

Now pushing $S^q P$ back through R recovers P. For each $a \in A$,

$$(R^{qS}(S^q P))(x \mid a) = \frac{\sum_{a' \in A'}(qS)(a')R(a \mid a')S^q P(x \mid a')}{((qS)R)(a)} = S^q P(x \mid S(a)) = P(x \mid a).$$

Therefore

$$R^{qS}(S^q P) = P.$$

Apply (DP-act) to $(qS, S^q P)$ with the refinement kernel R:

$$(qS)^0 \succeq_{S^q P \sqcup P} q^1.$$

By Condition (P3) applied to the two-block environment $P \sqcup S^q P$ with ordered pair $(1, 0)$, this is equivalent to

$$(qS)^1 \succeq_{P \sqcup S^q P} q^0.$$

Together the two inequalities give the desired indifference. $\square$

For a nonempty $B \subsetneq A$ and a finite law μ whose barycentre is supported on B, non-negativity implies that every atom p of μ is supported on B. Define its support projection by

$$\pi_B(\mu) := \sum_{p \in \operatorname{supp}(\mu)} \mu(\{p\})\, \delta_{p|_B}.$$

It is a finite law on $\Delta(B)$ with barycentre $b(\mu)|_B$.

Lemma 29 (Support restriction)**.** *Let $r \in \Delta(A)$ have support $B := \operatorname{supp}(r) \subsetneq A$. Then:*

(i) Posterior-law reduction. *For any channel $P : A \to \Delta(X)$, the posterior law $\mu_{r,P}$ depends only on the rows $P(\cdot \mid b)$ for $b \in B$. Rows $P(\cdot \mid a)$ with $a \notin B$ have zero contribution because $r(a) = 0$.*

(ii) Preference equivalence. *Two channels $P, Q : A \to \Delta(X)$ that agree on B (i.e. $P(\cdot \mid b) = Q(\cdot \mid b)$ for all $b \in B$) are indifferent at prior r.*

(iii) Face identification. *For every channel $P : A \to \Delta(X)$,*

$$(r, P) \sim (r|_B, P|_B)$$

as a two-block comparison; consequently the comparison between channels P, Q at prior r agrees with the comparison between $P|_B, Q|_B$ at the full-support prior $r|_B \in \Delta(B)$. This is proved by projecting A to B and embedding B back into A, using the all-completions form of clause (DP-act) *of Condition* (P4) *for the embedding.*

(iv) Representative. *Define $F_r : \mathcal{M}_r \to \mathbb{R}$ by $F_r(\mu) := F^B_{r|_B}(\pi_B(\mu))$, where π_B projects posterior laws from $\Delta(A)$ to $\Delta(B)$ and $F^B_{r|_B}$ is the Lemma 25 representative on $\Delta(B)$. This F_r represents continuation comparisons at prior r in action set A.*

Proof. In this proof, $(s, R) \succeq (t, T)$ abbreviates the block comparison $s^0 \succeq_{R \sqcup T} t^1$, and $\sim$ denotes both weak inequalities. These comparisons involve different priors and different action sets; they chain by transitivity in Lemma 20(a), which is stated across such pairs.

Part (i): The marginal probability $m_{r,P}(x) = \sum_a r(a)P(x \mid a)$ and the Bayes numerator $r(a)P(x \mid a)$ are both zero for $a \notin B$. Hence $\mu_{r,P}$ is determined by $\{P(\cdot \mid b) : b \in B\}$.

Part (ii): Let P and $\widetilde{P}$ be channels on A that agree on B. We first show $(r, P) \sim (r, \widetilde{P})$. Fix $b_0 \in B$, and let $S : A \to B$ be the deterministic map with $S(b) = b$ for $b \in B$ and $S(a) = b_0$ for $a \notin B$. Then $rS = r|_B$ and the Bayesian pushforward $S^r P$ is $P|_B$ on B. Clause (DP-act) gives

$$(r, P) \succeq (r|_B, P|_B).$$

Let $I : B \hookrightarrow A$ be the inclusion. Then $(r|_B)I = r$, and the Bayesian pushforward $I^{r|_B}(P|_B)$ is pinned down on B and unconstrained on $A \setminus B$. By the all-completions convention in (DP-act), choose the completion to be $\widetilde{P}$. Clause (DP-act) gives

$$(r|_B, P|_B) \succeq (r, \widetilde{P}).$$

Thus $(r, P) \succeq (r, \widetilde{P})$. Applying the same projection/inclusion argument with $\widetilde{P}$ as the original channel and choosing P as the inclusion completion gives $(r, \widetilde{P}) \succeq (r, P)$. Hence $(r, P) \sim (r, \widetilde{P})$.

If P, Q agree on B, choose a common extension $\widetilde{P} = \widetilde{Q}$ of this common restriction to all of A (for instance, use one fixed off-support row). The preceding paragraph gives $(r, P) \sim (r, \widetilde{P})$ and $(r, Q) \sim (r, \widetilde{Q})$; since $\widetilde{P} = \widetilde{Q}$, common-environment transitivity gives $(r, P) \sim (r, Q)$.

Part (iii): We first prove $(r, P) \sim (r|_B, P|_B)$ for every channel $P : A \to \Delta(X)$.

Forward direction ((DP-act) *projection).* Let $S : A \to B$ map all $a \notin B$ to some fixed $b_0 \in B$. The coarsened prior is $rS = r|_B$ (since $r(a) = 0$ for $a \notin B$), and the Bayesian pushforward $S^r P$ equals $P|_B$ on B. By (DP-act),

$$(r, P) \succeq (r|_B, P|_B).$$

Reverse direction ((DP-act) *inclusion with completion).* Let $I : B \hookrightarrow A$ be the inclusion. Then $(r|_B)I = r$. The Bayesian pushforward $I^{r|_B}(P|_B)$ is pinned down on B and arbitrary on $A \setminus B$. Choose the (DP-act) completion to be any extension $\widetilde{P} : A \to \Delta(X)$ of $P|_B$. Then

$$(r|_B, P|_B) \succeq (r, \widetilde{P}).$$

By Part (ii), $(r, \widetilde{P}) \sim (r, P)$ because $\widetilde{P}$ and P agree on $B = \mathrm{supp}(r)$. Hence $(r|_B, P|_B) \succeq (r, P)$.

Together the two directions give $(r, P) \sim (r|_B, P|_B)$.

Comparison equivalence. For any P, Q: $(r, P) \succeq (r, Q)$ iff (by the above indifferences $(r, P) \sim (r|_B, P|_B)$, $(r, Q) \sim (r|_B, Q|_B)$, and Condition (P1) transitivity) $(r|_B, P|_B) \succeq (r|_B, Q|_B)$.

Part (iv): If $\mu \in \mathcal{M}_r$, every posterior in the finite support of μ is supported on B; otherwise the barycentre r would assign positive mass to $A \setminus B$. Hence $\pi_B(\mu) \in \mathcal{M}_{r|_B}$ is well defined. By (iii), continuation comparisons at r are isomorphic to those at $r|_B$. The Lemma 25 representative $F^B_{r|_B}$ on $\Delta(B)$ therefore induces a representative on $\mathcal{M}_r$ via π_B. $\square$

Lemma 30 (Derived background inertness)**.** *Assume Conditions* (P1) *and* (P3)–(P5)*. Let A_1, A_2 be finite action sets and let $q_1 \in \Delta(A_1)$ and $q_2 \in \Delta(A_2)$ have full support. Then, for all channels P_1, Q_1 on A_1 and every background channel R_2 on A_2,*

$$(q_1 \otimes q_2)^0 \succeq_{(P_1 \otimes R_2) \sqcup (Q_1 \otimes R_2)} (q_1 \otimes q_2)^1 \iff q_1^0 \succeq_{P_1 \sqcup Q_1} q_1^1, \tag{BI}$$

and symmetrically for comparisons in the second component with a common first-component background. In particular the left-hand comparison does not depend on the common background: for all backgrounds R_2, S_2 on A_2,

$$(q_1 \otimes q_2)^0 \succeq_{(P_1 \otimes R_2) \sqcup (Q_1 \otimes R_2)} (q_1 \otimes q_2)^1 \iff (q_1 \otimes q_2)^0 \succeq_{(P_1 \otimes S_2) \sqcup (Q_1 \otimes S_2)} (q_1 \otimes q_2)^1.$$

Proof. Throughout, for a prior s on an action set A and channels V, W on A write $V \succeq_s W$ for the block comparison $s^0 \succeq_{V \sqcup W} s^1$, and $(s, V) \sim (t, W)$ for the two-block indifference between the pairs. By Lemma 20(a) these comparisons form a weak order for each fixed prior and chain transitively across pairs with different priors and action sets. By Lemma 22, at a full-support prior a channel may be replaced by any channel with the same posterior law, in weak and in strict comparisons alike; we call this *substitution*.

Preliminary reduction: a common foreground alphabet. Condition (P5) requires the two continuation channels used on a given branch to share a record alphabet. If $P_1 : A_1 \to \Delta(X)$ and $Q_1 : A_1 \to \Delta(X')$, replace them by their zero-padded versions on $X \sqcup X'$, as in Step 2 of Lemma 23. Padding adds only null records, so it changes neither $\mu_{q_1,\cdot}$ nor $\mu_{q_1 \otimes q_2, \cdot \otimes R_2}$; both sides of (BI) are therefore unaffected by substitution at the full-support priors q_1 and $q_1 \otimes q_2$. Assume from now on that P_1 and Q_1 share the record alphabet X_1.

For a channel $V : A_1 \to \Delta(X_1)$ and a finite set A_2, write $\widetilde{V}$ for its lift to the product action set,

$$\widetilde{V}(z \mid (a_1, a_2)) := V(z \mid a_1),$$

and let $\widetilde{R}_2(x_2 \mid (a_1, a_2)) := R_2(x_2 \mid a_2)$ be the lift of the background.

Step 1: a product channel is a two-stage channel with background first stage. Take $\widetilde{R}_2$ as first stage on the action set $A_1 \times A_2$ and the constant continuation profile $\widetilde{V}$ in every branch, so that every branch uses the same continuation alphabet X_1 and the compound record set $\bigsqcup_{x_2 \in X_2} (\{x_2\} \times X_1)$ is $X_2 \times X_1$. By the definition of two-stage composition,

$$(\widetilde{R}_2 \triangleright \{\widetilde{V}\}_{x_2})(x_2, z \mid (a_1, a_2)) = R_2(x_2 \mid a_2)\, V(z \mid a_1) = (V \otimes R_2)(z, x_2 \mid (a_1, a_2)).$$

The two channels therefore differ only by the bijection $(x_2, z) \mapsto (z, x_2)$ of record labels, which leaves every record probability and every posterior unchanged. Hence they induce the same posterior law at $q_1 \otimes q_2$, and substitution makes them interchangeable in every comparison at that prior. Applying this to $V = P_1$ and $V = Q_1$, the left-hand side of (BI) is equivalent to

$$\widetilde{R}_2 \triangleright \{\widetilde{P}_1\} \succeq_{q_1 \otimes q_2} \widetilde{R}_2 \triangleright \{\widetilde{Q}_1\}, \tag{$*$}$$

and likewise for the strict comparisons.

Step 2: every reached branch posterior is a product with first marginal q_1. Write $q := q_1 \otimes q_2$, $m(x_2) := m_{q, \widetilde{R}_2}(x_2)$ and, for $m(x_2) > 0$, $r_{x_2} := r_{x_2}^{q, \widetilde{R}_2}$. Since the rows of $\widetilde{R}_2$ do not depend on a_1,

$$m(x_2) = \sum_{a_2 \in A_2} q_2(a_2) R_2(x_2 \mid a_2) = m_{q_2, R_2}(x_2),$$

and Bayes' rule gives

$$r_{x_2}(a_1, a_2) = \frac{q_1(a_1) q_2(a_2) R_2(x_2 \mid a_2)}{m(x_2)} = q_1(a_1)\, q_2^{x_2}(a_2), \qquad q_2^{x_2} := r_{x_2}^{q_2, R_2}.$$

Thus $r_{x_2} = q_1 \otimes q_2^{x_2}$: the background signal may change beliefs about a_2, but in every reached branch the two components stay independent and the first marginal is still q_1. At least one branch has positive probability.

Step 3: a lifted foreground channel is evaluated as on its own component. We claim that for every $s_2 \in \Delta(A_2)$ and every channel $V : A_1 \to \Delta(X_1)$,

$$(q_1 \otimes s_2, \widetilde{V}) \sim (q_1, V). \tag{†}$$

First suppose s_2 has full support, so that $q_1 \otimes s_2$ has full support on $A_1 \times A_2$. Let $S : A_1 \times A_2 \to A_1$ be the coordinate projection, an onto deterministic map, and note that $\widetilde{V}$ is S-measurable, since $S(a) = S(b)$ implies $\widetilde{V}(\cdot \mid a) = \widetilde{V}(\cdot \mid b)$. Moreover $(q_1 \otimes s_2)S = q_1$ and, for every a_1,

$$(S^{q_1 \otimes s_2}\widetilde{V})(z \mid a_1) = \frac{\sum_{a_2 \in A_2} q_1(a_1)s_2(a_2)V(z \mid a_1)}{q_1(a_1)} = V(z \mid a_1),$$

so the Bayesian pushforward of $\widetilde{V}$ under S is V itself. Undetectable-distinctions neutrality (Lemma 28) gives (†).

If s_2 is not full support, let $B_2 := \mathrm{supp}(s_2)$, so that $B := A_1 \times B_2$ is the support of $q_1 \otimes s_2$ and $B \subsetneq A_1 \times A_2$. The set B *is* the product action set $A_1 \times B_2$, so no relabelling is involved: the restricted prior is $q_1 \otimes (s_2|_{B_2})$ and the restricted channel $\widetilde{V}|_B$ is the lift of V to $A_1 \times B_2$. Face identification (Lemma 29(iii)) gives $(q_1 \otimes s_2, \widetilde{V}) \sim (q_1 \otimes s_2|_{B_2}, \widetilde{V}|_B)$, and the full-support case just proved, applied on the action set $A_1 \times B_2$, gives $(q_1 \otimes s_2|_{B_2}, \widetilde{V}|_B) \sim (q_1, V)$. Transitivity in a common block environment yields (†) in general.

Fix a branch with $m(x_2) > 0$ and apply (†) at $s_2 = q_2^{x_2}$ to $V = P_1$ and to $V = Q_1$, using $r_{x_2} = q_1 \otimes q_2^{x_2}$ from Step 2. Place the four pairs $(r_{x_2}, \widetilde{P}_1)$, (q_1, P_1), $(r_{x_2}, \widetilde{Q}_1)$, (q_1, Q_1) as blocks of one finite block environment (Condition (P3), Lemma 20). The two indifferences and transitivity of the weak order in that environment (Condition (P1)) give

$$\widetilde{P}_1 \succeq_{r_{x_2}} \widetilde{Q}_1 \quad \iff \quad P_1 \succeq_{q_1} Q_1, \tag{‡}$$

and, since the relation is complete, the same equivalence for the strict comparisons. In particular the branch comparison is the same in every reached branch and does not involve R_2.

Step 4: forward implication. Assume $P_1 \succeq_{q_1} Q_1$. By (‡), the branch hypothesis $r_{x_2}^0 \succeq_{\widetilde{P}_1 \sqcup \widetilde{Q}_1} r_{x_2}^1$ of Condition (P5) holds for every branch with $m(x_2) > 0$, with the common first stage $\widetilde{R}_2$ and the two constant continuation profiles. Condition (P5) yields (∗), hence the left-hand side of (BI) by Step 1.

Step 5: converse, via the strictness clause. Assume the left-hand side of (BI), so that (∗) holds by Step 1, and suppose $P_1 \succeq_{q_1} Q_1$ fails. By completeness and the reverse-block equivalence of Lemma 20, $Q_1 \succ_{q_1} P_1$, so by the strict form of (‡) every reached branch satisfies $\widetilde{Q}_1 \succ_{r_{x_2}} \widetilde{P}_1$. At least one such branch exists and has positive probability, so the strictness clause of Condition (P5), applied with the two profiles exchanged, gives

$$\widetilde{R}_2 \triangleright \{\widetilde{Q}_1\} \succ_{q_1 \otimes q_2} \widetilde{R}_2 \triangleright \{\widetilde{P}_1\},$$

contradicting (∗) because $\succeq_{q_1 \otimes q_2}$ is a weak order (Lemma 20). Hence $P_1 \succeq_{q_1} Q_1$, proving (BI).

The second-component clause is the same argument with the roles of the two coordinates exchanged: the common first-component background is used as the first stage and the second-component channels as the continuations. Finally, since both product comparisons in the last display of the lemma are equivalent to the single background-free comparison $P_1 \succeq_{q_1} Q_1$, they are equivalent to each other. □

The lemma shows that separability across independent components is not an independent behavioural commitment. A product channel is a two-stage channel whose first stage is the background experiment: it can teach the observer about the background action, but it leaves the foreground component independent of it and its marginal unchanged, so in every reached branch the continuation comparison is the foreground comparison itself. Branchwise monotonicity then transmits that single comparison to the compound, and its strictness clause transmits it back. The derived statement is stronger than mere invariance to changing the background: it identifies the product comparison with the comparison of the foreground channels alone, which is the form used in Lemma 31 below.

For finite posterior laws $\mu = \sum_i m_i \delta_{r_i} \in \mathsf{X}_A$ and $\nu = \sum_j n_j \delta_{s_j} \in \mathsf{X}_B$, define

$$\mu \otimes \nu := \sum_{i,j} m_i n_j \, \delta_{r_i \otimes s_j}.$$

Bayes' rule then gives the product factorisation

$$\mu_{q_1 \otimes q_2, P_1 \otimes P_2} = \mu_{q_1, P_1} \otimes \mu_{q_2, P_2}. \tag{PL}$$

Lemma 31 (Coherent product quasi-additivity)**.** *There is a choice of positive rescalings of the zero-normalised fibrewise representatives F_q, still denoted F_q, and a scalar $\kappa \in \mathbb{R}$ such that, for all full-support priors $p \in \Delta(A)$ and $r \in \Delta(B)$ and all $\mu \in \mathcal{M}_p$, $\nu \in \mathcal{M}_r$,*

$$F_{p \otimes r}(\mu \otimes \nu) = F_p(\mu) + F_r(\nu) + \kappa F_p(\mu) F_r(\nu). \tag{QA}$$

If one factor is a singleton, the same formula holds with its identically zero representative. Moreover, for every nondegenerate r and every $\nu \in \mathcal{M}_r$,

$$1 + \kappa F_r(\nu) > 0. \tag{POS}$$

Proof. All representatives in this proof are zero-normalised: $F_q(\delta_q) = 0$. The proof first treats nondegenerate action sets, meaning action sets with at least two elements.

Step 1: product-coordinate independence. Fix full-support priors p and r on nondegenerate finite action sets and set

$$L_{p,r}(\mu, \nu) := F_{p \otimes r}(\mu \otimes \nu).$$

This map is continuous and separately affine. To identify a first-coordinate slice, let P, Q implement $\mu, \mu' \in \mathcal{M}_p$ and let R implement $\nu \in \mathcal{M}_r$. By the product factorisation (PL), $P \otimes R$ implements $\mu \otimes \nu$ and $Q \otimes R$ implements $\mu' \otimes \nu$ at the full-support prior $p \otimes r$. Hence Lemma 25, applied on the fibre $\mathcal{M}_{p \otimes r}$ and on the fibre $\mathcal{M}_p$, turns background inertness (BI) of Lemma 30 into

$$L_{p,r}(\mu, \nu) \geq L_{p,r}(\mu', \nu) \quad \iff \quad F_p(\mu) \geq F_p(\mu').$$

The symmetric clause of Lemma 30 identifies every second-coordinate slice with the F_r-order. Local non-triviality, the fibrewise representation, and the zero normalisation from Lemma 26 make both slice orders nonconstant: $F_p(\chi_p) > 0$ and $F_r(\chi_r) > 0$.

Step 2: the pair-specific bilinear form. Put $x(\mu) := F_p(\mu)$ and $y(\nu) := F_r(\nu)$. For fixed ν, affine-utility uniqueness gives

$$L_{p,r}(\mu, \nu) = \alpha(\nu) x(\mu) + \gamma(\nu), \qquad \alpha(\nu) > 0.$$

Taking $\mu = \delta_p$ shows $\gamma(\nu) = L_{p,r}(\delta_p, \nu)$. This is a continuous affine representative of the F_r-order and vanishes at δ_r, so

$$\gamma(\nu) = B_{p,r} y(\nu)$$

for some $B_{p,r} > 0$. Likewise, the slice at $\nu = \delta_r$ gives

$$L_{p,r}(\mu, \delta_r) = A_{p,r} x(\mu)$$

for some $A_{p,r} > 0$.

Let $\bar{\mu} := \chi_p$ and $h_p := F_p(\chi_p) > 0$. The function $\nu \mapsto L_{p,r}(\bar{\mu}, \nu)$ is again a continuous affine representative of the F_r-order. Hence there are $D_{p,r} > 0$ and a constant $E_{p,r}$ such that

$$L_{p,r}(\bar{\mu}, \nu) = D_{p,r} y(\nu) + E_{p,r}.$$

At $\nu = \delta_r$, $E_{p,r} = A_{p,r} h_p$. Comparing this identity with $L_{p,r}(\bar{\mu}, \nu) = \alpha(\nu) h_p + B_{p,r} y(\nu)$ gives

$$\alpha(\nu) = A_{p,r} + C_{p,r} y(\nu), \qquad C_{p,r} := \frac{D_{p,r} - B_{p,r}}{h_p}.$$

Therefore

$$L_{p,r}(\mu, \nu) = A_{p,r} x(\mu) + B_{p,r} y(\nu) + C_{p,r} x(\mu) y(\nu). \tag{BIL}$$

The coordinate-slice slopes are positive:

$$A_{p,r} + C_{p,r} y(\nu) > 0, \qquad B_{p,r} + C_{p,r} x(\mu) > 0.$$

Step 3: normalising the two linear coefficients. Adopt the strict-product convention: finite products are identified by tuple notation, so $(A \times B) \times C = A \times B \times C = A \times (B \times C)$ and similarly for product priors. This convention does not identify $A \times B$ with $B \times A$.

Expanding $F_{p \otimes r \otimes s}$ under the two groupings and comparing the coefficients of x, y, and z gives

$$\begin{aligned} A_{p \otimes r, s} A_{p,r} &= A_{p, r \otimes s}, && \text{(C1)} \\ A_{p \otimes r, s} B_{p,r} &= B_{p, r \otimes s} A_{r,s}, && \text{(C2)} \\ B_{p \otimes r, s} &= B_{p, r \otimes s} B_{r,s}. && \text{(C3)} \end{aligned}$$

Coefficient comparison is legitimate because the mixtures $(1-t)\delta_p + t\chi_p$ make x range over a nondegenerate interval, and similarly in the other coordinates.

The coordinate swap gives a scalar $d(p,r) > 0$ such that

$$F_{r \otimes p}(\nu \otimes \mu) = d(p,r) F_{p \otimes r}(\mu \otimes \nu).$$

Comparing coefficients in (BIL),

$$B_{r,p} = d(p,r) A_{p,r}, \qquad A_{r,p} = d(p,r) B_{p,r}, \qquad C_{r,p} = d(p,r) C_{p,r}.$$

Thus, with $\rho(p,r) := B_{p,r}/A_{p,r}$,

$$\rho(r,p) = \rho(p,r)^{-1}. \tag{SW}$$

Divide (C2) by (C1):

$$\rho(p,r) = \rho(p, r \otimes s) A_{r,s}.$$

Fix a nondegenerate reference prior q_0 on a finite action set A_0 and put $\phi(r) := \rho(q_0, r)$. Then

$$A_{r,s} = \frac{\phi(r)}{\phi(r \otimes s)}. \tag{G}$$

Rescale every representative by $F_p^* := \phi(p) F_p$. Under this positive rescaling,

$$A_{p,r}^* = A_{p,r} \frac{\phi(p \otimes r)}{\phi(p)}, \quad B_{p,r}^* = B_{p,r} \frac{\phi(p \otimes r)}{\phi(r)}, \quad C_{p,r}^* = C_{p,r} \frac{\phi(p \otimes r)}{\phi(p)\phi(r)}.$$

Equation (G) gives $A_{p,r}^* = 1$ for every pair. Write $\beta(p,r) := B_{p,r}^*$. The transformed versions of (C2)–(C3) and (SW) are

$$\begin{aligned} \beta(p,r) &= \beta(p, r \otimes s), \\ \beta(p \otimes r, s) &= \beta(p,r)\beta(r,s), \\ \beta(r,p) &= \beta(p,r)^{-1}. \end{aligned}$$

Moreover $\phi(q_0) = \rho(q_0, q_0) = 1$. Equation (G) gives $A_{q_0,p} = 1/\phi(q_0 \otimes p)$, while $B_{q_0,p} = A_{q_0,p}\rho(q_0, p) = \phi(p)/\phi(q_0 \otimes p)$. Therefore

$$\beta(q_0, p) = B_{q_0,p}\frac{\phi(q_0 \otimes p)}{\phi(p)} = 1.$$

By swap reciprocity, $\beta(p, q_0) = 1$. Taking $s = q_0$ in the second displayed identity gives

$$1 = \beta(p \otimes r, q_0) = \beta(p, r)\beta(r, q_0) = \beta(p, r).$$

Thus, after the positive gauge rescaling, $A_{p,r} = B_{p,r} = 1$ for every nondegenerate pair. Positivity of the slice slopes is preserved because the rescaling factors are positive.

Step 4: the interaction coefficient is universal. Write the remaining coefficient as $\kappa_{p,r}$. Comparing the xy, xz, yz, and xyz coefficients in the two associative expansions gives

$$\kappa_{p,r} = \kappa_{p,r\otimes s}, \tag{K1}$$
$$\kappa_{p\otimes r,s} = \kappa_{p,r\otimes s}, \tag{K2}$$
$$\kappa_{p\otimes r,s} = \kappa_{r,s}, \tag{K3}$$
$$\kappa_{p\otimes r,s}\kappa_{p,r} = \kappa_{p,r\otimes s}\kappa_{r,s}. \tag{K4}$$

The last equation follows from the first three, but it records the full coefficient comparison. After $A = B = 1$, the linear coefficient comparison under the coordinate swap forces the swap scalar to be one; the bilinear coefficient comparison then gives

$$\kappa_{p,r} = \kappa_{r,p}. \tag{KS}$$

Using (K1)–(K3) with the reference prior q_0,

$$\kappa_{p,r} = \kappa_{p,r\otimes q_0} = \kappa_{p\otimes r,q_0} = \kappa_{r,q_0}.$$

Hence $\kappa_{p,r}$ is independent of its first argument. By symmetry it is also independent of its second argument. Denote the common value by κ. This proves (QA) for nondegenerate factors.

The positive slice-slope condition now reads

$$1 + \kappa F_r(\nu) > 0,$$

which is (POS).

Step 5: singleton factors. Let $q_* = \delta_*$ be the prior on a singleton. Its zero-normalised representative is identically zero. If p is nondegenerate, the canonical bijection $A \to A \times \{*\}$ and Lemma 27 give a scalar $c > 0$ with

$$F_{p\otimes q_*}(\mu \otimes \delta_{q_*}) = cF_p(\mu).$$

Tensor both sides with the nondegenerate reference prior q_0. The canonical bijection between $(A \times \{*\}) \times A_0$ and $A \times A_0$ gives a scalar $d > 0$, and quasi-additivity for the two nondegenerate products gives, with $x = F_p(\mu)$ and $y = F_{q_0}(\nu)$,

$$cx + y + \kappa cxy = d(x + y + \kappa xy).$$

Taking $\mu = \delta_p$ and $\nu = \chi_{q_0}$ gives $x = 0$ and $y > 0$, hence $d = 1$. Taking $\mu = \chi_p$ and $\nu = \delta_{q_0}$ gives $x > 0$ and $y = 0$, hence $c = 1$. Thus (QA) also holds with a singleton factor. The reversed order is identical. □

Corollary 32 (Exact relabelling invariance)**.** *For every bijection $\pi : A \to A'$ and every full-support $q \in \Delta(A)$,*

$$F_{q\pi}(\mu\pi) = F_q(\mu) \qquad (\mu \in \mathcal{M}_q).$$

Proof. The singleton case is immediate because both representatives vanish. Suppose A is nondegenerate. Lemma 27 gives $F_{q\pi}(\mu\pi) = cF_q(\mu)$ for some $c > 0$. Apply the product bijection $\pi \times \mathrm{id}_{A_0}$ with the nondegenerate reference prior q_0. There is $d > 0$ such that, writing $x = F_q(\mu)$ and $y = F_{q_0}(\nu)$,

$$cx + y + \kappa cxy = d(x + y + \kappa xy).$$

Set $\mu = \delta_q$ and $\nu = \chi_{q_0}$, so $x = 0$ and $y > 0$, to obtain $d = 1$. Then set $\mu = \chi_q$ and $\nu = \delta_{q_0}$, so $x > 0$ and $y = 0$, to obtain $c = 1$. □

Lemma 33 (Cross-prior block representation)**.** *For arbitrary finite priors $q \in \Delta(A)$ and $r \in \Delta(B)$ and channels $P : A \to \Delta(X)$ and $Q : B \to \Delta(Y)$,*

$$q^0 \succeq_{P \sqcup Q} r^1 \iff F_q(\mu_{q,P}) \geq F_r(\mu_{r,Q}). \tag{BC}$$

Boundary representatives are interpreted by Lemma 29.

Proof. First suppose q and r have full support, allowing singleton action sets. At prior $q \otimes r$, quasi-additivity and zero normalisation give

$$F_{q\otimes r}(\mu_{q,P} \otimes \delta_r) = F_q(\mu_{q,P}), \qquad F_{q\otimes r}(\delta_q \otimes \mu_{r,Q}) = F_r(\mu_{r,Q}).$$

These values are in the same fibre $\mathcal{M}_{q\otimes r}$, so Lemma 25 gives

$$(q \otimes r)^0 \succeq_{(P\otimes U_B)\sqcup(U_A\otimes Q)} (q \otimes r)^1 \iff F_q(\mu_{q,P}) \geq F_r(\mu_{r,Q}).$$

It remains to transfer the product-lifted comparison back to the original two blocks. For the A-side pair, Clause (DP-act) of Condition (P4) applied to the projection $\pi_A : A \times B \to A$ gives

$$(q \otimes r, P \otimes U_B) \succeq (q, P)$$

in block comparisons. Conversely, let $T : A \to \Delta(A \times B)$ be

$$T((a, b) \mid a') = \mathbf{1}_{\{a=a'\}} r(b).$$

Then $qT = q \otimes r$, and the Bayesian pushforward $T^q P$ has row $P(\cdot \mid a)$ at every (a, b). Thus $T^q P$ is $P \otimes U_B$ after the canonical one-record identification of the B-experiment, and (DP-act) gives the reverse comparison. Equivalently, if the harmless record identification is not made, Lemma 22 identifies $T^q P$ and $P \otimes U_B$ because they induce the same posterior law at $q \otimes r$. Hence $(q \otimes r, P \otimes U_B)$ and (q, P) are indifferent in every relevant block comparison. The same projection/embedding argument identifies $(q \otimes r, U_A \otimes Q)$ with (r, Q).

Place the four pairs (q, P), $(q \otimes r, P \otimes U_B)$, (r, Q), and $(q \otimes r, U_A \otimes Q)$ as blocks of one finite block environment. Lemma 20 and transitivity transfer the two pairwise indifferences to the two-block comparison, proving (BC) for full-support priors.

For arbitrary priors, let $A_+ := \mathrm{supp}(q)$ and $B_+ := \mathrm{supp}(r)$. Lemma 29 identifies (q, P) with $(q|_{A_+}, P|_{A_+})$, identifies (r, Q) with $(r|_{B_+}, Q|_{B_+})$, and identifies the two values $F_q(\mu_{q,P})$ and $F_r(\mu_{r,Q})$ with their support-face values. Applying the full-support bridge on A_+ and B_+ and pulling back by Lemma 29 proves the claim. Singleton supports use the same zero-representative convention as Lemma 31. □

Lemma 34 (Branch aggregation)**.** *Let $P_1 : A \to \Delta(X_1)$ be a first-stage channel at full-support q, with branch probabilities $m(x)$ and posteriors r_x. There are positive branch coefficients $\beta(q, r_x)$, depending only on q and the reached posterior r_x, such that for every continuation profile $\{Q^x\}_{x:m(x)>0}$,*

$$F_q(\mu_{q,P_1 \triangleright \{Q^x\}}) = F_q(\mu_{q,P_1}) + \sum_{x:m(x)>0} m(x)\, \beta(q, r_x)\, F_{r_x}(\mu_{r_x,Q^x}).$$

If r_x has full support, $\beta(q, r_x)$ is the full-support tangent coefficient. If $1 < |\mathrm{supp}(r_x)| < |A|$, it is the full-to-face coefficient. If $|\mathrm{supp}(r_x)| = 1$, the value F_{r_x} is identically zero and $\beta(q, r_x)$ is an arbitrary positive convention.

Proof. Fix the first-stage channel P_1 and its branch structure $(m(x), r_x)_{x:m(x)>0}$. For any profile of continuation channels $\{Q^x\}$, the compound posterior law at prior q is

$$\mu_{q,P_1 \triangleright \{Q^x\}} = \sum_{x:m(x)>0} m(x)\, \mu_{r_x,Q^x}.$$

Setup: linear part of affine functional. Since F_q is affine on $\mathcal{M}_q$ (Lemma 25), it extends uniquely to an affine functional on the affine hull of $\mathcal{M}_q$. Write L_q for the linear part: for any $\mu, \mu' \in \mathcal{M}_q$,

$$F_q(\mu) - F_q(\mu') = L_q(\mu - \mu').$$

Branch-affine structure. Fix a positive-probability branch $\bar{x}$ and fix continuation channels $H^x : A \to \Delta(Z^x)$ in all other branches $x \neq \bar{x}$. Let $r := r_{\bar{x}}$ and $m := m(\bar{x})$. If $\operatorname{supp}(r)$ is a singleton, then $\mathcal{M}_r = \{\delta_r\}$ and $F_r(\delta_r) = 0$ by the singleton zero-normalisation convention; such branches contribute zero and their coefficient convention is recorded in Case 1 below. For the branch-affine argument in this paragraph, assume $|\operatorname{supp}(r)| \geq 2$. If r is a boundary posterior, F_r and continuation comparisons at r are understood through Lemma 29.

Let

$$\rho_H := \sum_{x \neq \bar{x}} m(x)\, \mu_{r_x,H^x}.$$

For any $\nu \in \mathcal{M}_r$, the aggregate posterior law produced by using branch law ν in branch $\bar{x}$ and the fixed continuations elsewhere is

$$T_H(\nu) := m\nu + \rho_H \in \mathcal{M}_q.$$

Define $g : \mathcal{M}_r \to \mathbb{R}$ by

$$g(\nu) := F_q(T_H(\nu)).$$

Thus g depends only on ν, and $g = F_q \circ T_H$ is continuous and affine.

We now show that g and F_r represent the same weak order on $\mathcal{M}_r$. For $\nu, \nu' \in \mathcal{M}_r$, choose finite continuation channels implementing them at prior r; when r is boundary these are implemented on $B = \operatorname{supp}(r)$ and extended arbitrarily to A, as licensed by Lemma 29. Padding record alphabets if needed, Condition (P5) compares the two continuation profiles that differ only in branch $\bar{x}$. If $F_r(\nu) > F_r(\nu')$, Condition (P5) gives $g(\nu) > g(\nu')$. Conversely, if $g(\nu) \geq g(\nu')$ but $F_r(\nu) < F_r(\nu')$, applying the strict part of Condition (P5) to the swapped branch comparison gives $g(\nu') > g(\nu)$, a contradiction. Hence $g(\nu) \geq g(\nu')$ iff $F_r(\nu) \geq F_r(\nu')$.

Since $|\operatorname{supp}(r)| \geq 2$, local non-triviality on the support face and Lemmas 29 and 26 imply F_r is nonconstant. The continuous affine functions g and F_r are therefore nonconstant affine representatives of the same weak order on $\mathcal{M}_r$. Herstein–Milnor uniqueness gives

$$g(\nu) = \alpha\, F_r(\nu) + \gamma$$

for some $\alpha > 0$ and $\gamma \in \mathbb{R}$.

The coefficient α is independent of the fixed continuations. Write $g^{(1)}$ and $g^{(2)}$ for the functionals arising from two different choices $\{H^x\}$ and $\{\widetilde{H}^x\}$ in the other branches. Both represent the same order as F_r, so $g^{(1)} = \alpha_1 F_r + \gamma_1$ and $g^{(2)} = \alpha_2 F_r + \gamma_2$ with $\alpha_1, \alpha_2 > 0$. To show $\alpha_1 = \alpha_2$, consider the difference $g^{(1)} - g^{(2)} = (\alpha_1 - \alpha_2)F_r + (\gamma_1 - \gamma_2)$. For any $\nu \in \mathcal{M}_r$,

$$g^{(1)}(\nu) - g^{(2)}(\nu) = F_q\Big(m(\bar{x})\nu + \textstyle\sum_{x \neq \bar{x}} m(x)\mu_{r_x,H^x}\Big) - F_q\Big(m(\bar{x})\nu + \textstyle\sum_{x \neq \bar{x}} m(x)\mu_{r_x,\widetilde{H}^x}\Big).$$

Both arguments are posterior laws in $\mathcal{M}_q$, differing only in the fixed non-$\bar{x}$ components. Since L_q has already been defined on the affine-hull difference space,

$$g^{(1)}(\nu) - g^{(2)}(\nu) = L_q(\rho_1 - \rho_2),$$

where ρ_i is the fixed non-$\bar{x}$ background law for $g^{(i)}$. This difference is independent of ν. Since F_r is nonconstant in the present nondegenerate case, $\alpha_1 = \alpha_2$.

Path-independence of the branch coefficient via tangent-space decomposition. We now prove that $\alpha/m(\bar{x})$ depends only on the pair $(q, r_{\bar{x}})$, not on the first-stage experiment. This is the key step that converts ordinal branchwise monotonicity into a cardinal aggregation formula.

Tangent space at a posterior. For a posterior $r \in \Delta(A)$ with support $B := \operatorname{supp}(r)$, define the *tangent space*:

$$V_r := \operatorname{span}\{\nu - \nu' : \nu, \nu' \in \mathcal{M}_r\}.$$

Elements of V_r are signed measures with total mass zero and barycentre zero.

Tangent-space identification. For any r with full support, we claim

$$V_r = W := \{\eta \in \operatorname{span} \Delta_f(\Delta(A)) : \eta(\Delta(A)) = 0,\ \textstyle\int p \, d\eta(p) = 0\},$$

the space of finite-support signed measures on $\Delta(A)$ with zero total mass and zero barycentre. Indeed, if $\nu, \nu' \in \mathcal{M}_r$, then $\nu - \nu'$ has total mass $1 - 1 = 0$ and barycentre $r - r = 0$, so $V_r \subset W$. Conversely, given any $\eta \in W$: if $\eta = 0$ then $\eta \in V_r$ trivially. Otherwise, since η has finite support, write its Jordan decomposition $\eta = \eta^+ - \eta^-$ where $\eta^+, \eta^- \geq 0$ are finite-support positive measures with equal total mass $c > 0$. Normalise $\zeta^+ := \eta^+/c$ and $\zeta^- := \eta^-/c$; both are finite-support probability measures on $\Delta(A)$ with the same barycentre $b := \int p \, d\zeta^+(p) = \int p \, d\zeta^-(p)$. Since $b \in \Delta(A)$, the set $\{a : b(a) > 0\}$ is nonempty. Because r has full support,

$$\theta := \min\Big\{1, \min_{a : b(a) > 0} \frac{r(a)}{b(a)}\Big\} > 0.$$

Choose $0 < \lambda < \theta$ and set

$$\tau := \frac{r - \lambda b}{1 - \lambda}.$$

Then $1 - \lambda > 0$, $\sum_a \tau(a) = 1$, and $\tau(a) \geq 0$ for every a: if $b(a) > 0$ this follows from $\lambda < r(a)/b(a)$, while if $b(a) = 0$ the numerator is $r(a) > 0$. Hence $\tau \in \Delta(A)$. Define

$$\nu := \lambda \zeta^+ + (1 - \lambda)\delta_\tau, \qquad \nu' := \lambda \zeta^- + (1 - \lambda)\delta_\tau.$$

Both are finite-support probability measures on $\Delta(A)$ with barycentre $\lambda b + (1 - \lambda)\tau = r$, so $\nu, \nu' \in \mathcal{M}_r$. Then $\nu - \nu' = \lambda(\zeta^+ - \zeta^-) = (\lambda/c)\eta$, hence $\eta = (c/\lambda)(\nu - \nu') \in V_r$. Thus $W \subset V_r$ and $V_r = W$. Since W depends only on the full-support property (not on the specific prior), $V_q = V_r = V_s = W$ for all full-support priors.

Case 1: Degenerate branch posterior ($|B| = 1$). If $r = \delta_a$ for some action a, then $\mathcal{M}_{\delta_a} = \{\delta_{\delta_a}\}$ is a singleton: the only Bayes-plausible law with barycentre δ_a is the point mass at δ_a itself. Hence $V_r = \{0\}$, and every continuation experiment at prior δ_a gives the same posterior law δ_{δ_a}. We set $F_{\delta_a}(\delta_{\delta_a}) = 0$. For such branches, the term $m(x)\beta(q, r_x)F_{r_x}(\mu_{r_x, Q^x})$ is zero under any convention for $\beta(q, r_x)$.

For the aggregation formula below, assign any positive value to $\beta(q, \delta_a)$; the choice is not behaviourally identified because the corresponding continuation value is always zero.

Case 2: Non-full-support branch posterior ($1 < |B| < |A|$). If r has support $B \subsetneq A$ with $|B| \geq 2$, apply Lemma 29. Continuation comparisons at prior r reduce to those at $r|_B \in \Delta(B)$, with representative $F_r := F^B_{r|_B} \circ \pi_B$.

For any finite action set C, write

$$W_C := \{\sigma \in \operatorname{span} \Delta_f(\Delta(C)) : \sigma(\Delta(C)) = 0,\ \textstyle\int_{\Delta(C)} p \, d\sigma(p) = 0\}.$$

Let $\jmath_B : \Delta(B) \to \Delta(A)$ be the face inclusion, extending each posterior by zero outside B, and let $i_B := (\jmath_B)_\#$ act on finite signed posterior laws. By Lemma 29(iv), $F^A_r = F^B_{r|_B} \circ \pi_B$, so on face directions

$$L^A_r(i_B \sigma) = L^B_{r|_B}(\sigma) \qquad (\sigma \in W_B).$$

Since $|B| \geq 2$ and $r|_B$ is full support on B, local non-triviality and Lemma 26 give

$$L^B_{r|_B}(\chi_{r|_B} - \delta_{r|_B}) > 0,$$

so $L^B_{r|_B} \neq 0$.

Full-to-face scalar. Fix any full-support $q \in \Delta(A)$. The boundary posterior r is reachable from q: choose

$$0 < \varepsilon < \min\Big\{1, \min_{b \in B} \frac{q(b)}{r(b)}\Big\}$$

and set $P_1(1 \mid a) = \varepsilon r(a)/q(a)$, with $r(a) = 0$ for $a \notin B$. Then branch 1 has positive probability ε and posterior r. For any nonzero $\sigma \in W_B$, the tangent-space identification on the face B gives $\sigma = t(\nu - \nu')$ for some $t > 0$ and $\nu, \nu' \in \mathcal{M}^B_{r|_B}$. Implement ν, ν' by continuation channels on B and extend them arbitrarily to A. Lemma 29 identifies their comparison at the boundary prior r with their comparison at $r|_B$.

Condition (P5) applied to the branch reaching r, with all other branches fixed, gives the positive sign direction for feasible differences. The negative direction follows by swapping the two continuation laws, and equality follows by applying the weak part of Condition (P5) in both directions. Extending by positive scalar multiples from feasible differences to all of W_B, $L^A_q \circ i_B$ and $L^B_{r|_B}$ have the same sign partition on W_B. Since $L^B_{r|_B} \neq 0$, both are nonzero and there is a unique $\beta_A(q, r) > 0$ such that

$$L^A_q(i_B\sigma) = \beta_A(q, r) L^B_{r|_B}(\sigma) \qquad (\sigma \in W_B).$$

This is the ambient full-to-face coefficient used for boundary branches.

Case 3: Full-support branch posterior ($B = A$). This is the main case. Suppose r has full support on A.

Claim 1: aggregate change from branch variation. If a first-stage branch reaches posterior r with probability m, and the branch continuation law changes from ν to ν' (both in $\mathcal{M}_r$), then the aggregate posterior law changes by $m(\nu - \nu') \in m \cdot V_r$. This difference depends only on (m, ν, ν', r)—not on which experiment produced the branch, nor on the other branches.

Proof of Claim 1. The aggregate changes from $m\nu + \rho$ to $m\nu' + \rho$, where ρ collects the other branches. The difference is $m(\nu - \nu')$.

Claim 2: $L_q|_{V_r}$ is a positive scalar multiple of $L_r|_{V_r}$.

Proof of Claim 2. We first show that every direction in V_r is proportional to a feasible difference. By the Tangent-space identification above, any nonzero $\sigma \in V_r$ can be written as $\sigma = t(\nu - \nu')$ for some $t > 0$ and $\nu, \nu' \in \mathcal{M}_r$. Hence every nonzero $\sigma \in V_r$ is a positive scalar multiple of some feasible difference.

By Condition (P5) (branchwise monotonicity), if $\nu \succ_r \nu'$ (i.e. $F_r(\nu) > F_r(\nu')$, equivalently $L_r(\nu - \nu') > 0$), then the aggregate comparison ranks the ν-continuation strictly higher. Since the aggregate value difference is $L_q(m(\nu - \nu')) = mL_q(\nu - \nu')$ with $m > 0$, we have $L_q(\nu - \nu') > 0$ whenever $L_r(\nu - \nu') > 0$.

Reverse and equality directions. If $L_r(\nu - \nu') < 0$, completeness of $\succeq_r$ gives $\nu' \succ_r \nu$, and Condition (P5) applied to the reversed comparison gives $L_q(\nu' - \nu) > 0$, i.e. $L_q(\nu - \nu') < 0$. If $L_r(\nu - \nu') = 0$, then $\nu \sim_r \nu'$, so both $\nu \succeq_r \nu'$ and $\nu' \succeq_r \nu$; applying Condition (P5) weakly in both directions gives $L_q(\nu - \nu') \geq 0$ and $L_q(\nu - \nu') \leq 0$, hence $L_q(\nu - \nu') = 0$.

Since every nonzero $\sigma \in V_r$ is proportional to some feasible difference, the sign agreement extends from feasible differences to all of V_r: for any $\sigma \in V_r$, $L_q(\sigma) > 0$ iff $L_r(\sigma) > 0$, $L_q(\sigma) < 0$ iff $L_r(\sigma) < 0$, and $L_q(\sigma) = 0$ iff $L_r(\sigma) = 0$.

Two nonzero linear functionals on a vector space that have the same positive half-space must be positive scalar multiples of each other. (If $L_r \neq 0$, then $\ker L_q = \ker L_r$ and they lie in the same half-space, so $L_q = cL_r$ for some $c > 0$.) Since r has full support and local non-triviality gives $F_r(\chi_r) > F_r(\delta_r) = 0$, we have $L_r \neq 0$. Hence there exists $\beta(q,r) > 0$ such that

$$L_q(\sigma) = \beta(q,r)\, L_r(\sigma) \qquad \text{for all } \sigma \in V_r.$$

Path-independence. Consider any first-stage experiment reaching posterior r in some branch with probability m. Changing the branch continuation from ν to ν' changes the aggregate value by

$$L_q(m(\nu - \nu')) = m\, \beta(q,r)\, L_r(\nu - \nu') = m\, \beta(q,r)\, (F_r(\nu) - F_r(\nu')).$$

The coefficient $m\,\beta(q,r)$ depends only on (q,r,m). In particular, $\beta(q,r)$ is independent of which experiment reached r and independent of the other branches.

Comparing with the earlier notation $g(\nu) = \alpha F_r(\nu) + \gamma$: the value difference is $g(\nu) - g(\nu') = \alpha(F_r(\nu) - F_r(\nu'))$, so $\alpha = m\,\beta(q,r)$ and hence $\alpha/m = \beta(q,r)$ universally. The coefficient $\beta(q, r_{\bar{x}})$ is defined as the scalar from $L_q|_{V_{r_{\bar{x}}}} = \beta(q, r_{\bar{x}}) L_{r_{\bar{x}}}|_{V_{r_{\bar{x}}}}$; it depends only on q and $r_{\bar{x}}$, not on the experiment that produced the branch.

Aggregation formula. Let

$$\mu^0 := \sum_{x:m(x)>0} m(x)\delta_{r_x} = \mu_{q,P_1}.$$

For each positive-probability branch let $B_x := \operatorname{supp}(r_x)$. If $B_x = A$, set $v_x := \mu_{r_x,Q^x} - \delta_{r_x} \in W_A$. If $1 < |B_x| < |A|$, set

$$\bar{v}_x := \pi_{B_x}(\mu_{r_x,Q^x}) - \delta_{r_x|B_x} \in W_{B_x}, \qquad v_x := i_{B_x}\bar{v}_x \in W_A.$$

If $|B_x| = 1$, set $v_x := 0$. Then

$$\mu_{q,P_1 \triangleright \{Q^x\}} = \mu^0 + \sum_{x:m(x)>0} m(x) v_x.$$

Since $V_q = W_A$, affinity of F_q gives

$$F_q(\mu_{q,P_1 \triangleright \{Q^x\}}) - F_q(\mu^0) = \sum_{x:m(x)>0} m(x) L_q^A(v_x).$$

For a full-support branch $B_x = A$, Case 3 gives

$$L_q^A(v_x) = \beta(q,r_x) L_{r_x}^A(v_x) = \beta(q,r_x) F_{r_x}(\mu_{r_x,Q^x}).$$

For a nondegenerate boundary branch $1 < |B_x| < |A|$, Case 2 gives

$$L_q^A(v_x) = L_q^A(i_{B_x}\bar{v}_x) = \beta_A(q,r_x) L_{r_x|B_x}^{B_x}(\bar{v}_x) = \beta_A(q,r_x) F_{r_x}(\mu_{r_x,Q^x}).$$

The last equalities use zero normalisation: Lemma 26 in full support, Lemma 29 plus Lemma 26 on the support face in the boundary case, and the singleton convention when $|B_x| = 1$. Therefore

$$F_q(\mu_{q,P_1 \triangleright \{Q^x\}}) = F_q(\mu_{q,P_1}) + \sum_{x:m(x)>0} m(x)\, \beta(q,r_x)\, F_{r_x}(\mu_{r_x,Q^x}),$$

where $\beta(q, r_x)$ denotes the Case 3 coefficient when $B_x = A$, the Case 2 full-to-face coefficient when $1 < |B_x| < |A|$, and any positive singleton convention when $|B_x| = 1$. □

Lemma 35 (Branch-coefficient cocycle and normalised chain rule)**.** *Let the full-support and full-to-face coefficients be those constructed in Lemma 34, applied in the relevant ambient simplex or intrinsically on the relevant support face. If q has full support and r, s are nondegenerate priors with*

$$\operatorname{supp}(s) \subseteq \operatorname{supp}(r) \subseteq \operatorname{supp}(q),$$

then the coefficients satisfy the cocycle identity

$$\beta(q,s) = \beta(q,r)\beta(r,s).$$

Hence $\beta(q,r) = a_q/a_r$ for positive scales a_q, with singleton scales fixed by convention. The rescaled values

$$\widehat{F}_q(\mu) := \frac{F_q(\mu)}{a_q}$$

satisfy the exact chain rule: for every first-stage channel P_1 at full-support q and every continuation profile $\{Q^x\}_{x:m(x)>0}$,

$$\widehat{F}_q(\mu_{q,P_1\triangleright\{Q^x\}}) = \widehat{F}_q(\mu_{q,P_1}) \; + \sum_{x:m(x)>0} m(x)\,\widehat{F}_{r_x}(\mu_{r_x,Q^x}).$$

Proof. Full-support cocycle. First consider reachable full-support $q, r, s \in \operatorname{int}\Delta(A)$ with $|A| \geq 2$. For such priors, all three tangent spaces coincide. By the tangent-space identification in Lemma 34, $V_q = V_r = V_s = W$ for all full-support priors. The full-support coefficient identities constructed there share a common domain W:

$$L_q|_W = \beta(q,r)L_r|_W, \qquad L_r|_W = \beta(r,s)L_s|_W.$$

Substituting the second into the first:

$$L_q|_W = \beta(q,r)\beta(r,s)L_s|_W.$$

Comparing with $L_q|_W = \beta(q,s)L_s|_W$ and using $L_s \neq 0$:

$$\beta(q,s) = \beta(q,r)\beta(r,s).$$

Reachability. The above uses the full-support coefficient construction for the pairs (q,r), (r,s), and (q,s). Each invocation requires the pair to be reachable: the second prior must be a possible posterior of some experiment at the first prior. For any full-support q, r, construct a two-record experiment P_1 with $P_1(1 \mid a) = \varepsilon r(a)/q(a)$ for $0 < \varepsilon < \min\{1, \min_a q(a)/r(a)\}$. Record 1 has probability ε and posterior r; record 0 has posterior $(q - \varepsilon r)/(1-\varepsilon)$. Hence any full-support r is reachable from any full-support q, so the construction applies to all full-support pairs. The cocycle therefore holds for all full-support triples.

Fix a full-support basepoint q_0, set $a_{q_0} = 1$, and for full-support q define

$$a_q := \frac{1}{\beta(q_0,q)}.$$

By the full-support cocycle, this is equivalently $a_q = \beta(q,q_0)$, and

$$\beta(q,r) = \frac{\beta(q,q_0)}{\beta(r,q_0)} = \frac{a_q}{a_r}.$$

Boundary coefficients and face scales. If r is nondegenerate with support $B \subsetneq A$, Lemma 34 constructs the ambient full-to-face coefficient $\beta_A(q,r) > 0$ by

$$L_q^A i_B = \beta_A(q,r) L^B_{r|_B}.$$

When the ambient action set is clear, write this coefficient as $\beta(q,r)$.

When nondegenerate boundary priors r, s have common support B, $\beta_A(r,s)$ means the unique $c > 0$ satisfying

$$L_r^A(i_B\sigma) = c\, L_s^A(i_B\sigma) \qquad (\sigma \in W_B).$$

This is a scalar on the common face, not on all of W_A. Since

$$L_r^A \circ i_B = L^B_{r|_B}, \qquad L_s^A \circ i_B = L^B_{s|_B},$$

and the intrinsic face scalar satisfies $L^B_{r|_B}(\sigma) = \beta_B(r|_B, s|_B) L^B_{s|_B}(\sigma)$, we have

$$\beta_A(r,s) = \beta_B(r|_B, s|_B).$$

Fix the same full-support basepoint q_0 on $\Delta(A)$. If r_0 is any nondegenerate face basepoint with support B, then for every nondegenerate r with support B,

$$\beta_A(q_0, r) = \beta_A(q_0, r_0)\, \beta_B(r_0|_B, r|_B).$$

Indeed, both sides are the coefficient multiplying $L^B_{r|_B}$ to obtain $L^A_{q_0} \circ i_B$, and $L^B_{r|_B} \neq 0$. Define the ambient boundary scale by

$$a_r := \frac{1}{\beta_A(q_0, r)}.$$

If the face scale is based at r_0 with $a_{r_0} := 1/\beta_A(q_0, r_0)$, then

$$a^B_{r|_B} := \frac{a_{r_0}}{\beta_B(r_0|_B, r|_B)} = \frac{1}{\beta_A(q_0, r_0)\beta_B(r_0|_B, r|_B)} = \frac{1}{\beta_A(q_0, r)} = a_r.$$

Thus the ambient and intrinsic scales agree: treating r as a boundary posterior in $\Delta(A)$ or as a full-support prior on $\Delta(B)$ yields the same normalised representative $\widehat{F}_r$.

Extension to boundary posteriors. If r is nondegenerate with support $B \subsetneq A$, Lemma 34 defines $\beta(q, r) = \beta_A(q, r)$ by

$$L_q^A i_B = \beta_A(q, r) L^B_{r|_B}.$$

Using the full-support identity $L_q^A = \beta(q, q_0) L^A_{q_0}$ and restricting to $i_B(W_B)$,

$$L_q^A i_B = \beta(q, q_0) L^A_{q_0} i_B = \beta(q, q_0)\beta_A(q_0, r) L^B_{r|_B}.$$

Comparing with the definition of $\beta_A(q, r)$ and using $L^B_{r|_B} \neq 0$ gives

$$\beta(q, r) = \beta(q, q_0)\beta(q_0, r) = \frac{a_q}{a_r},$$

where $a_r = 1/\beta(q_0, r)$ by the boundary-scale definition above.

Singleton scale conventions. If $r = \delta_a$ is a singleton prior, set $a_{\delta_a} := a_{q_0} = 1$ and $\widehat{F}_{\delta_a} \equiv 0$. Define $\beta(q, \delta_a) := a_q / a_{\delta_a}$ as a convention. No behavioural branch coefficient is identified at a singleton target, but singleton terms in the aggregation and chain formulas are always zero.

Mixed-support cocycle. Let r, s be nondegenerate with nested supports $C := \operatorname{supp}(s) \subseteq B := \operatorname{supp}(r)$, and write $s|_B$ for s viewed as a boundary prior on the action set B. Interpret

$$\beta(r, s) := \beta_B(r|_B, s|_B),$$

the intrinsic full-to-face coefficient on action set B. Let $i_C^A : W_C \to W_A$ and $i_C^B : W_C \to W_B$ be the face inclusions. For every $\sigma \in W_C$,

$$L_q^A i_C^A \sigma = \beta(q, r) L^B_{r|_B} i_C^B \sigma = \beta(q, r)\beta(r, s) L^C_{s|_C} \sigma.$$

The direct full-to-face definition from Lemma 34 for the pair (q, s) gives

$$L_q^A i_C^A \sigma = \beta(q, s) L^C_{s|_C} \sigma.$$

Since $L^C_{s|_C} \neq 0$, $\beta(q,s) = \beta(q,r)\beta(r,s)$. If $\operatorname{supp}(s) \not\subseteq \operatorname{supp}(r)$, then $\beta(r,s)$ is not behaviourally defined; singleton targets are also excluded from the behavioural cocycle because their scales are conventions only.

Chain rule. Applying Lemma 34 and substituting $\beta(q,r) = a_q/a_r$ gives

$$F_q(\mu_{q,P_1 \triangleright \{Q^x\}}) = F_q(\mu_{q,P_1}) + \sum_{x:m(x)>0} m(x)\,\frac{a_q}{a_{r_x}}\,F_{r_x}(\mu_{r_x,Q^x}).$$

Dividing by a_q and writing $\widehat{F}_q(\mu) := F_q(\mu)/a_q$ yields

$$\widehat{F}_q(\mu_{q,P_1 \triangleright \{Q^x\}}) = \widehat{F}_q(\mu_{q,P_1}) + \sum_{x:m(x)>0} m(x)\,\widehat{F}_{r_x}(\mu_{r_x,Q^x}).$$ □

Lemma 36 (Coherent relabelling and face scales)**.** *The scales produced by Lemma 35 can be chosen simultaneously across all finite action sets so that:*

1. *for every bijection $\pi : A \to A'$,*
$$a^{A'}_{q\pi} = a^A_q;$$
2. *if $\iota : B \hookrightarrow A$ is an injection with $|B| \geq 2$, r has full support on B, and q has full support on A, then the full-to-face branch coefficient for reaching the face $\iota(B)$ is*
$$\beta_\iota(q,r) = \frac{a^A_q}{a^B_r}. \tag{FS}$$

Singleton target scales remain a convention because every continuation value on a singleton fibre is zero.

Proof. For every nondegenerate finite action set A, first choose the full-support scales supplied by the cocycle part of Lemma 35, so that

$$\beta_A(q,q') = \frac{a^A_q}{a^A_{q'}} \qquad (q,q' \in \operatorname{int}\Delta(A)).$$

They are unique up to multiplication by one positive constant depending on A. Normalise initially by $a^A_{u_A} = 1$, where u_A is uniform. Exact relabelling invariance of the representatives implies relabelling invariance of the uniquely determined branch coefficients. Since bijections carry uniform priors to uniform priors, the initial scales are exactly invariant under bijections.

Let $\iota : B \hookrightarrow A$ be an injection between nondegenerate finite sets. By relabelling B onto its image $\iota(B)$ and using Corollary 32, it is enough to define the coefficient for the inclusion of the face $\iota(B)$; the notation below keeps the original symbol ι. Let i_ι denote the induced map on signed posterior-law tangent spaces. For full-support $q \in \Delta(A)$ and $r \in \Delta(B)$, let $\beta_\iota(q,r) > 0$ be the full-to-face scalar characterised by

$$L^A_q \circ i_\iota = \beta_\iota(q,r) L^B_r \quad \text{on } W_B.$$

The number

$$K(\iota) := \beta_\iota(q,r)\frac{a^B_r}{a^A_q} \tag{K}$$

is independent of both q and r. Independence of q follows by comparing two full-support ambient priors and using $L^A_q = (a^A_q/a^A_{q'})L^A_{q'}$ on W_A. Independence of r follows from same-face compatibility: if r,s have full support on B, then $L^B_r = (a^B_r/a^B_s)L^B_s$, and hence $\beta_\iota(q,s) = \beta_\iota(q,r)a^B_r/a^B_s$.

The defects multiply under composition. If $C \overset{\jmath}{\hookrightarrow} B \overset{\iota}{\hookrightarrow} A$ and all three sets in the comparison are nondegenerate, then restricting first to B and then to C gives, for full-support q, r, s on A, B, C respectively,

$$\beta_{\iota\circ\jmath}(q, s) = \beta_\iota(q, r)\beta_\jmath(r, s).$$

Substituting the definition of K and cancelling the intermediate scale a_r^B gives

$$K(\iota \circ \jmath) = K(\iota)K(\jmath). \tag{KC}$$

Exact relabelling invariance shows that $K(\iota)$ depends only on $n := |A|$ and $m := |B|$; write it as $K_{n,m}$. Bijections give $K_{n,n} = 1$, and

$$K_{n,\ell} = K_{n,m}K_{m,\ell} \qquad (2 \le \ell \le m \le n).$$

Put $t_n := K_{n,2}$, with $t_2 = 1$. Then $K_{n,m} = t_n/t_m$. Finally replace every scale on an n-point action set by

$$\widetilde{a}_q^A := t_n a_q^A.$$

Under this change, the defect in (K) becomes

$$\widetilde{K}_{n,m} = K_{n,m}\frac{t_m}{t_n} = 1.$$

The rescaling is common to all priors on a fixed action set, so it leaves all within-set cocycle ratios unchanged; because t_n depends only on cardinality, it also preserves exact relabelling invariance. Dropping tildes gives (FS). Singleton target scales may be set arbitrarily because all singleton continuation terms are zero. □

Lemma 37 (Interaction collapse and universal chain scale)**.** *The universal interaction coefficient in* (QA) *is zero. Moreover, there exists $a > 0$ such that the coherently chosen chain scales satisfy $a_q = a$ for every nondegenerate full-support prior on every finite action set. The same value may be assigned to singleton priors. Consequently,*

$$F_{p\otimes r}(\mu \otimes \nu) = F_p(\mu) + F_r(\nu) \tag{PA}$$

and the branch formula has a universal chain scale.

Proof. Define, for every finite prior q read on its support,

$$H(q) := F_q(\chi_q), \qquad Z(q) := 1 + \kappa H(q).$$

For a nondegenerate prior, local non-triviality gives $H(q) > 0$, and (POS) gives

$$Z(q) > 0. \tag{Z+}$$

For a singleton, $H(q) = 0$ and $Z(q) = 1$.

Step 1: product revelation links the chain scale to Z. Let $q \in \Delta(A)$ and $r \in \Delta(B)$ be nondegenerate and full support. Quasi-additivity applied to full revelation gives

$$H(q \otimes r) = H(q) + H(r) + \kappa H(q)H(r). \tag{QH}$$

Compute the same value sequentially. First reveal the A-coordinate and then reveal the B-coordinate in every branch. The first-stage value is $F_{q\otimes r}(\chi_q \otimes \delta_r) = H(q)$ by (QA). After observing a, the posterior is $e_a \otimes r$ on the face $\{a\} \times B$; support restriction and relabelling identify its continuation full-revelation value with $H(r)$. For the injection

$$\iota_a : B \hookrightarrow A \times B, \qquad b \mapsto (a, b),$$

Lemma 36 gives the branch coefficient

$$\beta_{\iota_a}(q \otimes r, r) = \frac{a^{A\times B}_{q\otimes r}}{a^B_r}.$$

The branch weights are $q(a)$ and the same coefficient and continuation value appear in every branch, so $\sum_a q(a) = 1$. Lemma 34 therefore yields

$$H(q \otimes r) = H(q) + \frac{a_{q\otimes r}}{a_r} H(r). \tag{SA}$$

Comparing (QH) and (SA), and using $H(r) > 0$,

$$\frac{a_{q\otimes r}}{a_r} = Z(q). \tag{R1}$$

Revealing B first gives symmetrically

$$\frac{a_{q\otimes r}}{a_q} = Z(r). \tag{R2}$$

Hence $a_r Z(q) = a_q Z(r)$, so there is a universal constant $C > 0$ such that

$$a_q = CZ(q) \tag{AS}$$

for every nondegenerate full-support prior on every finite action set. Assign $a_q := C$ at singleton priors, where $Z(q) = 1$ and all continuation values are zero.

If $\kappa = 0$, then $Z \equiv 1$, equation (AS) already gives universal scales, and (QA) is exact product additivity. For the remaining argument suppose $\kappa \neq 0$.

Step 2: the induced weight equation. Let q be full support on A, let $\mathcal{P} = \{A_1, \ldots, A_m\}$ be a partition into nonempty cells, and put

$$p_i := q(A_i), \qquad q^i := q(\cdot \mid A_i), \qquad p := (p_1, \ldots, p_m).$$

Let $G_{\mathcal{P}}$ reveal the block. Undetectable-distinctions neutrality gives $(q, G_{\mathcal{P}}) \sim (p, \mathrm{Id}_{\{1,\ldots,m\}})$. The early cross-prior representation Lemma 33, importantly for the *unscaled* F's, therefore gives

$$F_q(\mu_{q,G_{\mathcal{P}}}) = H(p). \tag{BG}$$

Append full revelation in every branch. The compound posterior law is χ_q, so the chain formula and (AS) give

$$H(q) = H(p) + Z(q) \sum_{i=1}^{m} p_i \frac{H(q^i)}{Z(q^i)}. \tag{GR}$$

Define

$$w(q) := \frac{1}{Z(q)} > 0.$$

Using $\kappa H(x) = Z(x) - 1$ in (GR) gives

$$\frac{Z(p)}{Z(q)} = \sum_{i=1}^{m} p_i \frac{1}{Z(q^i)}.$$

Equivalently,

$$w(q) = w(p) \sum_{i=1}^{m} p_i w(q^i). \tag{W}$$

No continuity assertion is needed here.

Step 3: a two-grouping argument forces $w \equiv 1$. All distributions in this paragraph are read on their positive-probability supports, so the weight equation applies after support restriction. First, (W) applied to a product distribution gives

$$w(u \otimes v) = w(u)w(v), \tag{M}$$

because the first-coordinate block probabilities are u and every conditional distribution is v.

Take arbitrary finite probability vectors u and v, possibly on different alphabets, and write

$$x := w(u), \qquad y := w(v), \qquad p := (1/2, 1/2).$$

On the labelled disjoint union $(U \times U)^0 \sqcup (V \times V)^1$, form the distribution

$$T := \tfrac{1}{2}(u \otimes u)^0 \ \sqcup\ \tfrac{1}{2}(v \otimes v)^1.$$

Grouping first by the top block and using (M),

$$w(T) = w(p)\frac{x^2 + y^2}{2}. \tag{E1}$$

Now group first by the pair consisting of the top-block label and the first within-block coordinate. Its marginal is

$$S := \tfrac{1}{2}u^0 \ \sqcup\ \tfrac{1}{2}v^1,$$

and (W) gives

$$w(S) = w(p)\frac{x + y}{2}.$$

Conditional on a point in the first part of S, the remaining coordinate has distribution u; conditional on a point in the second part, it has distribution v. A second application of (W) therefore gives

$$w(T) = w(S)\frac{x + y}{2} = w(p)\left(\frac{x + y}{2}\right)^2. \tag{E2}$$

Since $w(p) > 0$, comparison of (E1) and (E2) yields

$$\frac{x^2 + y^2}{2} = \left(\frac{x + y}{2}\right)^2,$$

so $(x - y)^2 = 0$. As u and v were arbitrary, w is constant on all finite probability vectors. At a singleton prior, $H = 0$, hence $w = 1$. Therefore $w(q) \equiv 1$.

Thus $Z(q) = 1$ and $\kappa H(q) = 0$ for every q. Local non-triviality gives a nondegenerate q with $H(q) > 0$, hence $\kappa = 0$, contradicting the temporary alternative unless the interaction coefficient is zero. Equation (AS) now gives $a_q = C$ universally. Finally (QA) reduces to exact product additivity (PA). □

Lemma 38 (Entropy reduction and Faddeev)**.** *By Lemma 37, $a_q \equiv a$ is universal. Define*

$$\widehat{F}_q := F_q/a, \qquad H(q) := \widehat{F}_q(\chi_q),$$

the rescaled value of full revelation. Then, for every channel P,

$$\widehat{F}_q(\mu_{q,P}) = H(q) - \sum_{x : m_{q,P}(x) > 0} m_{q,P}(x)\, H(r_x^{q,P}). \tag{ER}$$

Moreover H satisfies Faddeev's recursion, so

$$H(q) = \alpha H(q)$$

for some $\alpha > 0$. Consequently,

$$\widehat{F}_q(\mu_{q,P}) = \alpha I_{q,P}(A; X).$$

Proof. Entropy-reduction formula. Boundary values of H are read through support restriction: if q has support B, then $H(q) := H(q|_B)$, and $H(\delta_a) = 0$. Let $P : A \to \Delta(X)$ be any channel at full-support prior q. Append full revelation $\mathrm{Id}_A : A \to \Delta(A)$ after P: the compound $P \triangleright \{\mathrm{Id}_A\}_x$ (using Id_A in every branch) has $\mu_{r_x,\mathrm{Id}_A} = \chi_{r_x}$ and the overall posterior law is full revelation χ_q. Applying the chain rule (Lemma 35) gives

$$\widehat{F}_q(\chi_q) = \widehat{F}_q(\mu_{q,P}) + \sum_{x:m(x)>0} m(x)\, \widehat{F}_{r_x}(\chi_{r_x}).$$

Hence

$$\widehat{F}_q(\mu_{q,P}) = H(q) - \sum_{x:m(x)>0} m(x)\, H(r_x).$$

If q is on the boundary, restrict q and P to $B = \mathrm{supp}(q)$, apply the full-support formula on $\Delta(B)$, and pull the value and comparison back by Lemma 29; zero-probability rows and records do not affect either side of the displayed identity.

Scaled cross-prior bridge. Lemma 33 was proved for the unscaled values F_q. Since Lemma 37 has made $a_q \equiv a$ universal, it is equivalently valid for

$$V(q, P) := \widehat{F}_q(\mu_{q,P}) = F_q(\mu_{q,P})/a :$$

for all finite priors and channels,

$$q^0 \succeq_{P \sqcup Q} r^1 \iff V(q, P) \geq V(r, Q).$$

Faddeev recursion. First let q be full support and let $\mathcal{P} = \{A_1, \ldots, A_m\}$ be a partition into nonempty cells. Then $p_i := q(A_i) > 0$ for every i. Let $G_{\mathcal{P}} : A \to \Delta(\{1, \ldots, m\})$ reveal only the block index: $G_{\mathcal{P}}(i \mid a) = \mathbf{1}_{a \in A_i}$. Write

$$p_i := q(A_i), \qquad q^i := q(\cdot \mid A_i),$$

and let $p = (p_1, \ldots, p_m) \in \Delta(\{1, \ldots, m\})$. Consider the deterministic kernel $S : A \to \Delta(\{1, \ldots, m\})$ given by $S(i \mid a) = \mathbf{1}_{a \in A_i}$, so $qS = p$.

By Lemma 28, since $G_{\mathcal{P}}$ is S-measurable,

$$(q, G_{\mathcal{P}}) \sim (p, \mathrm{Id}_{\{1,\ldots,m\}})$$

in the two-block comparison. Applying the scaled form of Lemma 33 to the forward comparison gives $V(q, G_{\mathcal{P}}) \geq V(p, \mathrm{Id}_{\{1,\ldots,m\}})$. Applying the reverse-block form from Lemma 20 and then the same bridge to the reverse comparison gives $V(p, \mathrm{Id}_{\{1,\ldots,m\}}) \geq V(q, G_{\mathcal{P}})$. Therefore

$$\widehat{F}_q(\mu_{q,G_{\mathcal{P}}}) = \widehat{F}_p(\chi_p) = H(p).$$

The entropy-reduction formula applied to $G_{\mathcal{P}}$ gives

$$H(q) = \widehat{F}_q(\chi_q) = \widehat{F}_q(\mu_{q,G_{\mathcal{P}}}) + \sum_{i=1}^m p_i \widehat{F}_{q^i}(\chi_{q^i}) = H(p) + \sum_{i=1}^m p_i H(q^i).$$

This is Faddeev's recursion. For boundary q, restrict to $\mathrm{supp}(q)$, apply the same full-support recursion to the nonempty positive-mass intersections, and pull back by Lemma 29; equivalently, the recursion is read over positive-mass blocks only.

We now verify that H is continuous on each $\Delta_n := \{q \in \Delta(A) : |A| = n\}$.

Binary continuity via erasure calibrators. Define the binary entropy function $h : [0, 1] \to \mathbb{R}$ by $h(t) := H(t, 1 - t)$. We show h is continuous using prior-independent calibrators. By Lemma 26, support restriction, and $a > 0$, $H(q) \geq 0$ for every q, so $h(t) \geq 0$ on $[0, 1]$. If $c < 0$, then

$$\{t : h(t) \geq c\} = [0, 1], \qquad \{t : h(t) \leq c\} = \varnothing,$$

both closed. Now fix $c \geq 0$. By local non-triviality, there exists a binary prior $\theta = (s, 1-s)$ with $u := H(\theta) > 0$. Choose $n \in \mathbb{N}$ and $\lambda \in [0,1]$ such that $c = \lambda n u$ (take n large enough that $c \leq nu$, then set $\lambda := c/(nu)$).

Let $w_n := \theta^{\otimes n}$ be the n-fold product prior on $B_n := \{0,1\}^n$. By product additivity, $H(w_n) = nH(\theta) = nu$. Define the erasure experiment $E_{n,\lambda} : B_n \to \Delta(\{*, 1, \ldots, 2^n\})$ which reveals the state fully with probability λ and reveals nothing with probability $1 - \lambda$. By the entropy-reduction formula,

$$\widehat{F}_{w_n}(\mu_{w_n, E_{n,\lambda}}) = \lambda H(w_n) = \lambda n u = c.$$

Now consider the block action space $A_c := \{0,1\} \sqcup B_n$ and the block-diagonal channel $K_c := \mathrm{Id}_{\{0,1\}} \sqcup E_{n,\lambda}$. For $t \in [0,1]$, let q_t^0 be the prior $(t, 1-t)$ supported on block $\{0,1\}$, and let w_n^1 be w_n supported on block B_n. By the scaled form of Lemma 33 (which extends to boundary priors via Lemma 29),

$$q_t^0 \succeq_{K_c} w_n^1 \quad \Longleftrightarrow \quad V(q_t, \mathrm{Id}_{\{0,1\}}) = h(t) \geq c = V(w_n, E_{n,\lambda}).$$

Since K_c is fixed and the map $t \mapsto q_t^0$ is continuous in ℓ_1, Condition (P2) implies that the set $\{t \in [0,1] : h(t) \geq c\}$ is closed. Similarly, $\{t \in [0,1] : h(t) \leq c\}$ is closed. Thus all real superlevel and sublevel sets of h are closed, so h is both upper and lower semicontinuous on $[0,1]$, hence continuous.

Continuity on higher simplices by induction. For $n \geq 3$, write any $q \in \Delta_n$ as $q = (q_1, (1-q_1)\bar{q})$ where $\bar{q} := (q_2/(1-q_1), \ldots, q_n/(1-q_1)) \in \Delta_{n-1}$ for $q_1 < 1$. The Faddeev recursion with the binary partition $\{\{1\}, \{2, \ldots, n\}\}$ gives

$$H(q) = h(q_1) + (1 - q_1) H(\bar{q}).$$

On the region $0 < q_1 < 1$, this is continuous: h is continuous by binary continuity, $H|_{\Delta_{n-1}}$ is continuous by the induction hypothesis, and $q \mapsto (q_1, \bar{q})$ is continuous. If $q_1 = 0$, support restriction, equivalently the positive-mass-block convention in the recursion, gives $H(q) = H(\bar{q})$ and $h(0) = 0$, so continuity at that face follows from the induction hypothesis. At $q_1 = 1$, the induction hypothesis and compactness of Δ_{n-1} imply H is bounded on Δ_{n-1}, so $(1-q_1)H(\bar{q}) \to 0$ as $q_1 \to 1$. Thus $H(q) \to h(1) = 0 = H(\delta_1)$. The other boundary faces are handled by support restriction, relabelling on the support, and pulling back. This completes the induction.

We have verified the hypotheses of the strong-additivity form of Faddeev's finite entropy characterisation, as stated for example by Baez–Fritz–Leinster [7, Theorem 6], following Faddeev's original theorem [23]: a function on finite probability vectors that is continuous on each closed simplex, invariant under bijections, expansive under adjoining zero-probability states, zero on point masses, and strongly additive is a nonnegative multiple of Shannon entropy. Here invariance under bijections follows from Corollary 32, with boundary cases obtained by restricting to the support and pulling back; expansibility follows from support restriction; and the grouping recursion above is precisely strong additivity, read over positive-mass blocks. Faddeev's theorem therefore gives $H(q) = \alpha H(q)$ for some $\alpha \geq 0$. Local non-triviality (Condition (P1)) forces $\alpha > 0$.

Mutual information. Substituting $H = \alpha H$ into the entropy-reduction formula yields

$$\widehat{F}_q(\mu_{q,P}) = \alpha H(q) - \sum_{x : m_{q,P}(x) > 0} m_{q,P}(x) \alpha H(r_x^{q,P}) = \alpha(H(q) - \mathbb{E}[H(r_X)]) = \alpha I_{q,P}(A; X). \quad \square$$

Lemma 39 (Fixed-environment representation)**.** *For every finite channel $P : A \to \Delta(X)$ and all $q, q' \in \Delta(A)$,*

$$q \succeq_P q' \quad \Longleftrightarrow \quad I_{q,P}(A; X) \geq I_{q',P}(A; X).$$

Proof. Let

$$V(s,R) := \widehat{F}_s(\mu_{s,R}),$$

with boundary priors interpreted by Lemma 29. Lemma 33, divided by the universal scale from Lemma 37, gives the cross-prior block-comparison bridge: for finite action sets, priors s, t, and channels R, S,

$$s^0 \succeq_{R \sqcup S} t^1 \quad \Longleftrightarrow \quad V(s,R) \geq V(t,S),$$

including boundary priors by support restriction. Lemma 38 gives, for every finite prior/channel pair,

$$V(s,R) = \alpha I_{s,R}$$

with $\alpha > 0$.

Fix $P : A \to \Delta(X)$ and $q, q' \in \Delta(A)$. By the first clause of Condition (P3),

$$q \succeq_P q' \quad \Longleftrightarrow \quad q^0 \succeq_{P \sqcup P} (q')^1.$$

Applying this bridge with $(s,R) = (q,P)$ and $(t,S) = (q',P)$ gives

$$q^0 \succeq_{P \sqcup P} (q')^1 \quad \Longleftrightarrow \quad V(q,P) \geq V(q',P).$$

Substituting $V = \alpha I$ and using $\alpha > 0$ yields

$$q \succeq_P q' \quad \Longleftrightarrow \quad I_{q,P}(A;X) \geq I_{q',P}(A;X).$$

If q or q' is on the boundary, the bridge is read after restricting each pair to its positive-probability support and then pulling the comparison back by Lemma 29. Deleting zero-probability actions does not change mutual information, and the Condition (P3) equivalence remains the ambient fixed-channel comparison for P. □

Proof of Theorem 16. That (ii) implies (i) is the necessity verification at the start of this appendix. That (i) implies (ii) is Lemma 39. For the moreover clause, apply (ii) to the fixed block environment $\bigsqcup_{k \in K} P_k$: a lottery supported on block i has constant block label, so

$$I_{q_i^i, \bigsqcup_{k \in K} P_k} = I_{q_i, P_i}(A_i; X_i),$$

and similarly for block j. □

C.3 Uniqueness

Proof of Proposition 17. First note that the range of $(q,P) \mapsto I_{q,P}$ is $[0,\infty)$. For $q \in \Delta(A)$ and $t \in [0,1]$, let $E_t : A \to \Delta(A \sqcup \{*\})$ reveal the realised action with probability t and otherwise report $*$:

$$E_t(a' \mid a) = t\mathbf{1}_{a'=a}, \qquad E_t(* \mid a) = 1 - t.$$

Then $I_{q,E_t} = tH(q)$. In particular, if u_n is uniform on $n \geq 2$ actions, the values at u_n fill $[0, \log n]$. These intervals are unbounded, so every non-negative value is attained; mutual information is non-negative, so there are no other values.

For part (i), let V be a common-scale representation and define $\varphi : [0,\infty) \to \mathbb{R}$ by $\varphi(v) := V(q,P)$ for any pair satisfying $I_{q,P} = v$. The definition is independent of the chosen pair. Indeed, if $I_{q,P} = I_{r,Q}$, the moreover clause of Theorem 16 gives $q^0 \sim_{P \sqcup Q} r^1$, and the common-scale property gives $V(q,P) = V(r,Q)$. If $I_{q,P} > I_{r,Q}$, the block comparison is strict—the reverse weak comparison would require the opposite information inequality—so $V(q,P) > V(r,Q)$. Thus φ is strictly increasing and $V = \varphi \circ I$. Conversely, Theorem 16 shows that every $\varphi \circ I$ with φ strictly increasing is a common-scale representation.

For part (ii), every aI with $a > 0$ is a common-scale representation by part (i) and is branch-additive by the mutual-information chain rule (N3). Conversely, let the common-scale representation $V = \varphi \circ I$ be branch-additive. Composing the one-record uninformative channel with itself gives $\varphi(0) = 2\varphi(0)$, hence $\varphi(0) = 0$.

Fix $n \geq 2$, put $q = u_n$, and write $h_n := H(u_n) = \log n$. For $0 < \lambda < 1$, let the first stage be a public coin with two records and identical rows $(\lambda, 1 - \lambda)$. Its information is zero, each posterior is q, and its branch weights are λ and $1 - \lambda$. For any $x \in [0, h_n]$, use the erasure channel E_{x/h_n} in the first branch and the uninformative channel in the second. The chain rule (N3) makes the compound information λx, while branch additivity gives

$$\varphi(\lambda x) = \varphi(0) + \lambda\varphi(x) + (1 - \lambda)\varphi(0) = \lambda\varphi(x).$$

The same identity holds at $\lambda = 0, 1$ because $\varphi(0) = 0$. Taking $x = h_n$ and $\lambda = z/h_n$ therefore shows that

$$\varphi(z) = \frac{\varphi(h_n)}{h_n} z \qquad (0 \leq z \leq h_n).$$

The intervals $[0, h_n]$ are nested and unbounded; since they carry the same function φ, their slopes agree. Hence $\varphi(z) = az$ on $[0, \infty)$ for a single constant a, and $a > 0$ because φ is strictly increasing. □

C.4 Independence of the pure-record conditions

Proof of Proposition 19. For each condition, we construct a family of weak orders that satisfies the other four and violates it.

Condition (P1). Let every pair of lotteries be indifferent in every environment. The graphs in (P2) are then the whole parameter space, and (P3)–(P5) hold trivially; the strict clause of (P5) is vacuous. Local non-triviality fails.

Condition (P2). A discontinuous refinement of mutual information gives the required family. Choose an additive map $g : \mathbb{R} \to \mathbb{R}$ that is not of the form $g(x) = cx$, and, for a finite distribution p, set

$$H_g(p) := -\sum_{a:p(a)>0} p(a)g(\log p(a)).$$

Because $g(x + y) = g(x) + g(y)$, this entropy obeys the grouping identity and is additive on products, although it need not be continuous, concave, or non-negative. Define the associated information reduction by

$$I^g_{q,P} := H_g(q) - \sum_{x:m_{q,P}(x)>0} m_{q,P}(x)H_g(r^{q,P}_x),$$

and compare procedures lexicographically:

$$q \succeq_P q' \quad \Longleftrightarrow \quad (I_{q,P}, I^g_{q,P}) \geq_{\text{lex}} (I_{q',P}, I^g_{q',P}).$$

This is a weak order, and the first coordinate gives local non-triviality. A block-supported lottery has the same two coordinates on its own block, so (P3) holds. Both coordinates obey the chain rule; hence branchwise lexicographic improvements aggregate, with strictness, and (P5) holds.

For (P4), let (A, X, A', X') carry the joint law in (DP). Shannon data processing settles the comparison whenever $I(A; X) > I(A'; X')$. In the equality case,

$$I(A; X) - I(A'; X') = I(A; X \mid A') + I(A'; X \mid X') = 0.$$

Thus $A \perp X \mid A'$ and $A' \perp X \mid X'$. Product additivity makes the corresponding two conditional I^g terms zero as well, so $I^g(A; X) = I^g(A'; X')$. The second coordinate therefore ties whenever the first one ties, which proves (P4).

It remains to choose g so that continuity fails. Let $p = (1/2, 1/4, 1/8, 1/8)$, and choose an interior $r \in \Delta(\{1, 2, 3, 4\})$ on the same Shannon-entropy level set such that $\log 2, \log r_1, \ldots, \log r_4$ are linearly independent over $\mathbb{Q}$; generic r on that level set has this property. Extend these numbers to a Hamel basis and prescribe

$$g(\log 2) = 0, \qquad g(\log r_1) = -1/r_1, \qquad g(\log r_i) = 0 \quad (i = 2, 3, 4).$$

Then $H(p) = H(r)$ but $H_g(p) = 0 < 1 = H_g(r)$. Under full revelation, therefore, $p \not\succeq_{\mathrm{Id}_4} r$. If $p_n \to p$ moves from p toward the uniform distribution, strict concavity gives $H(p_n) > H(p) = H(r)$, so $p_n \succ_{\mathrm{Id}_4} r$ for every n. The graph in (P2) is not closed.

Condition (P3). The labelled copies used in block environments make a simple label-dependent family possible. Treat labels as syntactic tags, with the outermost tag taking precedence, and define

$$w(a^0) = 1, \qquad w(a^i) = 0 \quad (i \neq 0),$$

with $w(a) = 0$ for untagged actions. For a fixed $\varepsilon > 0$, represent each environment by

$$V_\varepsilon(q, P) := I_{q,P} + \varepsilon \sum_a q(a)w(a).$$

This continuous criterion defines a weak order and satisfies local non-triviality. Every comparison in (P4) places the original procedure in block 0 and the processed procedure in block 1, so its value difference is

$$I(A; X) - I(A'; X') + \varepsilon > 0.$$

For (P5), write $\Delta_x := I_{r_x,Q^x} - I_{r_x,R^x}$. A branch comparison is weak precisely when $\Delta_x + \varepsilon \geq 0$, while the aggregate value difference is

$$\sum_x m(x)\Delta_x + \varepsilon = \sum_x m(x)(\Delta_x + \varepsilon).$$

The weak and strict conclusions of (P5) follow immediately. Duplication coherence fails: take an untagged binary action set, $P = \mathrm{Id}_2$, a degenerate q, and a non-degenerate q' close enough to q that $0 < H(q') < \varepsilon$. Then $q \not\succeq_P q'$, whereas $q^0 \succ_{P \sqcup P} (q')^1$ because the former receives the block premium.

Condition (P4). Represent every environment by $V(q, P) := (1 + H(q))I_{q,P}$. Continuity and block coherence are immediate. In each branch comparison the entropy factor is common to the two continuations, so the mutual-information chain rule gives (P5), including strictness. To violate (P4), take $A = \{1, 2\}$, $q = (0.99, 0.01)$, $P = \mathrm{Id}_A$, and split action 1 uniformly between two synonymous reported labels while sending action 2 to a third. The reported act still determines the original act, so mutual information is unchanged, but $H(qS) > H(q)$. The processed procedure is strictly preferred, contrary to (DP-act).

Condition (P5). Let $V(q, P) := \min\{I_{q,P}, c\}$ for a fixed $c > 0$. This continuous, weakly increasing transform of mutual information satisfies (P1)–(P4). For $c = 1$ bit, take q uniform on four actions and let the first stage reveal which of two pairs contains the realised act. Full revelation within each pair strictly improves every branch from zero to one bit, but the two compound procedures carry respectively one and two bits before capping and are therefore indifferent. The strict conclusion of (P5) fails. $\square$